

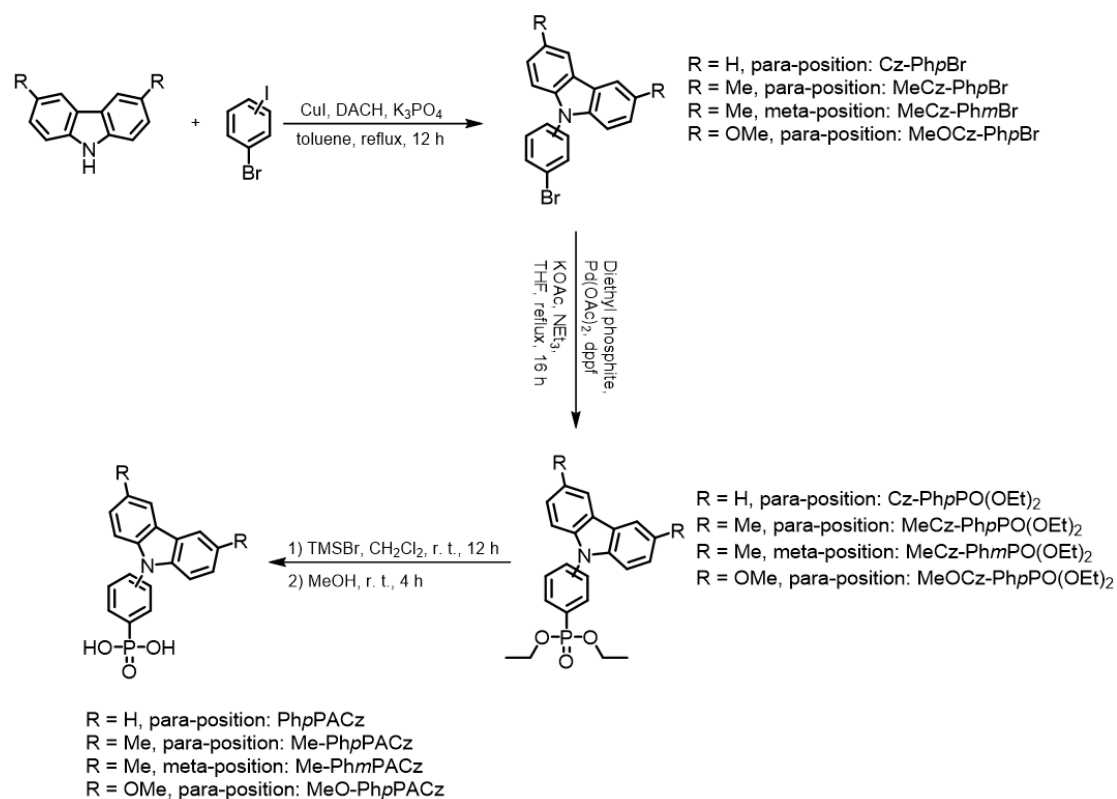
Joule, Volume 8

Supplemental information

Conjugated linker-boosted self-assembled monolayer molecule for inverted perovskite solar cells

Geping Qu, Siyuan Cai, Ying Qiao, Deng Wang, Shaokuan Gong, Danish Khan, Yu Wang, Kui Jiang, Qian Chen, Letian Zhang, Yang-Gang Wang, Xihan Chen, Alex K.-Y Jen, and Zong-Xiang Xu

1 **Supplemental Figures and Tables**



2

3 **Figure S1.** Synthetic routes for phenylene-linked phosphonic acid based self-assembled
 4 monolayer materials with different carbazole derivatives.

5

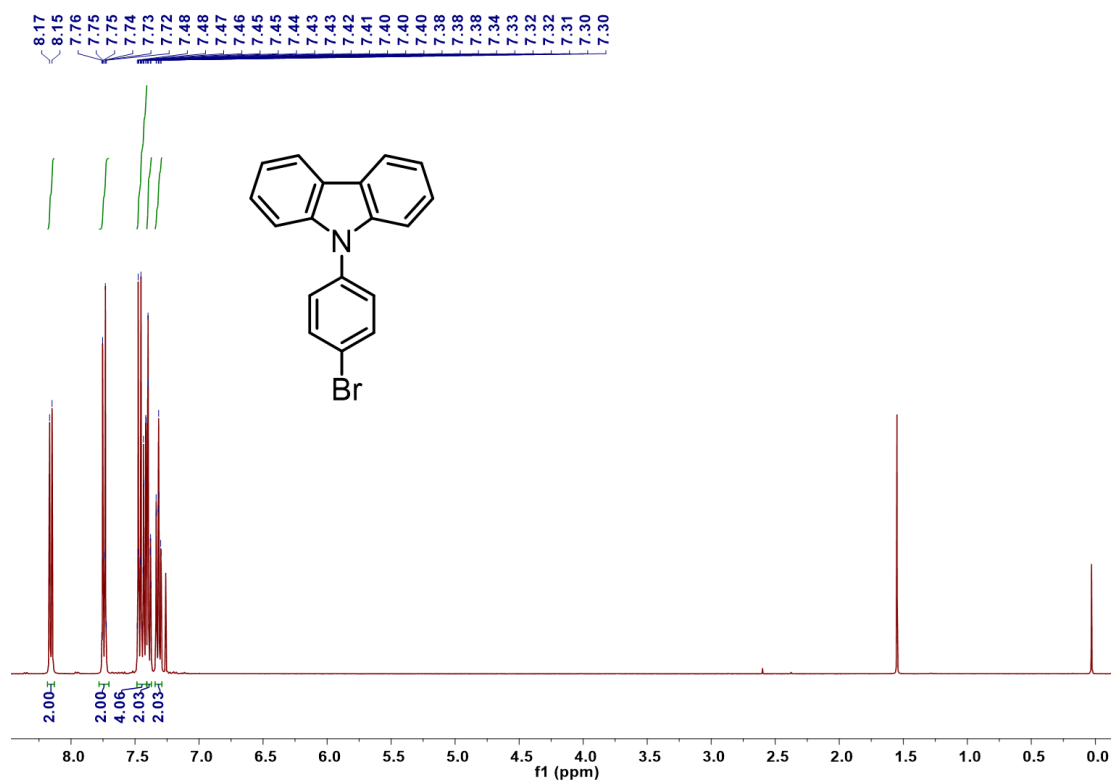


Figure S2. ¹H NMR spectrum of **Cz-PhBr** (400 MHz, CDCl₃).

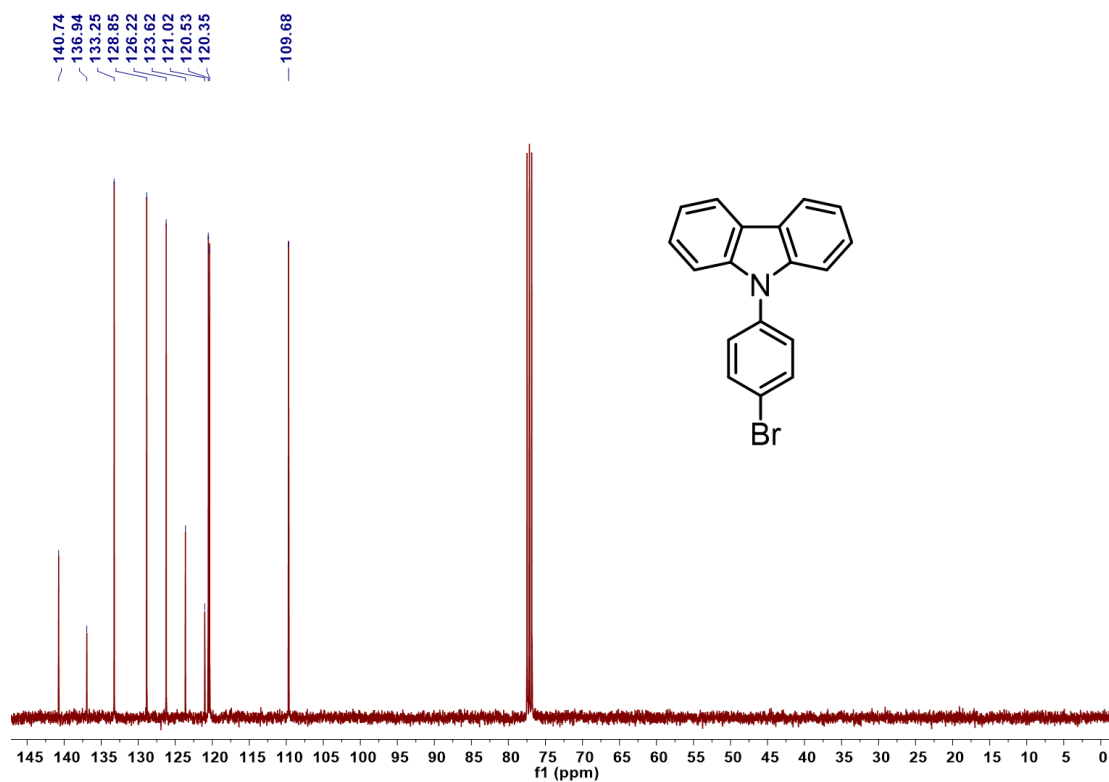


Figure S3. ^{13}C NMR spectrum of **Cz-PhBr** (101 MHz, CDCl_3).

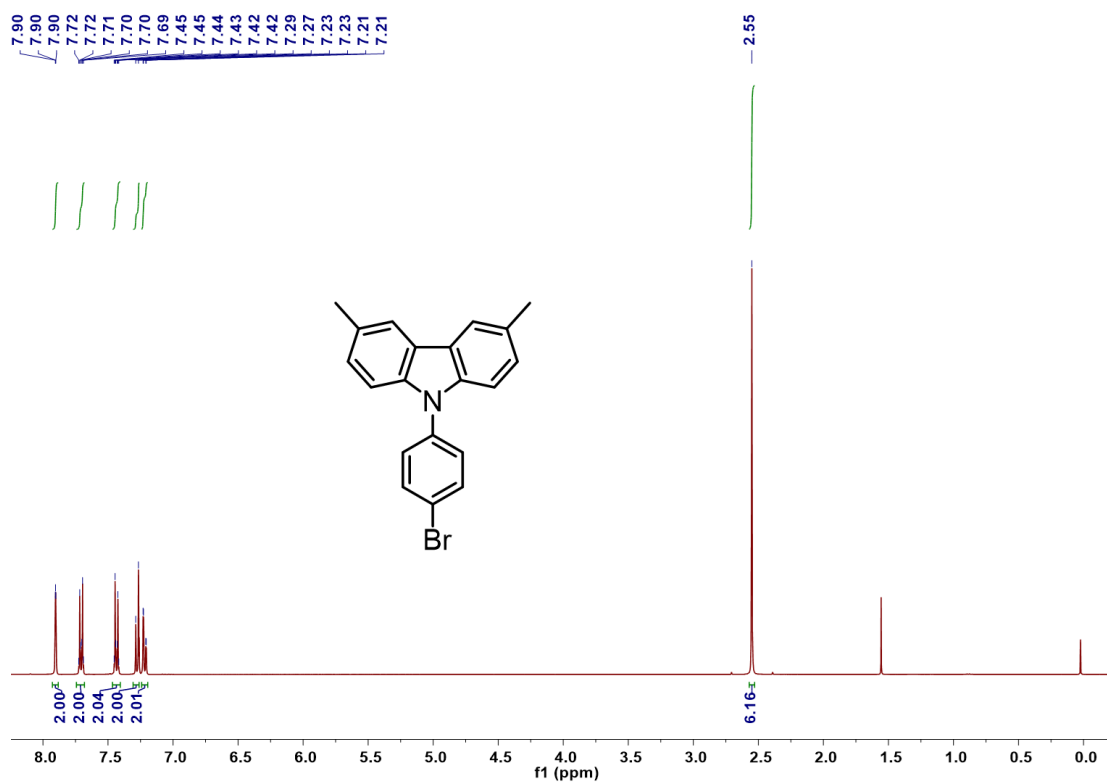


Figure S4. ¹H NMR spectrum of **MeCz-PhBr** (400 MHz, CDCl₃).

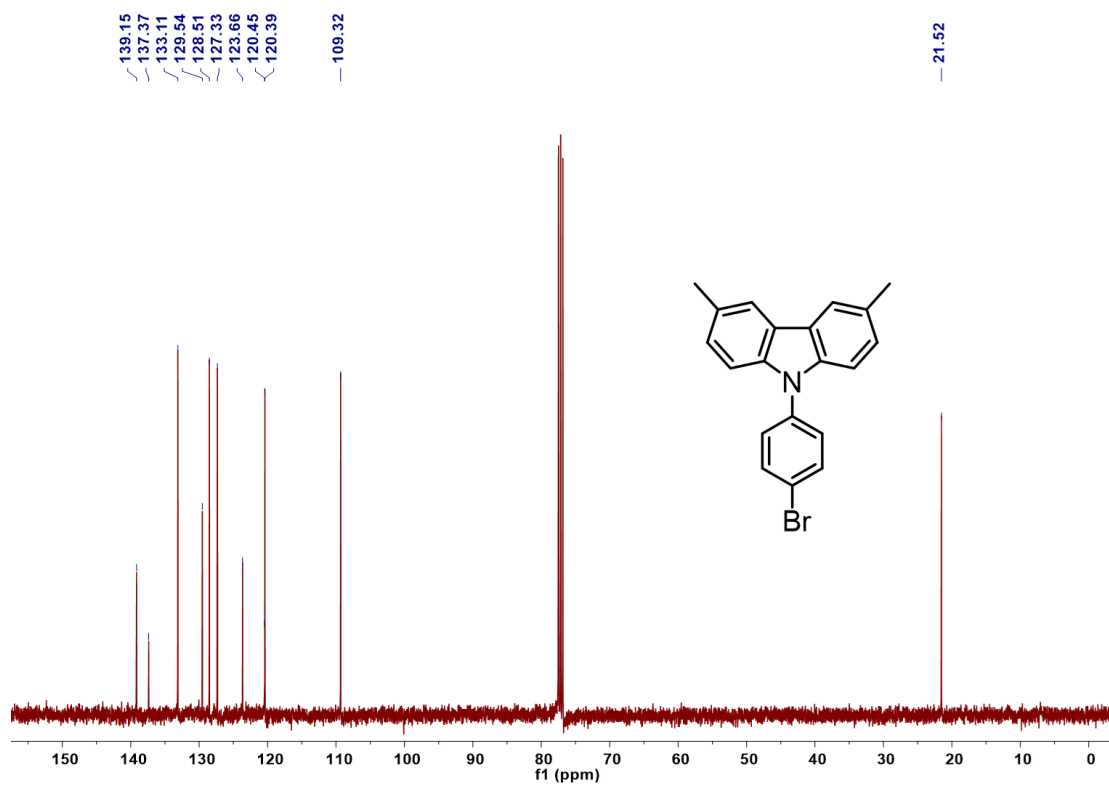


Figure S5. ¹³C NMR spectrum of **MeCz-PhpBr** (101 MHz, CDCl₃).

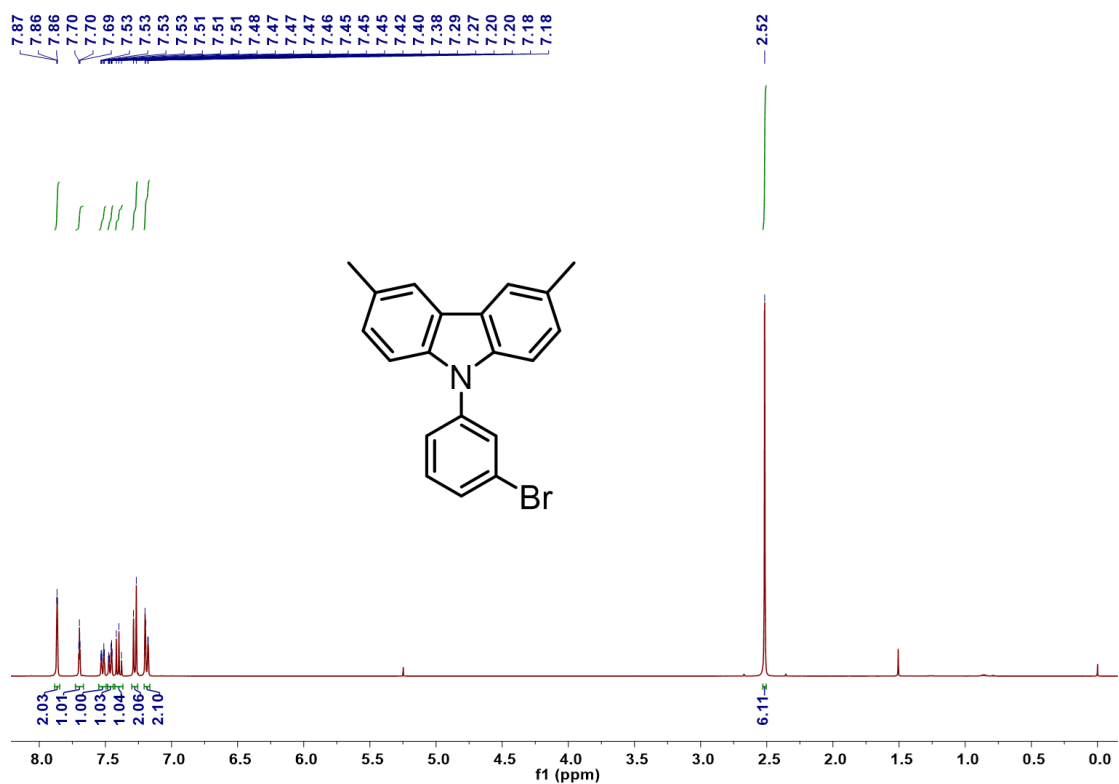


Figure S6. ^1H NMR spectrum of **MeCz-PhmBr** (400 MHz, CDCl_3).

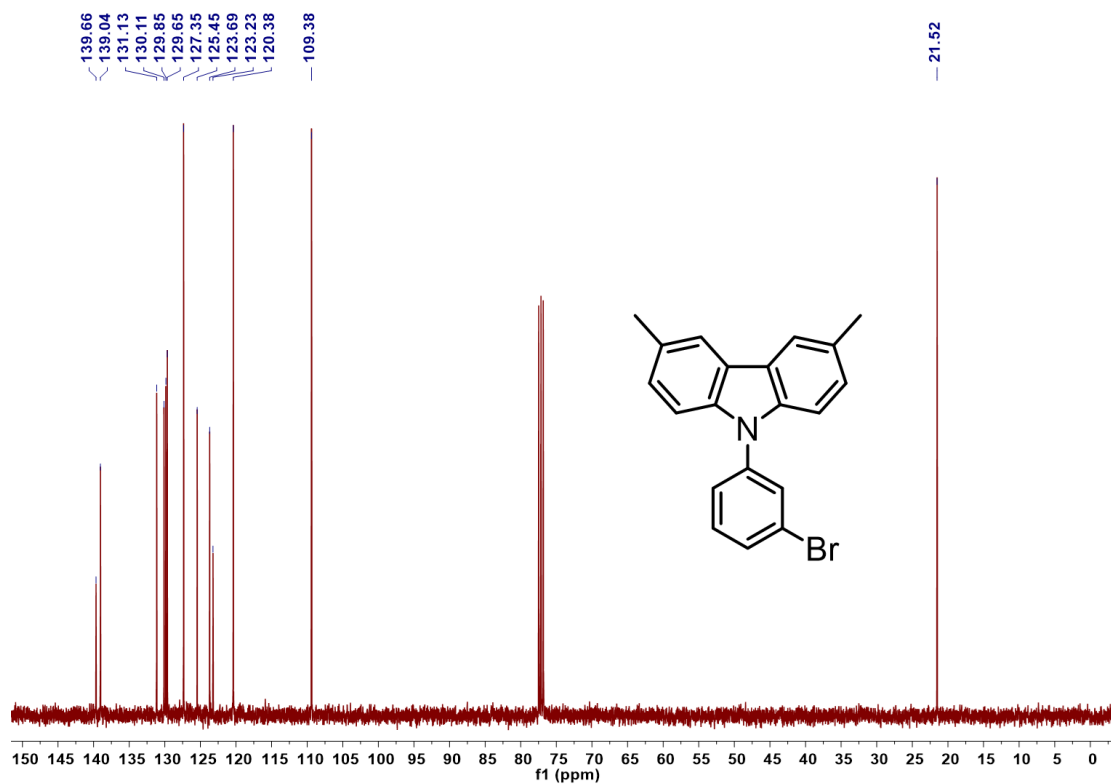


Figure S7. ¹³C NMR spectrum of **MeCz-PhmBr** (101 MHz, CDCl₃).

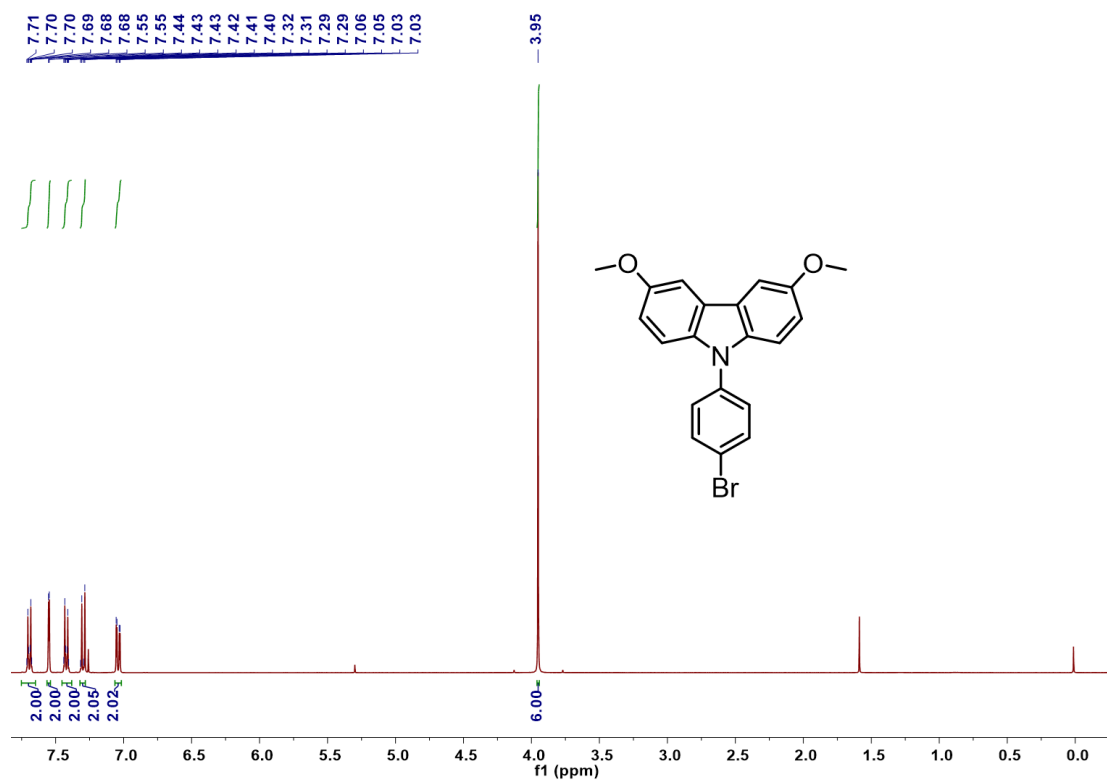


Figure S8. ¹H NMR spectrum of **MeOCz-PhBr** (400 MHz, CDCl₃).

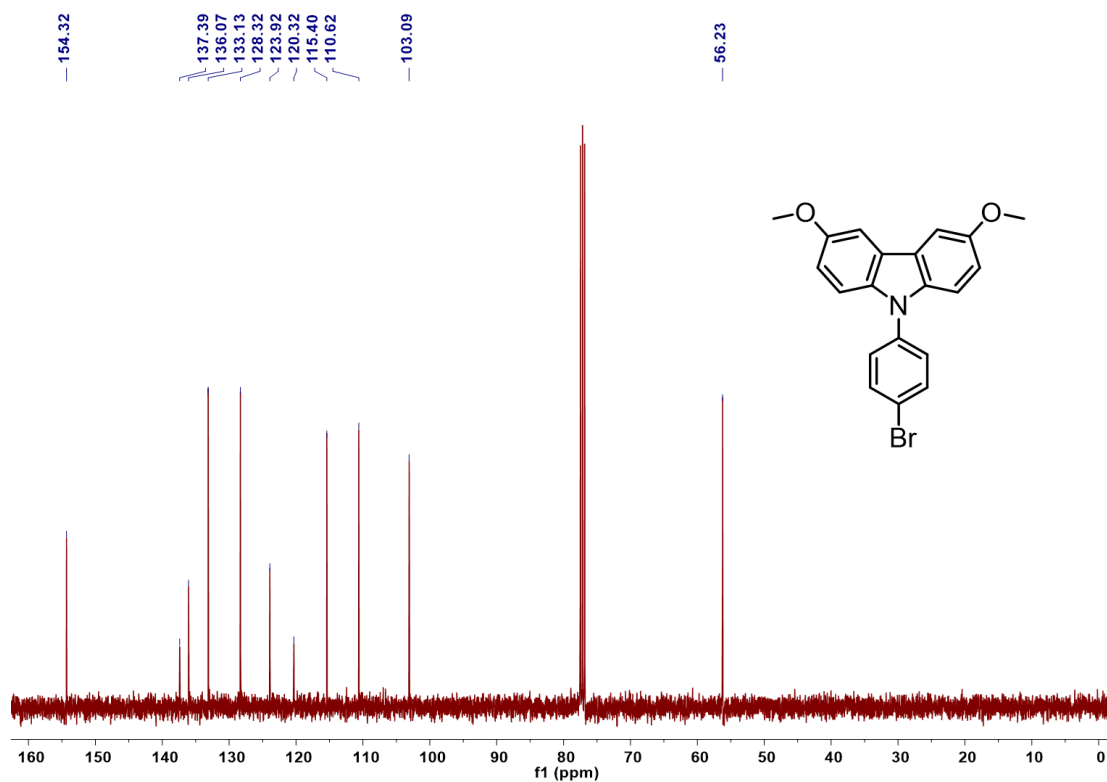


Figure S9. ^{13}C NMR spectrum of MeOCz-PhBr (101 MHz, CDCl_3).

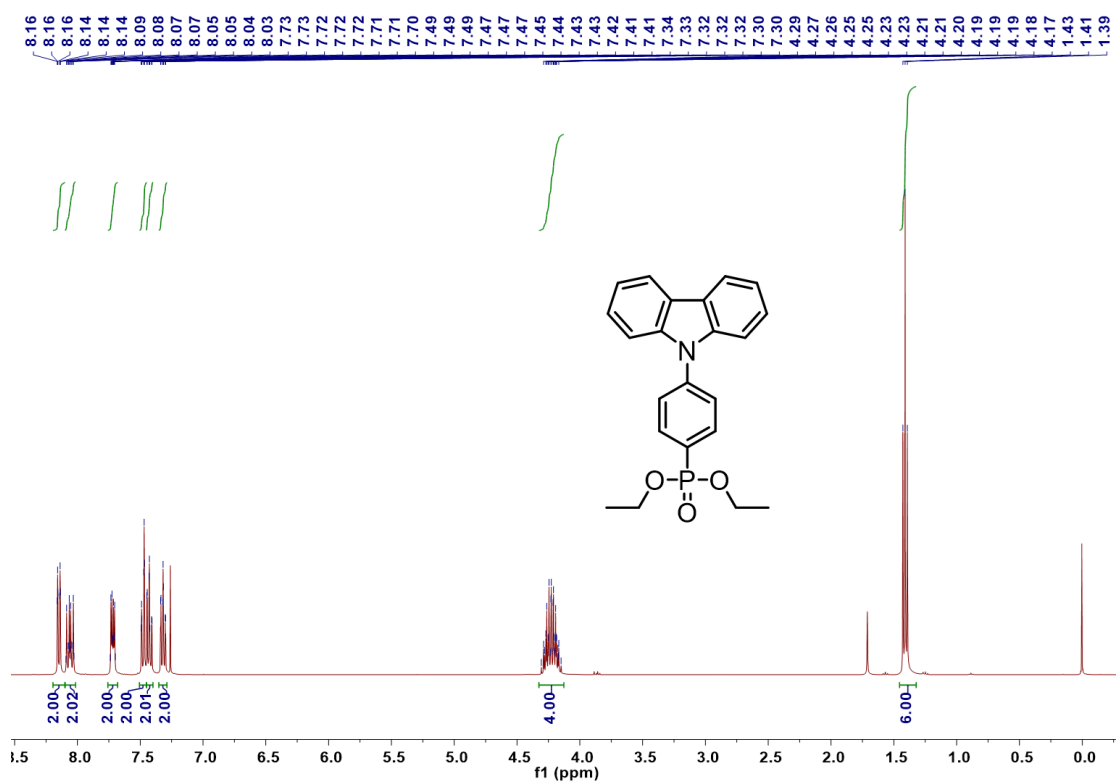


Figure S10. ¹H NMR spectrum of **Cz-PhpPO(OEt)₂** (400 MHz, CDCl₃).

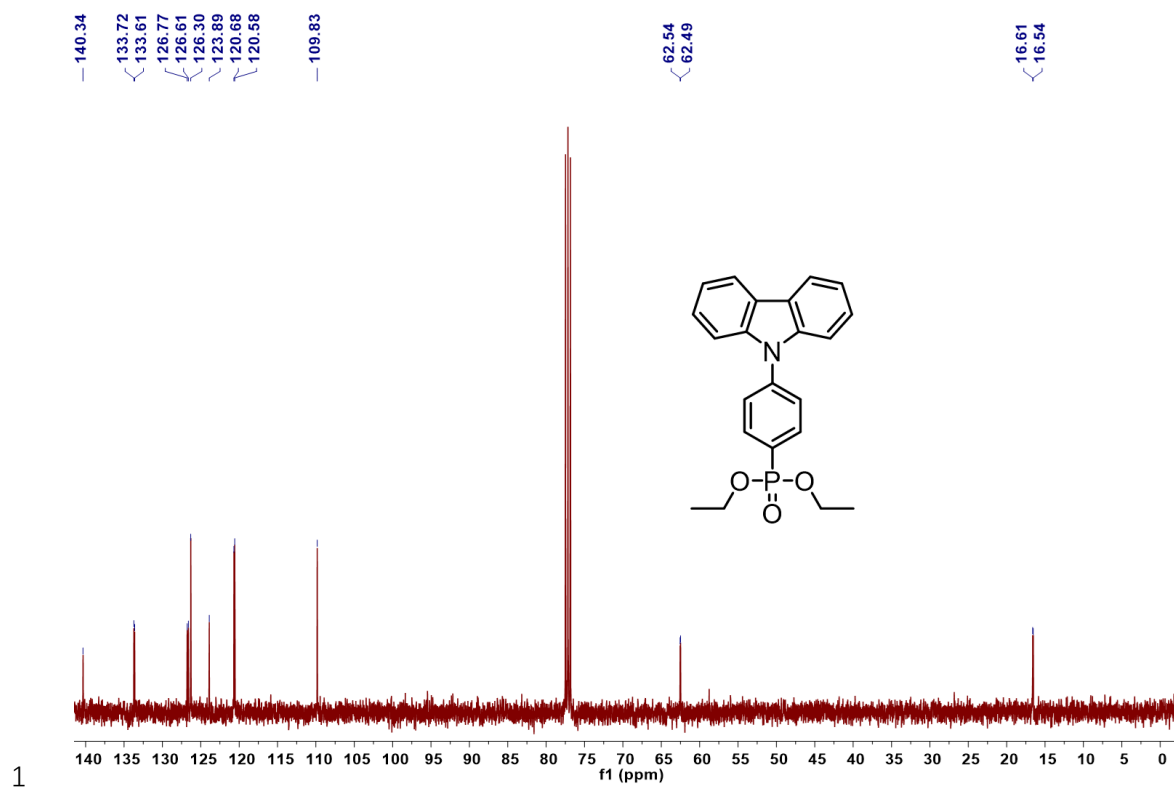


Figure S11. ¹³C NMR spectrum of **Cz-PhPO(OEt)₂** (101 MHz, CDCl₃).

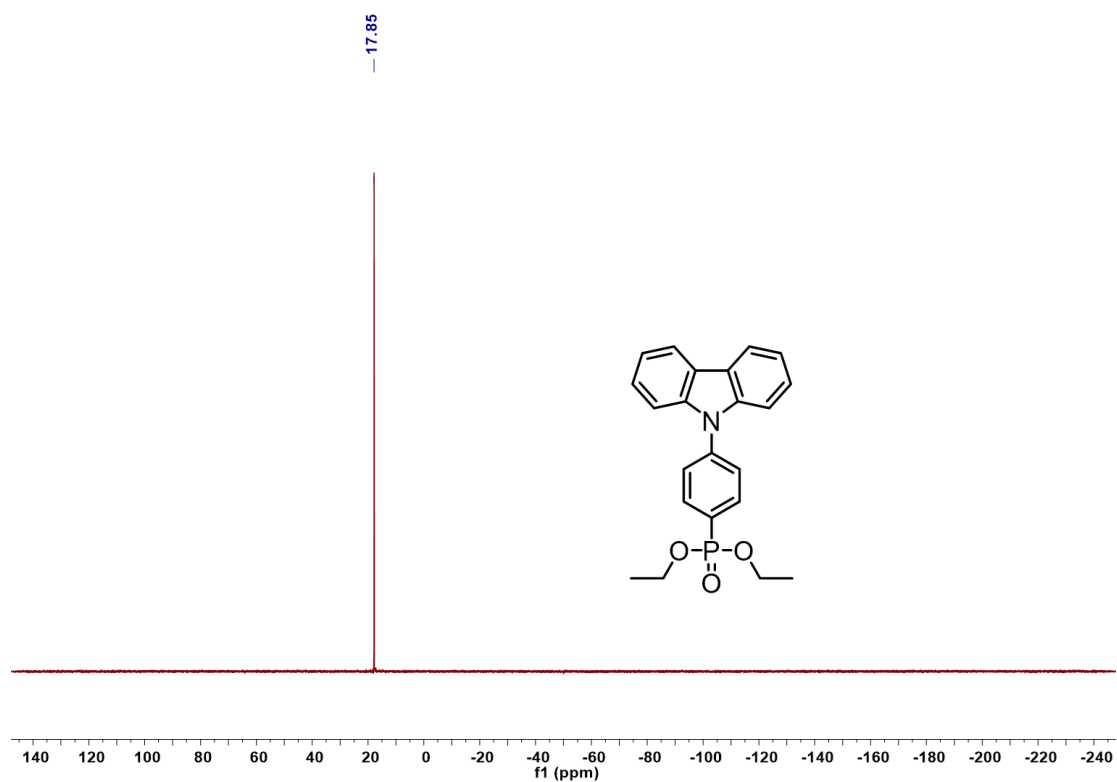


Figure S12. ^{31}P NMR spectrum of **Cz-PhPO(OEt)₂** (162 MHz, CDCl_3).

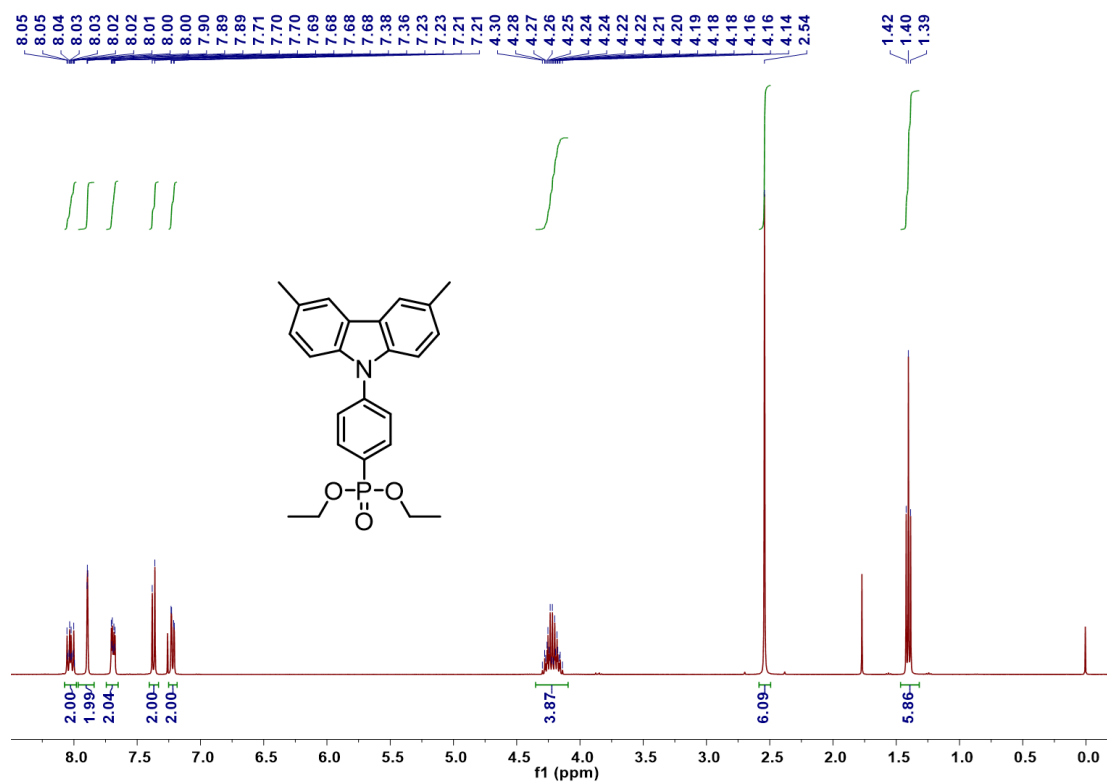


Figure S13. ¹H NMR spectrum of MeCz-PhpPO(OEt)₂ (400 MHz, CDCl₃).

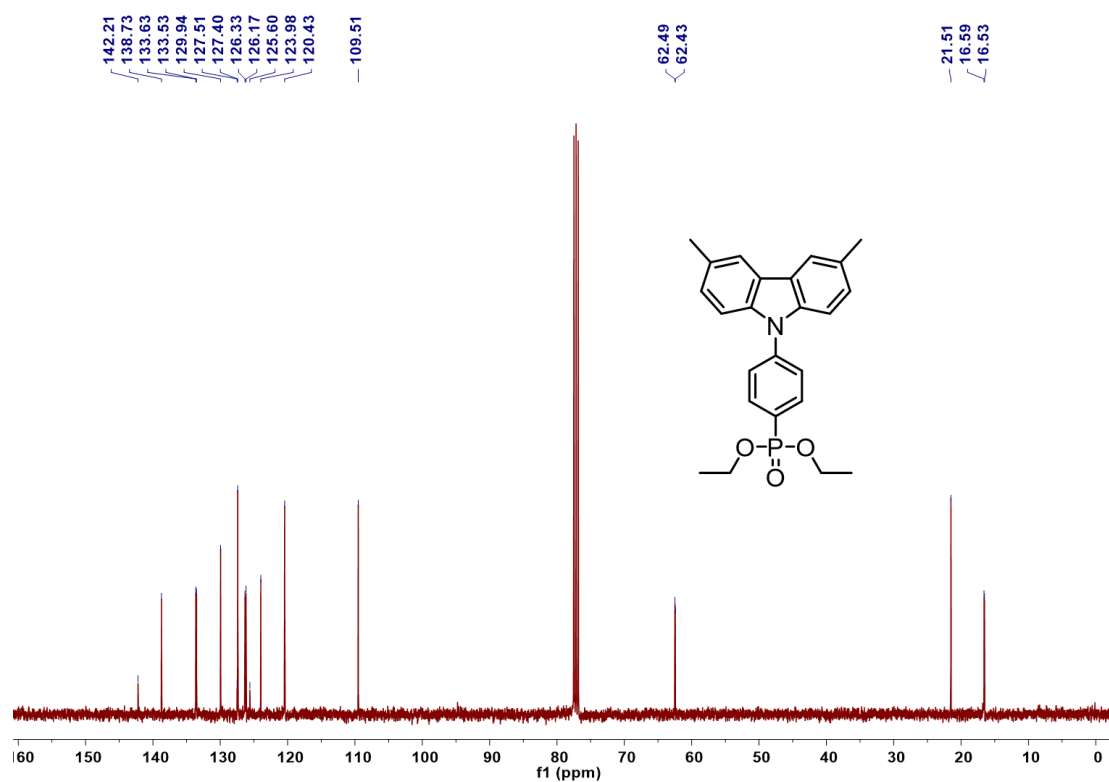


Figure S14. ¹³C NMR spectrum of MeCz-PhpPO(OEt)₂ (101 MHz, CDCl₃).

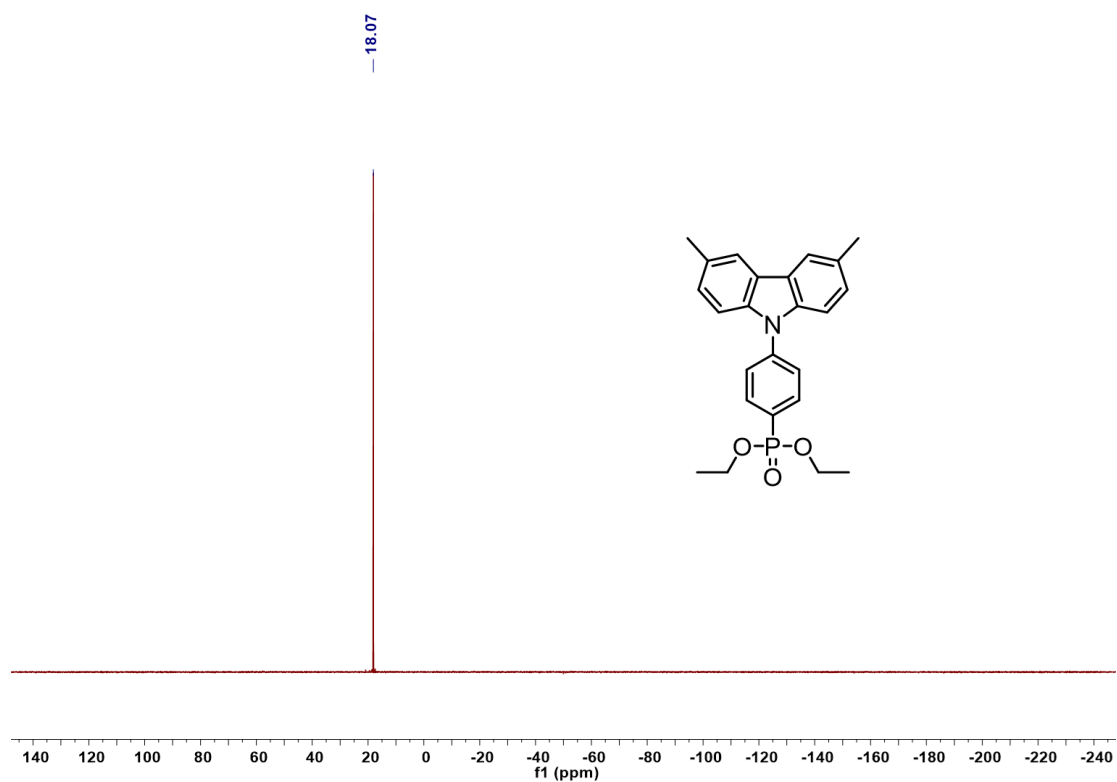


Figure S15. ^{31}P NMR spectrum of **MeCz-PhpPO(OEt)₂** (162 MHz, CDCl_3).

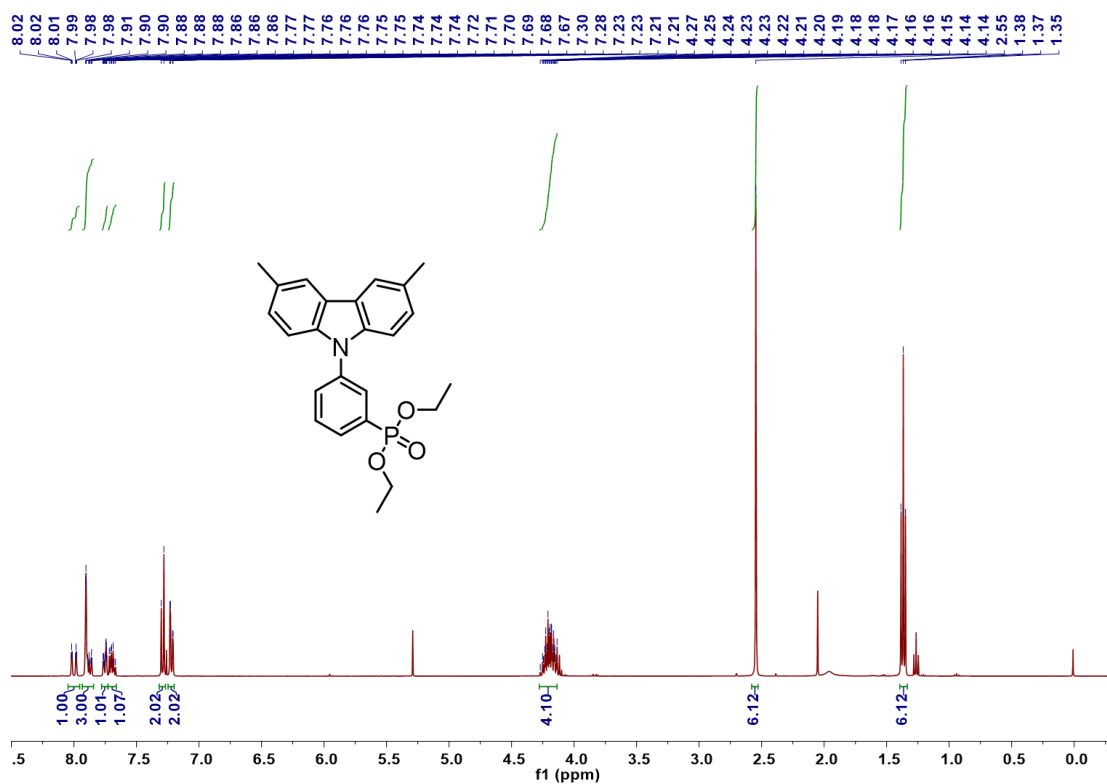


Figure S16. ¹H NMR spectrum of MeCz-PhmPO(OEt)₂ (400 MHz, CDCl₃).

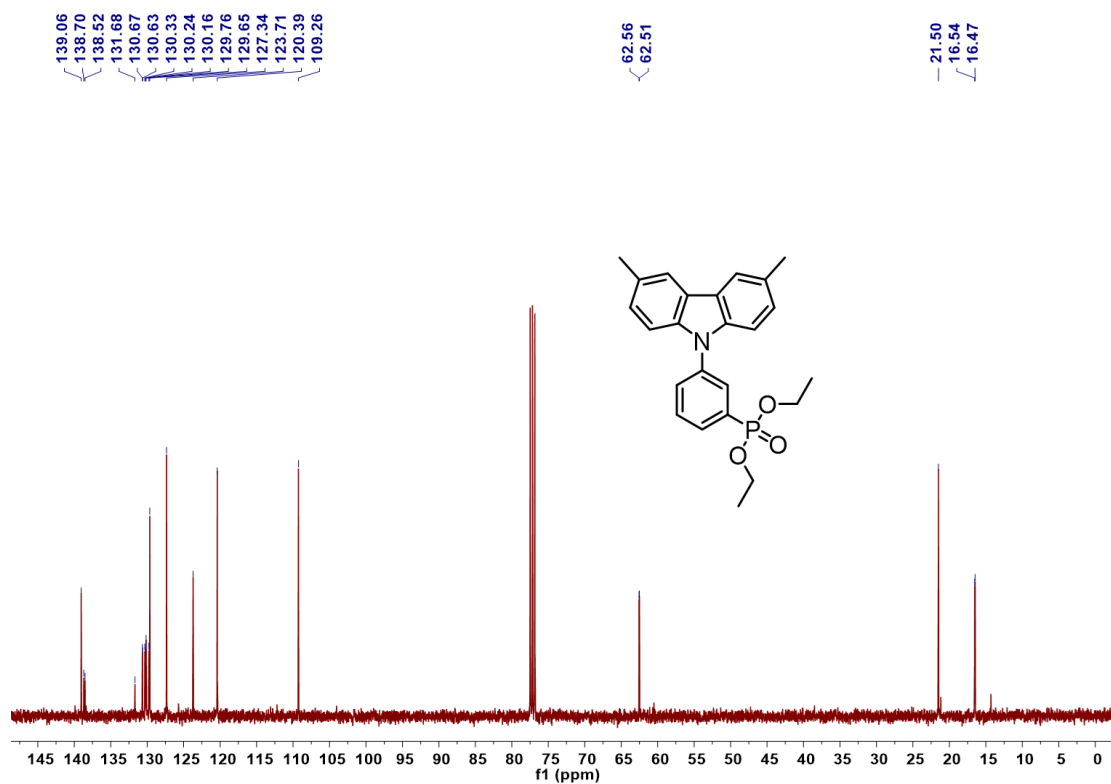


Figure S17. ^{13}C NMR spectrum of **MeCz-PhmPO(OEt)₂** (101 MHz, CDCl_3).

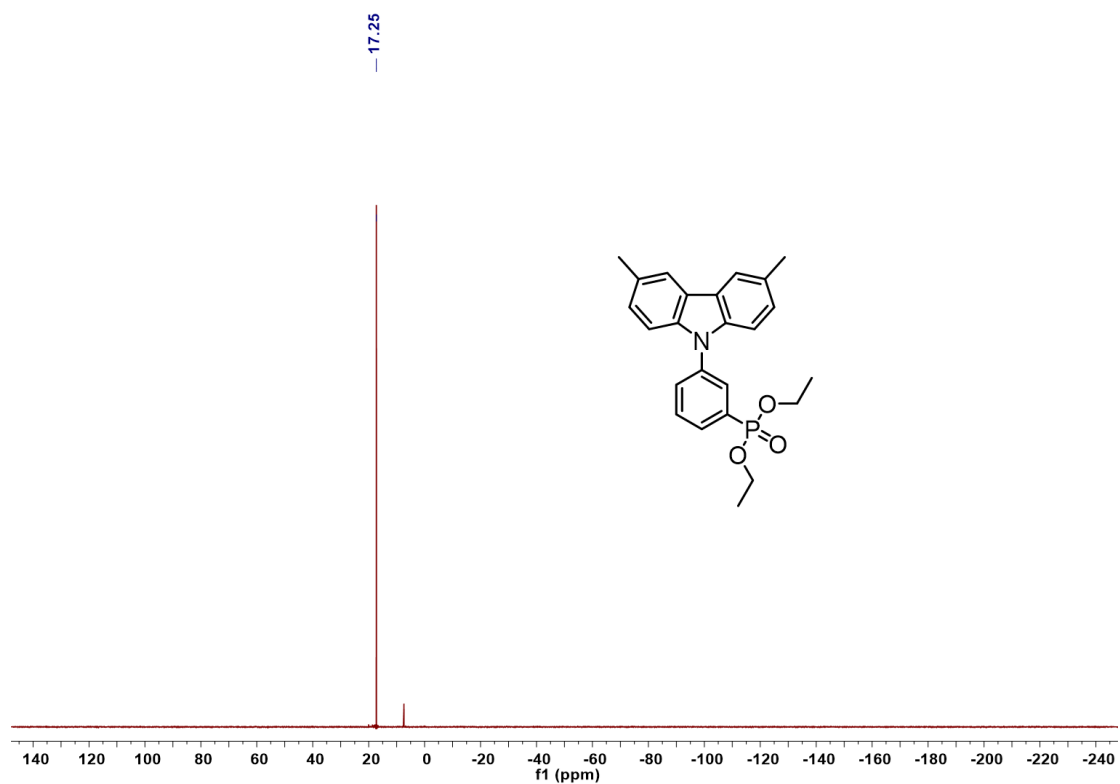


Figure S18. ^{31}P NMR spectrum of MeCz-PhmPO(OEt) $_2$ (162 MHz, CDCl $_3$).

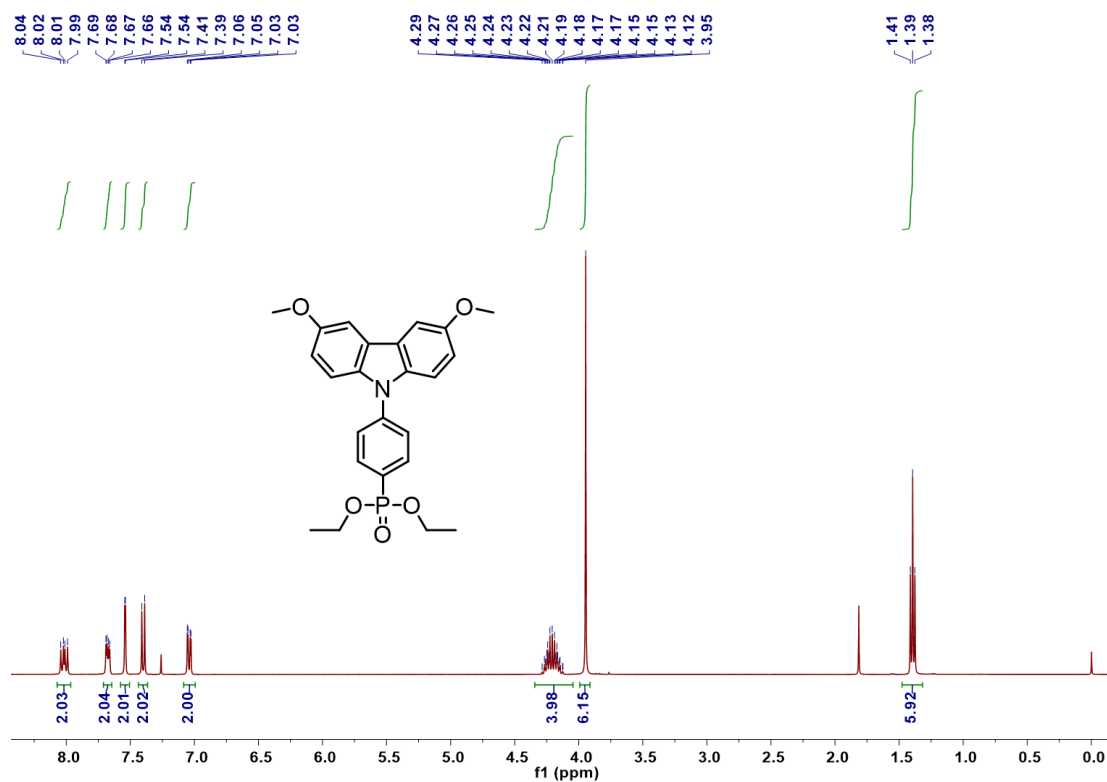


Figure S19. ¹H NMR spectrum of MeOCz-PhpPO(OEt)₂ (400 MHz, CDCl₃).

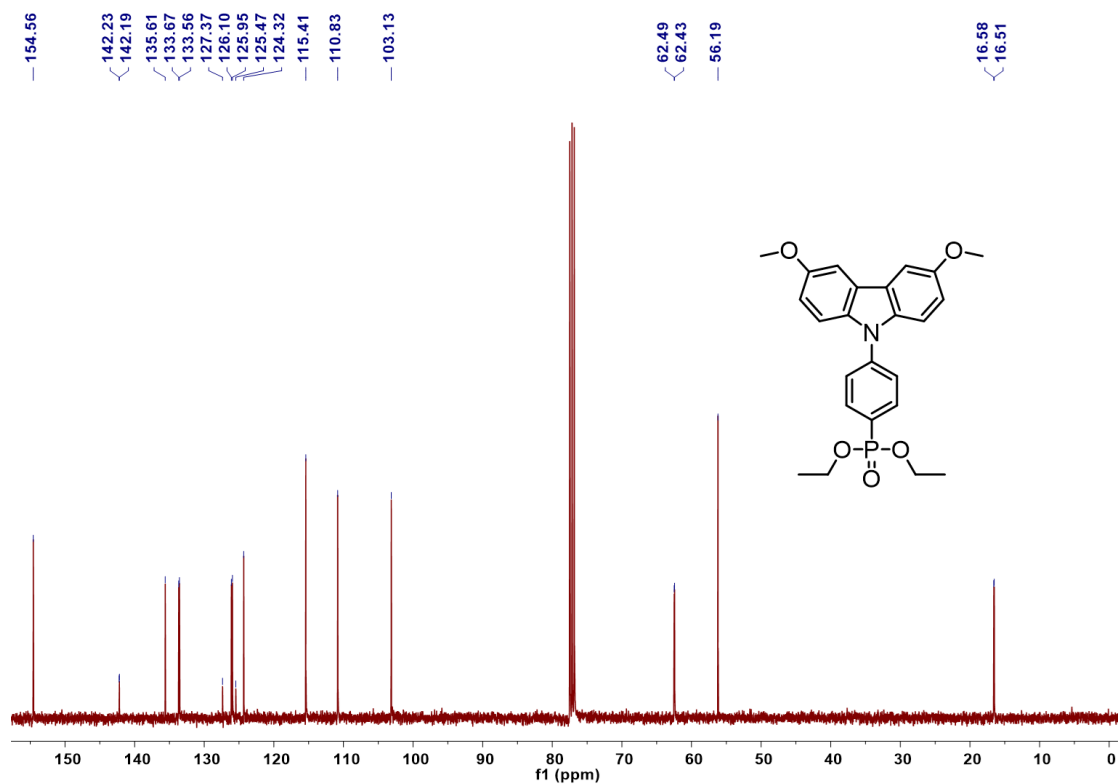


Figure S20. ^{13}C NMR spectrum of MeOCz-PhPO(OEt)_2 (101 MHz, CDCl_3).

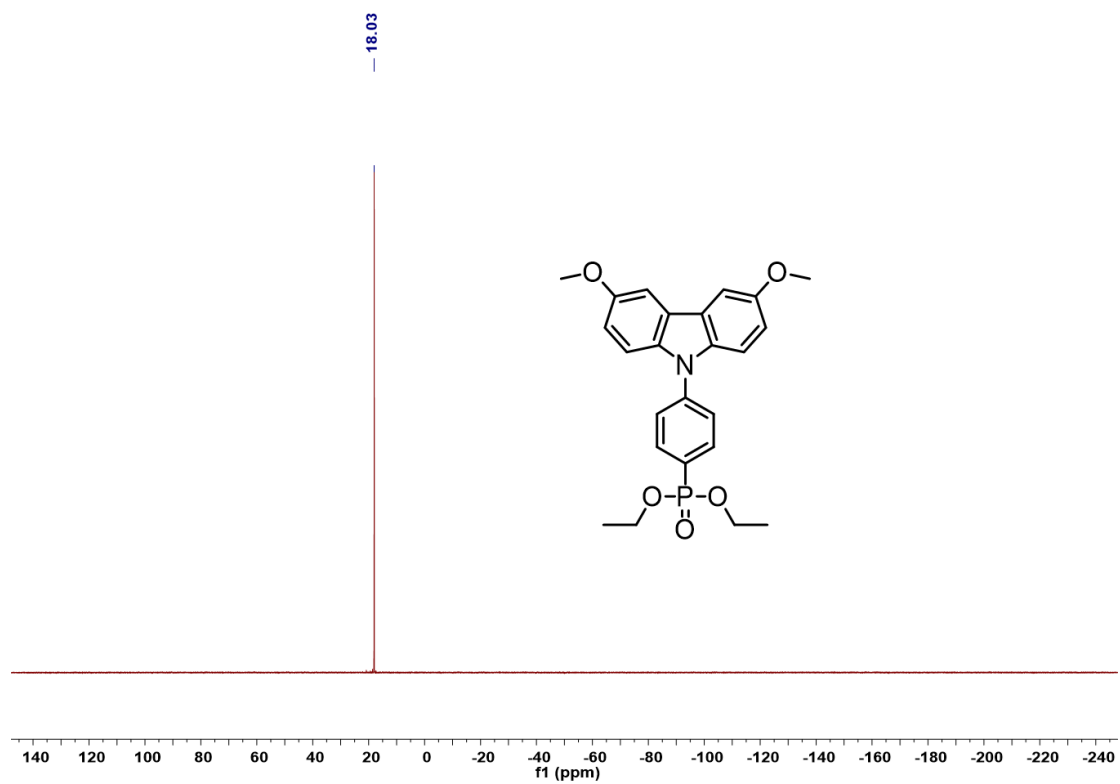


Figure S21. ^{31}P NMR spectrum of **MeOCz-PhPO(OEt)₂** (162 MHz, CDCl_3).

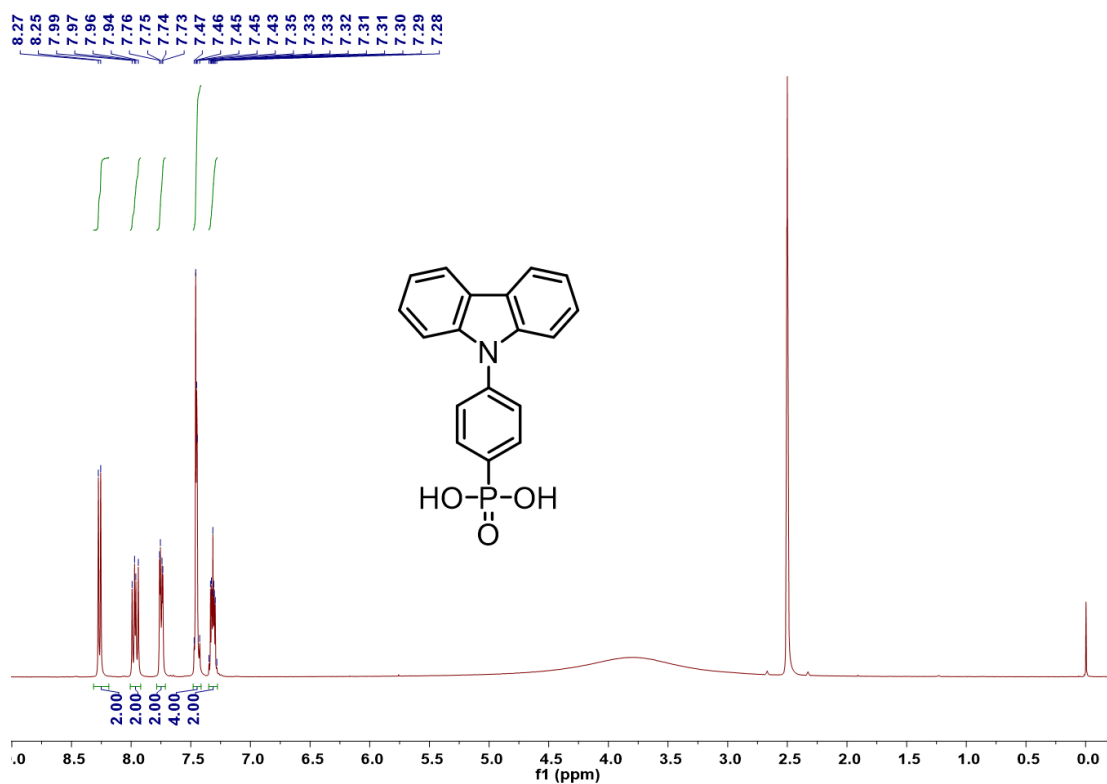


Figure S22. ^1H NMR spectrum of **PhPACz** (400 MHz, DMSO- d_6).

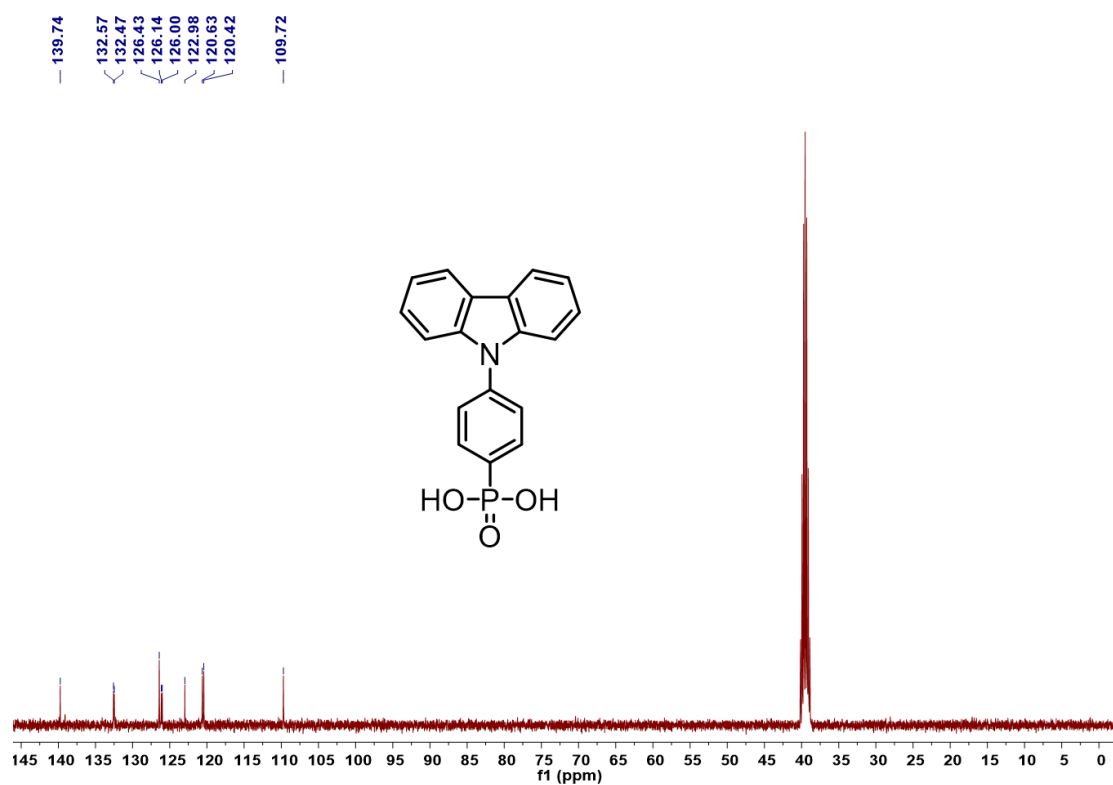


Figure S23. ¹³C NMR spectrum of **PhpPACz** (101 MHz, DMSO-*d*₆).

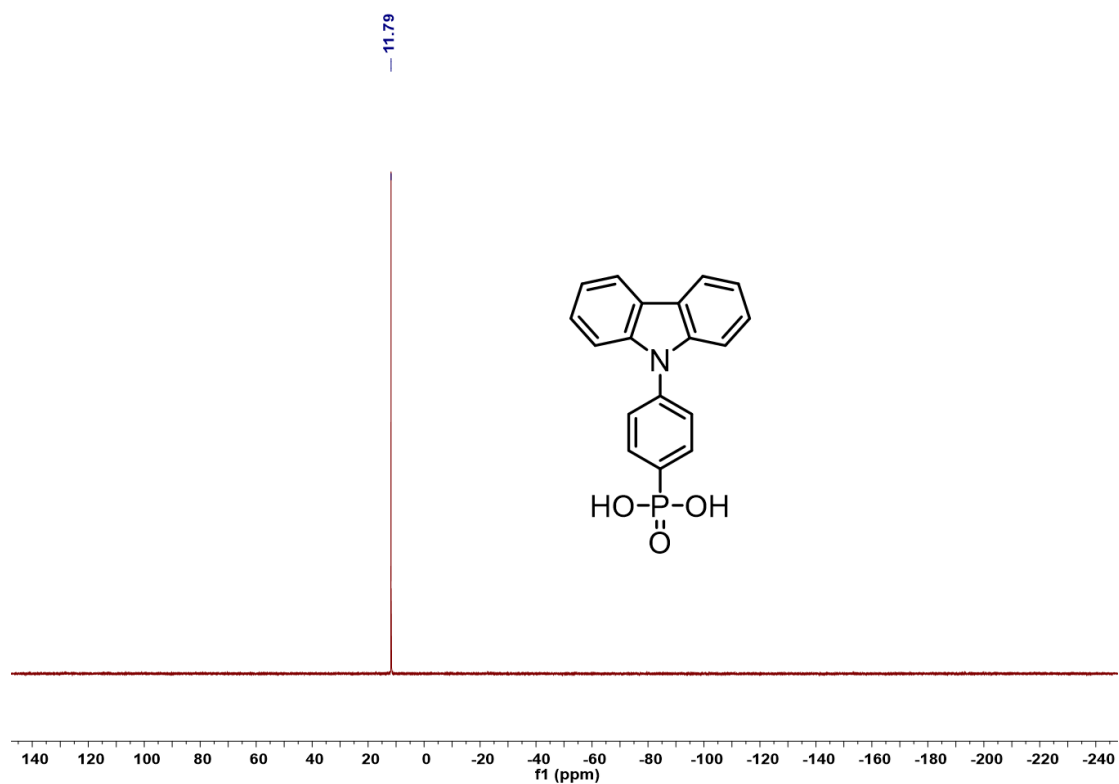


Figure S24. ³¹P NMR spectrum of **PhpPACz** (162 MHz, DMSO-*d*₆).

caisiyuan-20230531-1 #3-18 RT: 0.03-0.16 AV: 8 NL: 3.30E8
T: FTMS - p ESI Full ms [100.0000-1500.0000]

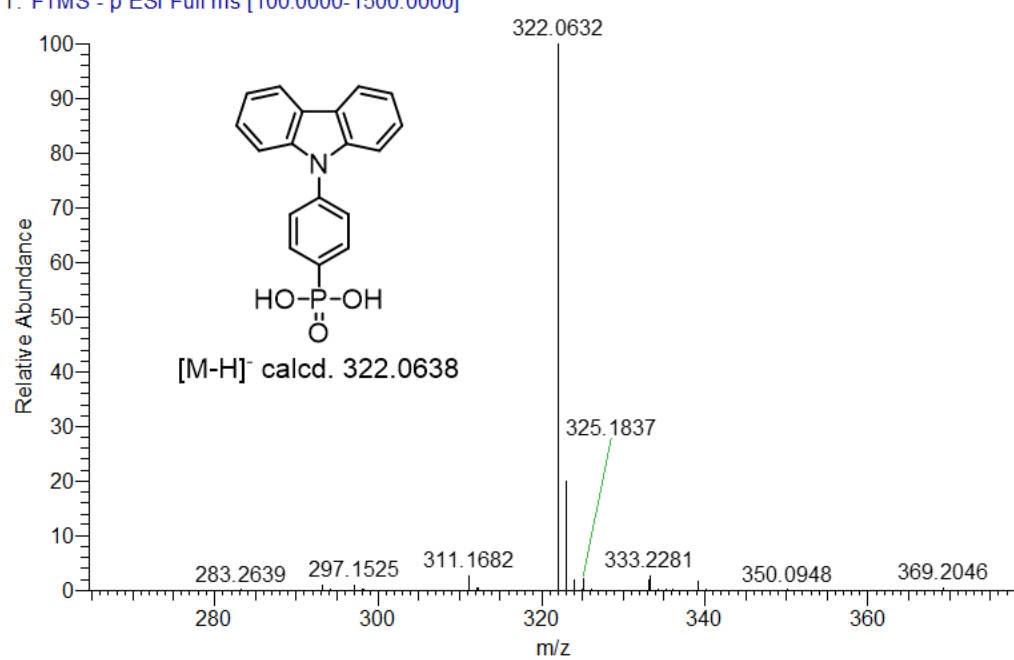


Figure S25. High-resolution mass spectrum of **PhpPACz**.

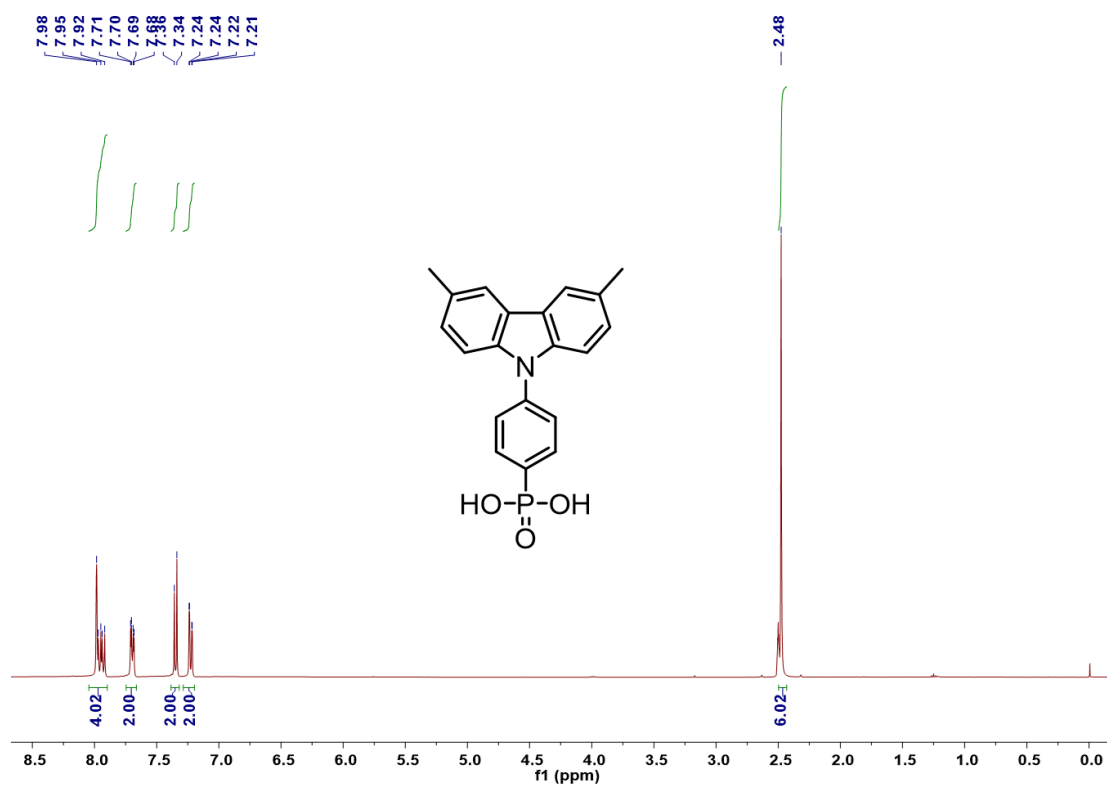


Figure S26. ¹H NMR spectrum of **Me-PhpPACz** (400 MHz, DMSO-*d*₆).

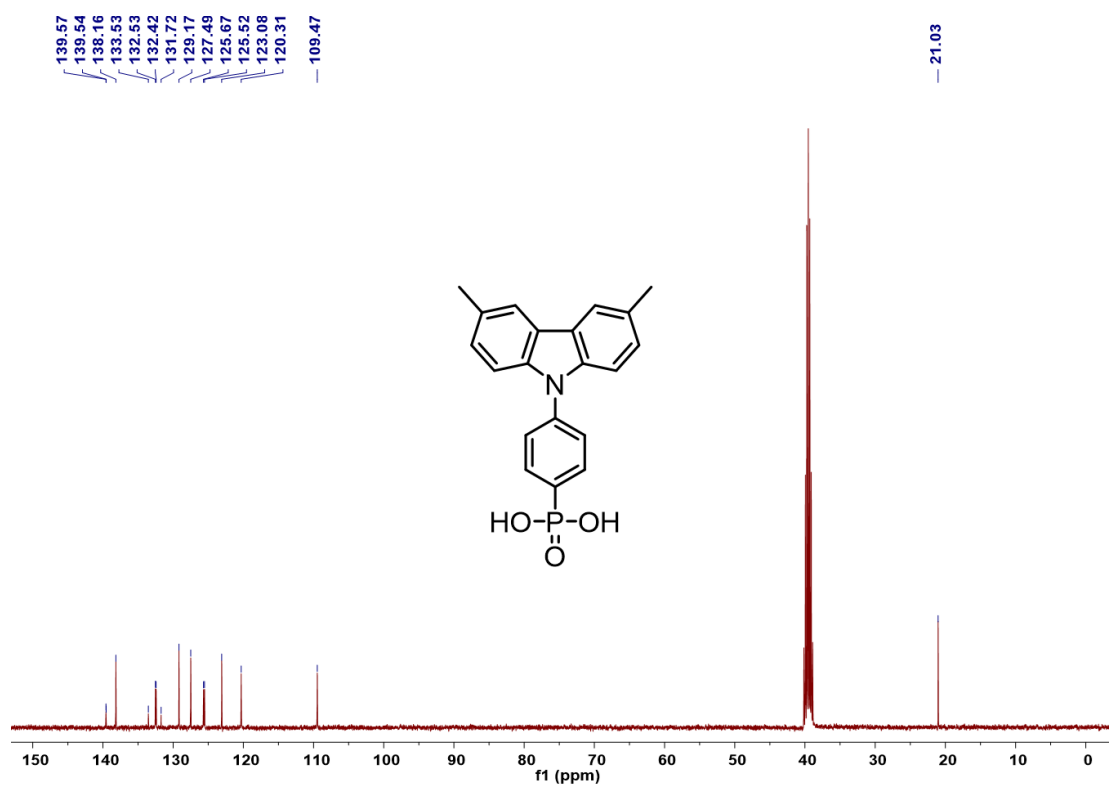


Figure S27. ¹³C NMR spectrum of **Me-PhpPACz** (101 MHz, DMSO-*d*₆).

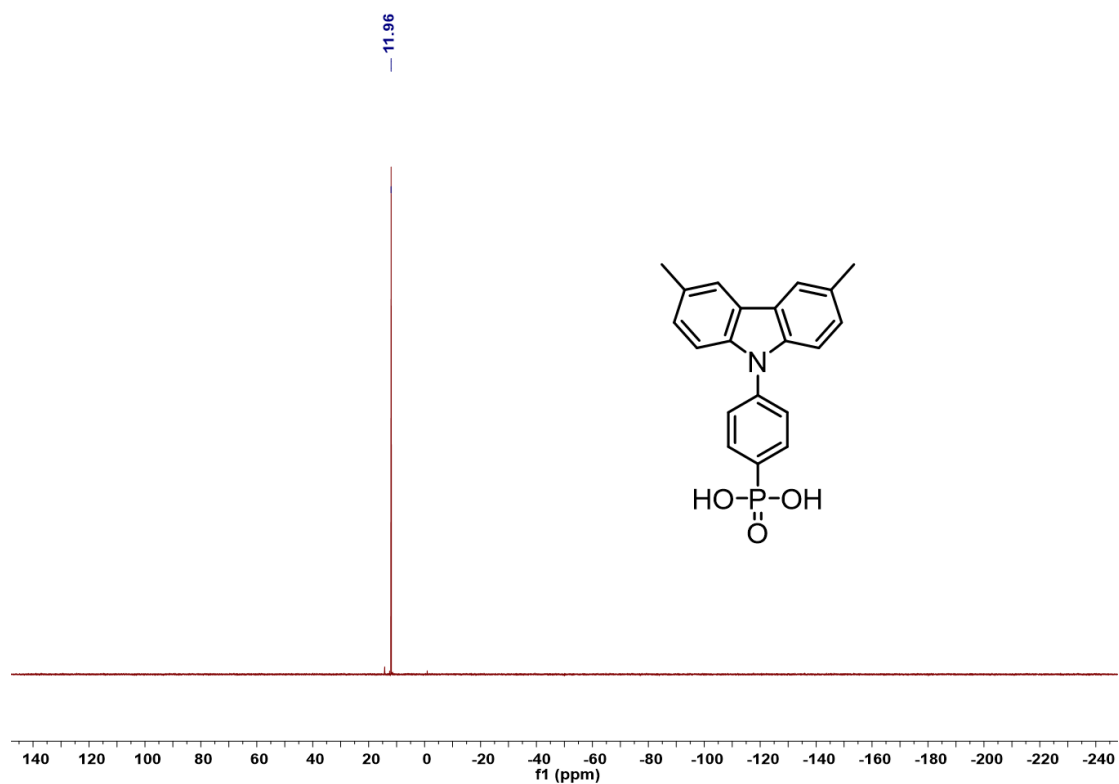


Figure S28. ³¹P NMR spectrum of **Me-PhpPACz** (162 MHz, DMSO-*d*₆).

caisiyuan-20230531-3 #4-21 RT: 0.03-0.18 AV: 9 NL: 1.95E8
T: FTMS - p ESI Full ms [100.0000-1500.0000]

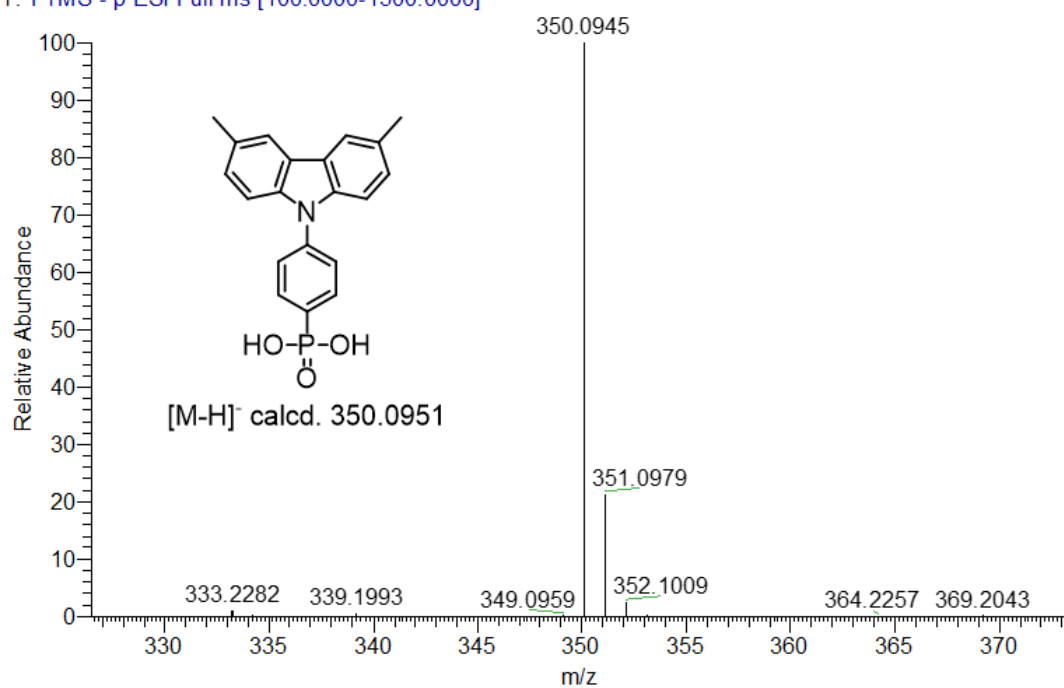


Figure S29. High-resolution mass spectrum of **Me-PhpPACz**.

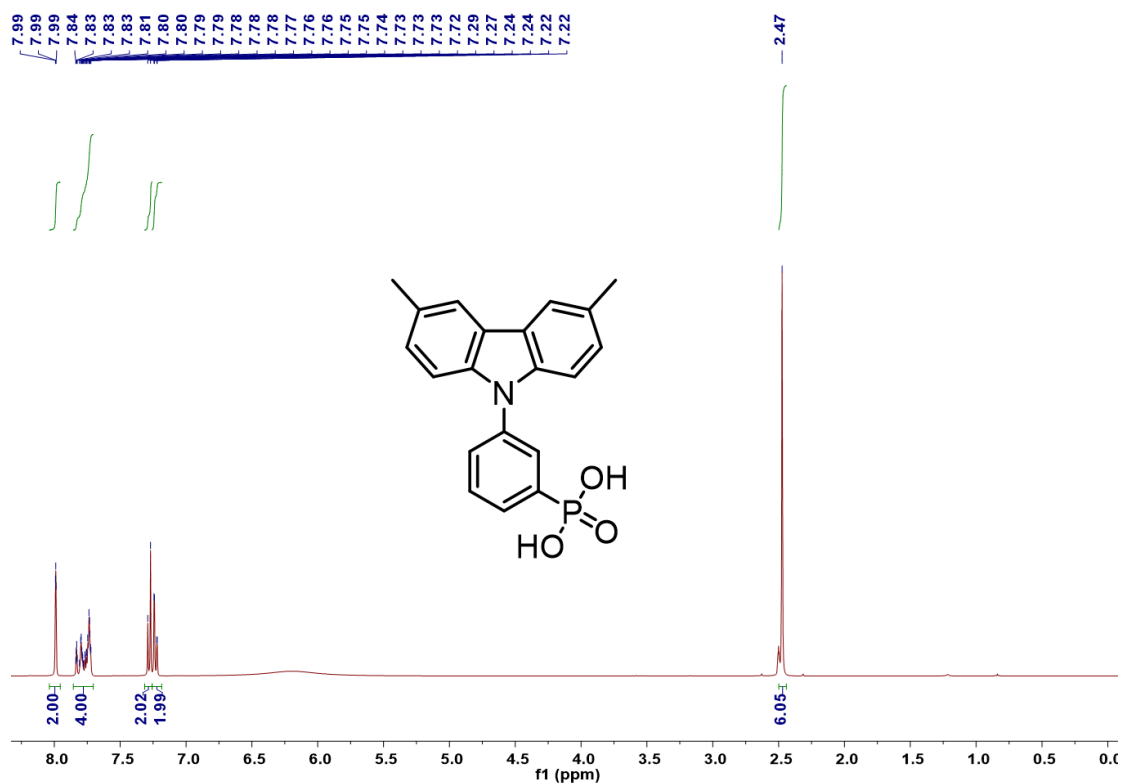


Figure S30. ¹H NMR spectrum of **Me-PhmPACz** (400 MHz, DMSO-*d*₆).

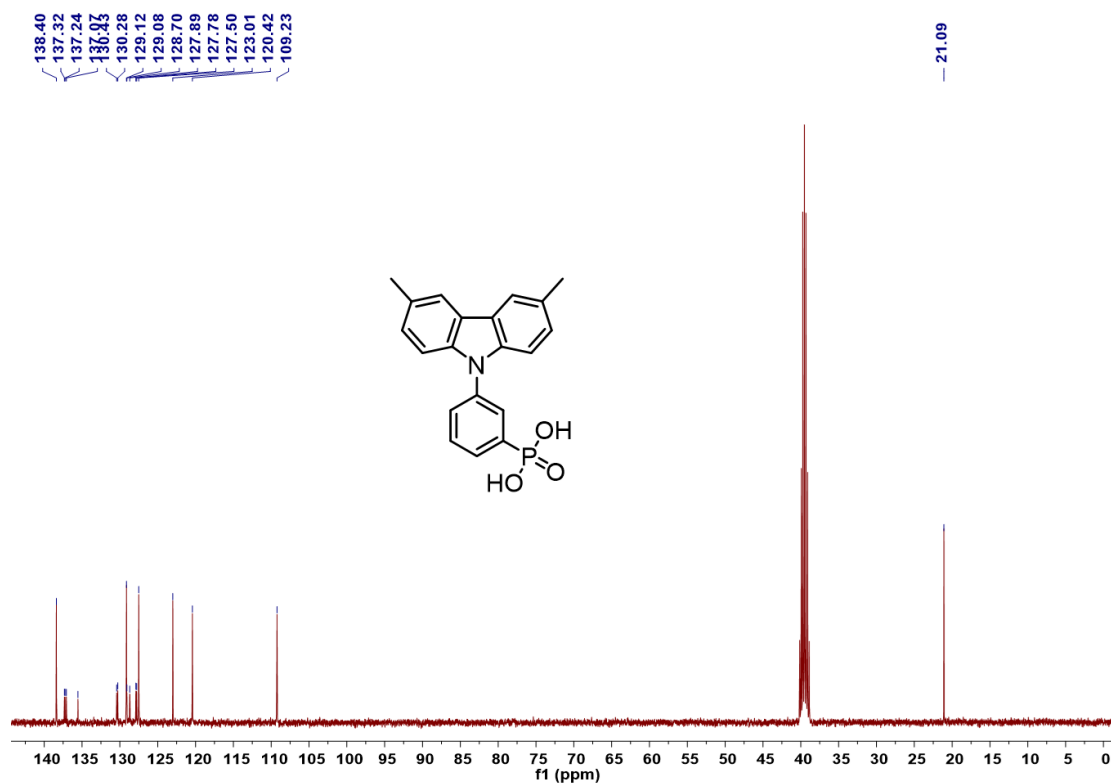


Figure S31. ¹³C NMR spectrum of **Me-PhmPACz** (101 MHz, DMSO-*d*₆).

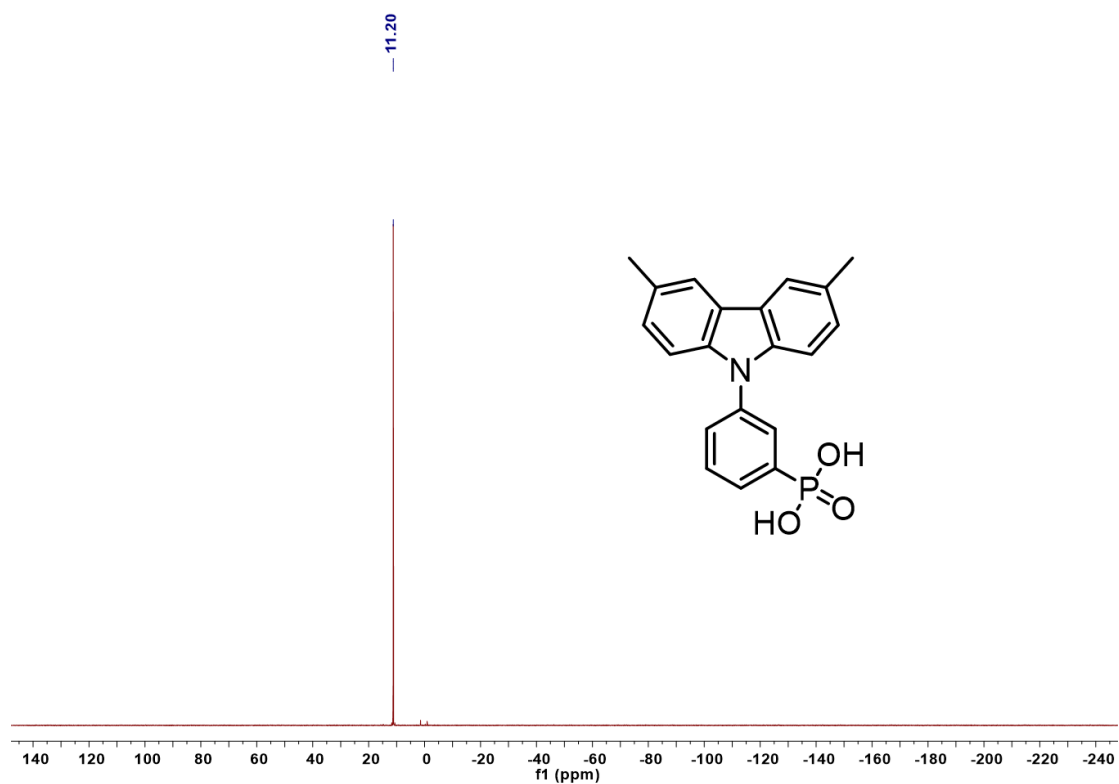


Figure S32. ^{31}P NMR spectrum of **Me-PhmPACz** (162 MHz, $\text{DMSO-}d_6$).

caisiyuan-20230719-1 #5-23 RT: 0.05-0.20 AV: 9 NL: 3.50E8
T: FTMS - p ESI Full ms [50.0000-600.0000]

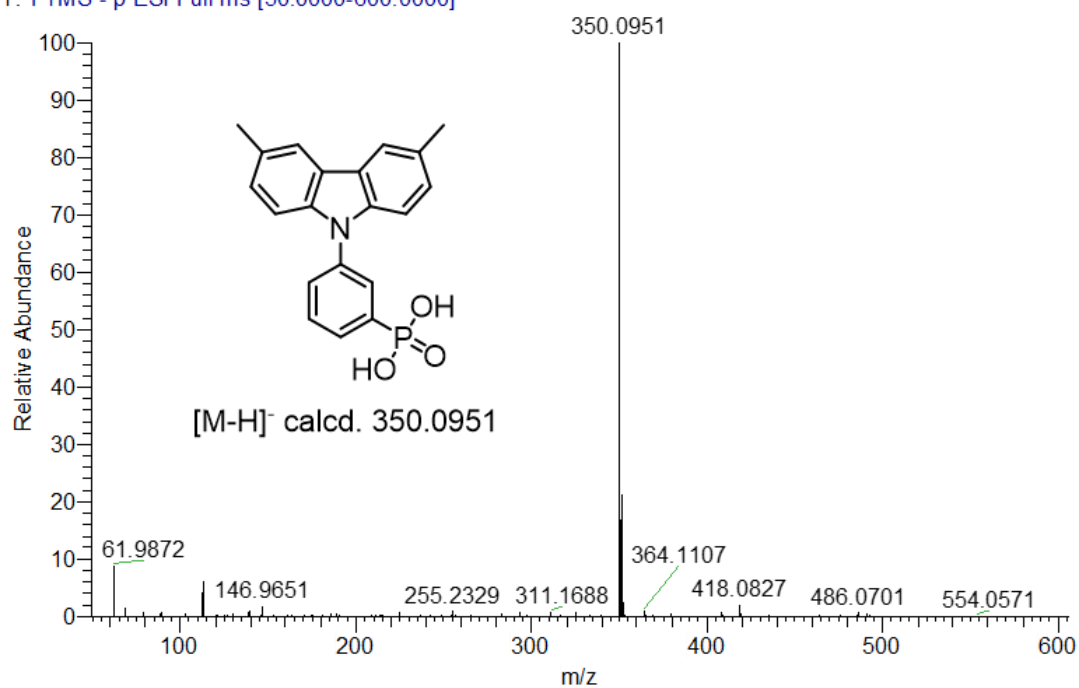


Figure S33. High-resolution mass spectrum of **Me-PhmPACz**.

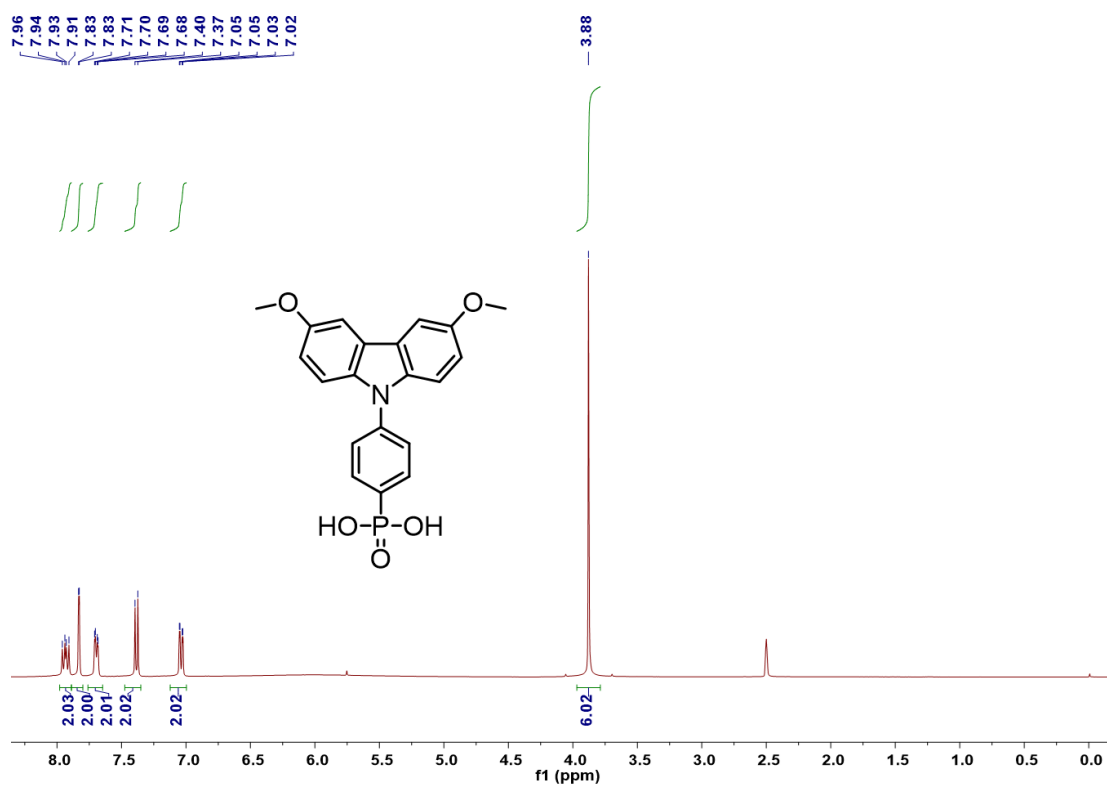


Figure S34. ¹H NMR spectrum of MeO-PhpPACz (400 MHz, DMSO-*d*₆).

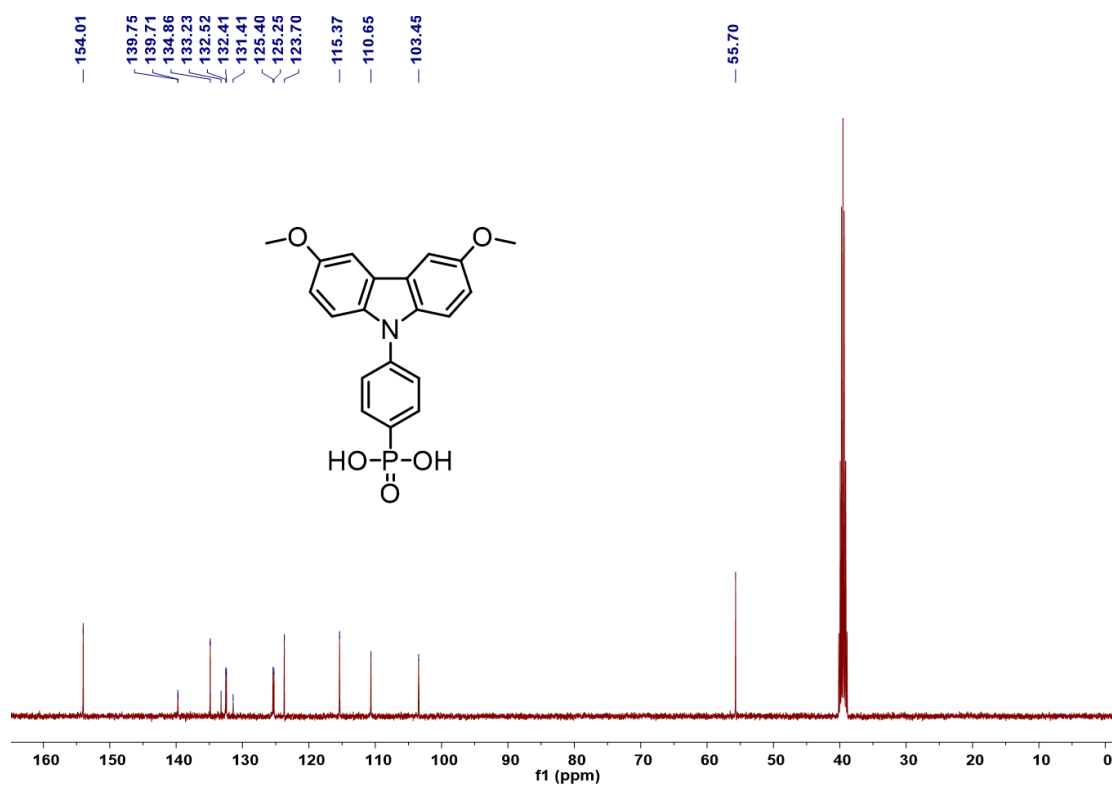


Figure S35. ¹³C NMR spectrum of **MeO-PhpPACz** (101 MHz, DMSO-*d*₆).

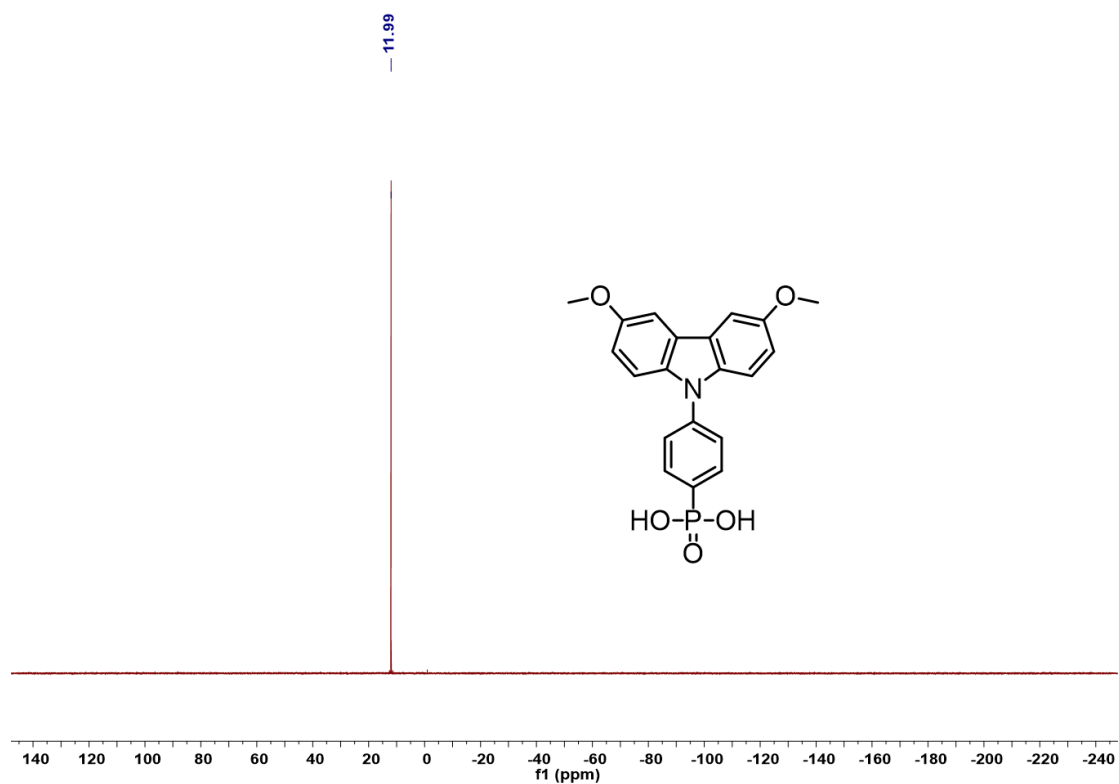


Figure S36. ^{31}P NMR spectrum of MeO-PhpPACz (162 MHz, $\text{DMSO}-d_6$).

caisiyuan-20230531-4 #4-22 RT: 0.03-0.20 AV: 10 NL: 1.93E8
T: FTMS - p ESI Full ms [100.0000-1500.0000]

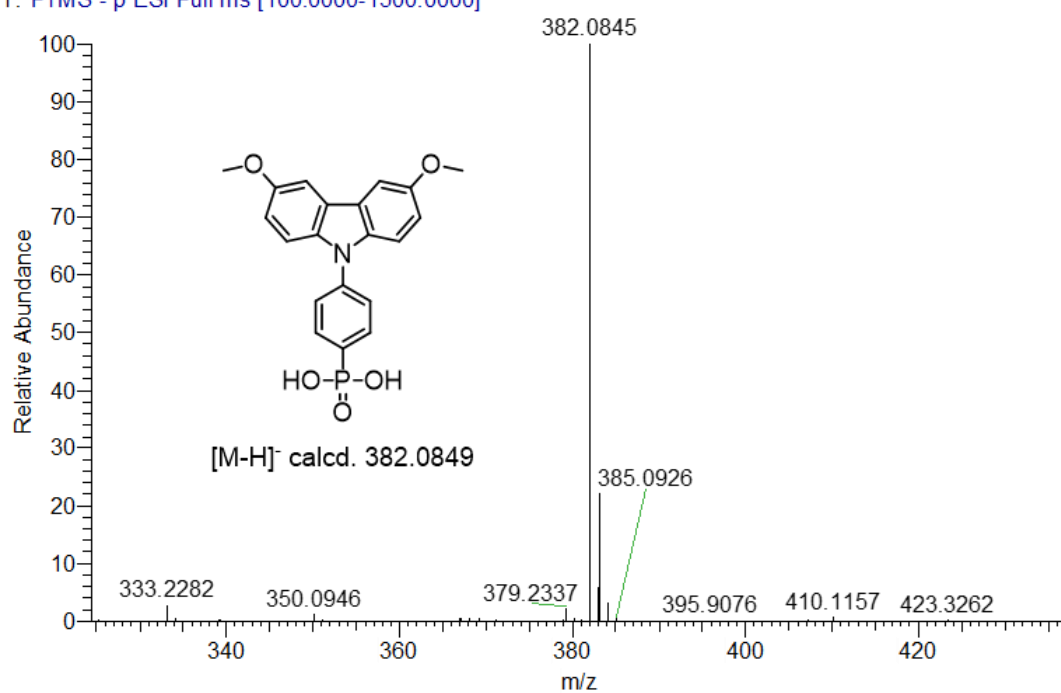


Figure S37. High-resolution mass spectrum of **MeO-PhpPACz**.

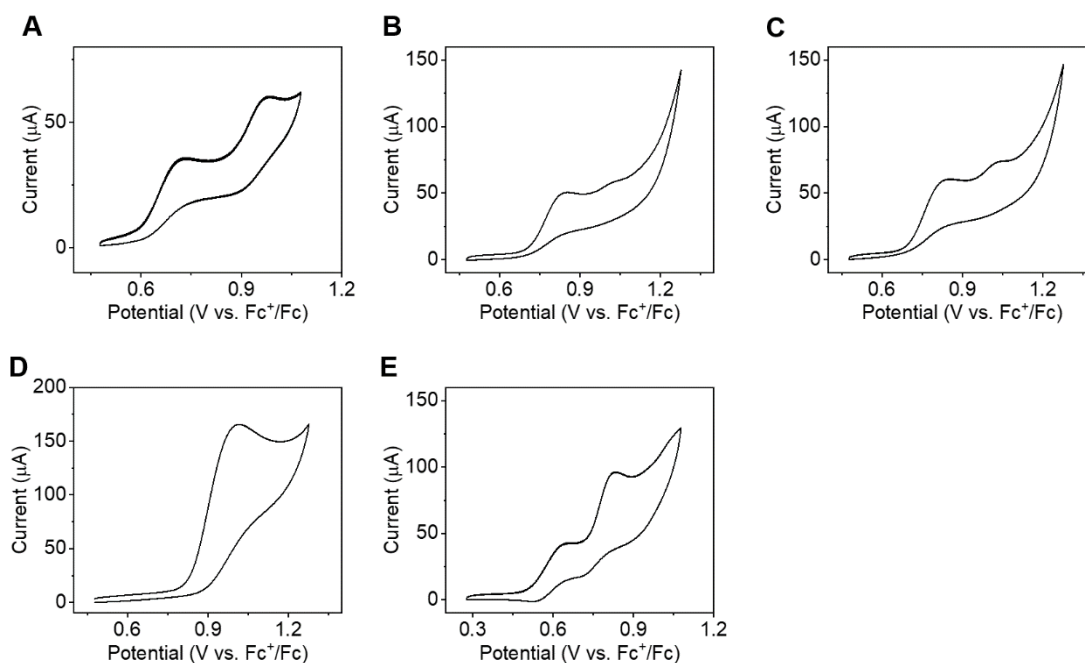


Figure S38. Cyclic voltammograms curves of (A) Me-4PACz, (B) Me-PhpPACz, (C) Me-PhmPACz, (D) PhpPACz, and (E) MeO-PhpPACz.

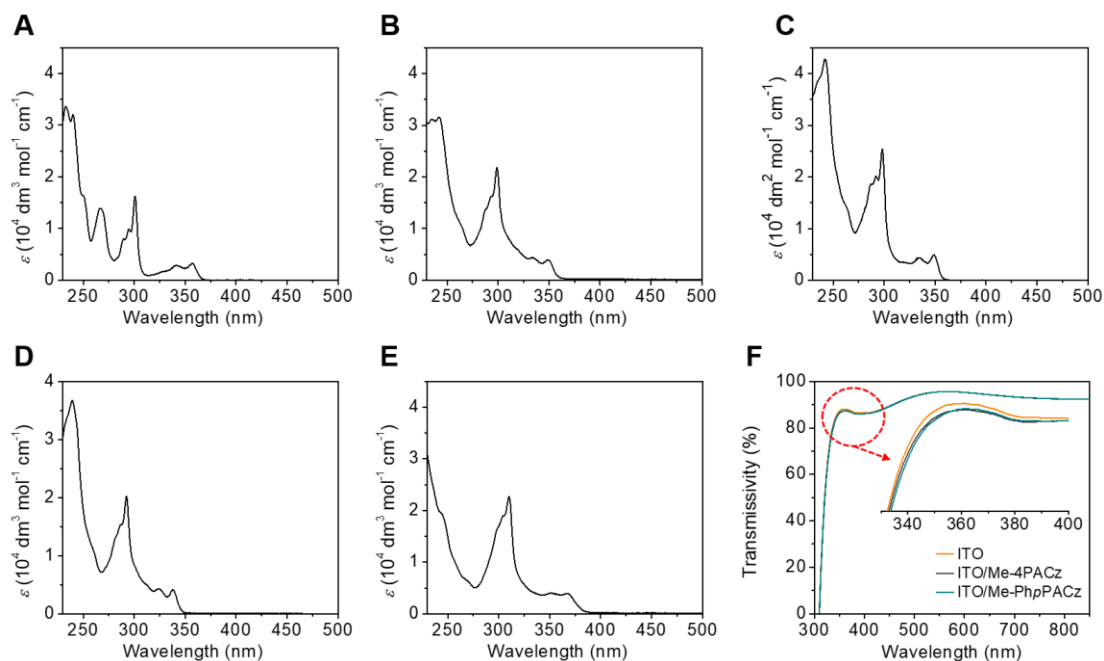


Figure S39. Solution UV-vis spectra of (A) Me-4PACz, (B) Me-PhpPACz, (C) Me-PhmPACz, (D) PhpPACz, (E) MeO-PhpPACz, and (F) thin film UV-vis spectra of ITO and ITO with SAMs.

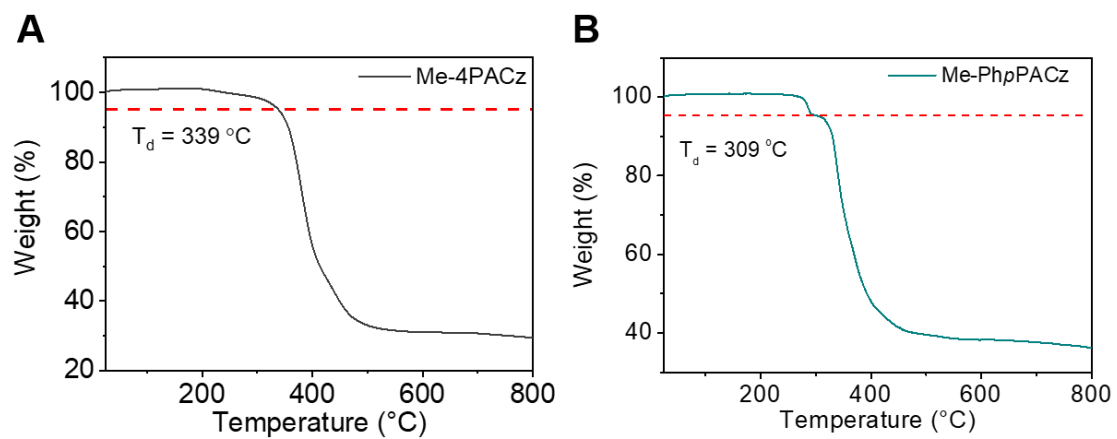


Figure S40. TGA curves of (A) Me-4PACz and (B) Me-PhpPACz. The decomposition temperature of Me-4PACz, Me-PhpPACz are 339 °C and 309 °C, respectively.

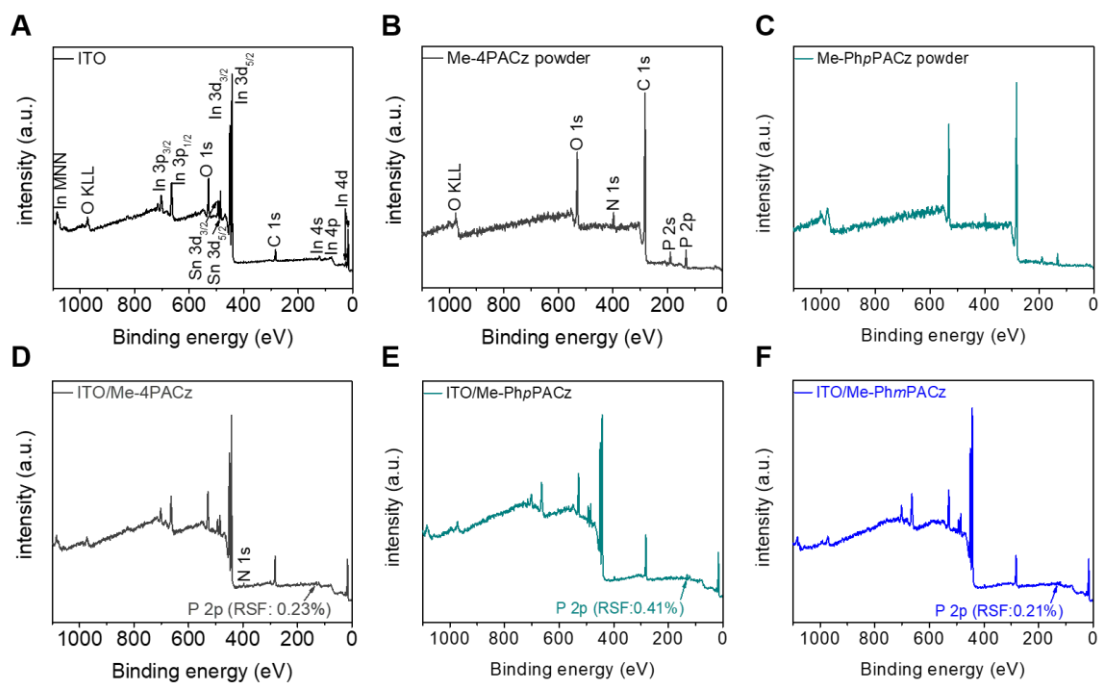


Figure S41. The XPS full spectra of (A) ITO, (B) Me-4PACz powder, (C) Me-PhpPACz powders and SAMs deposited on ITO for (D) Me-4PACz, (E) Me-PhpPACz and (F) Me-PhmPACz.

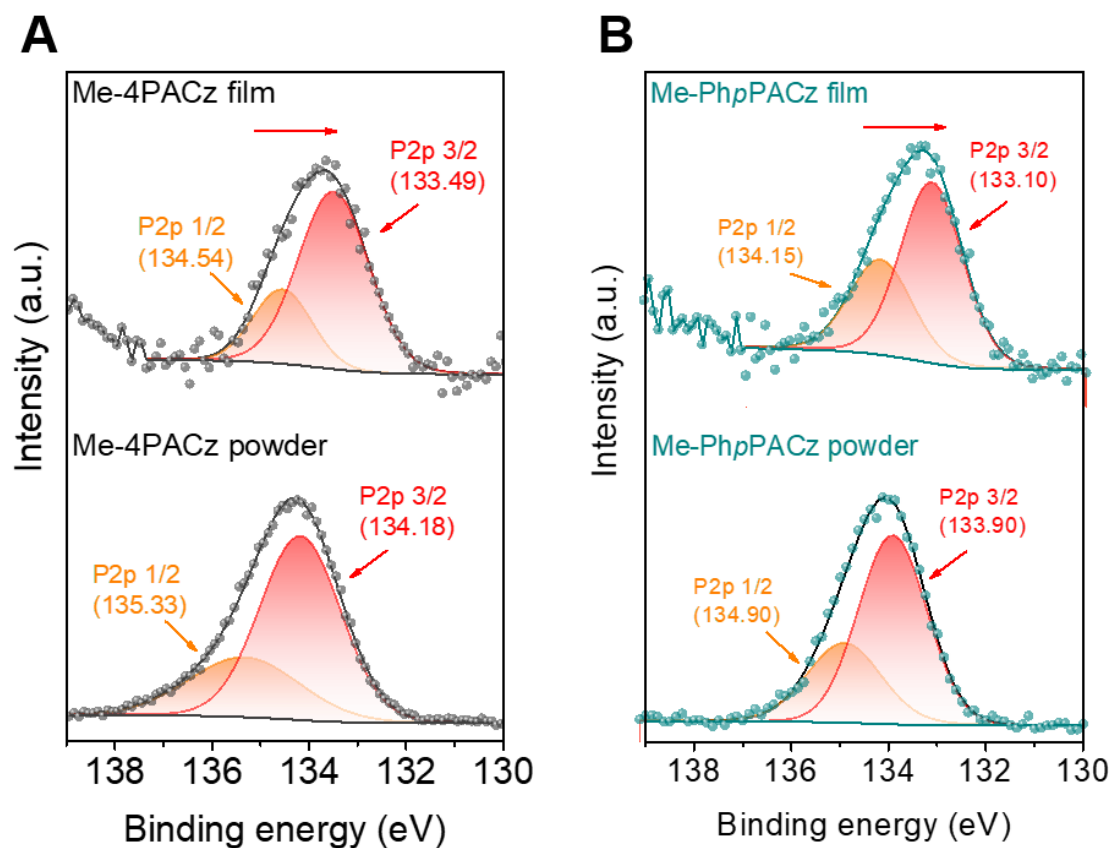


Figure S42. The XPS spectra of P 2p for powder and thin film deposited on ITO of (A) Me-4PACz and (B) Me-PhpPACz.

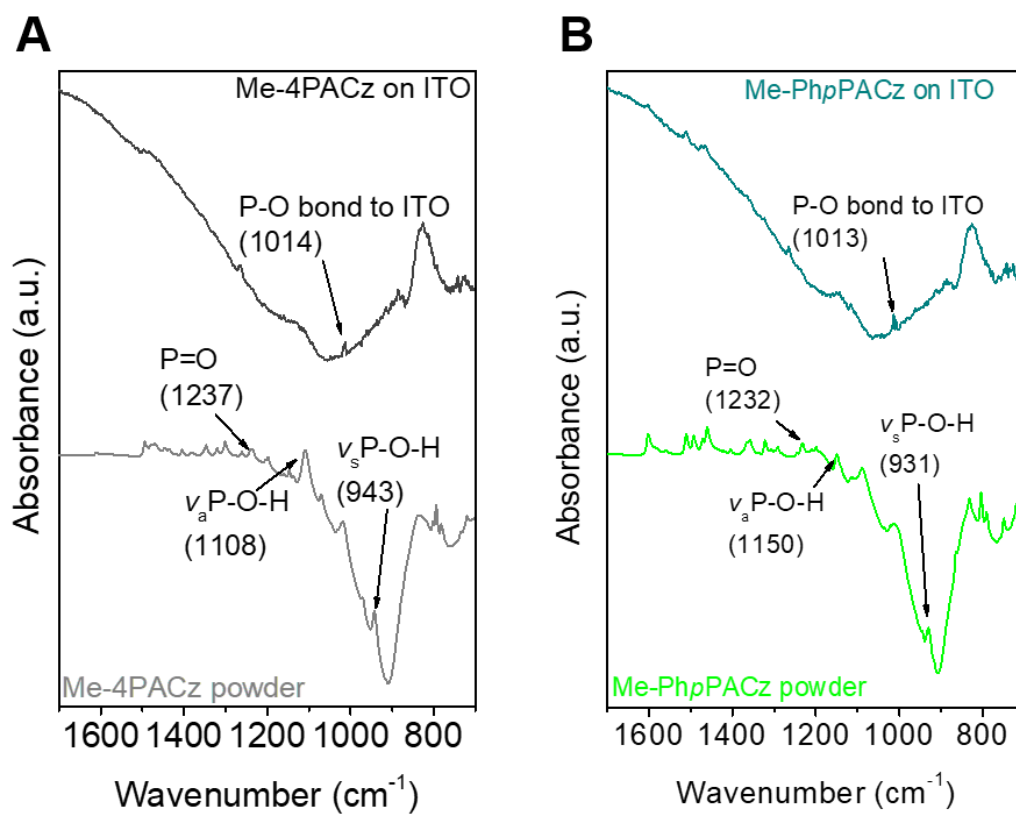


Figure S43. The FTIR absorption spectra for power and thin film of (A) Me-4PACz and (B) Me-PhpPACz.

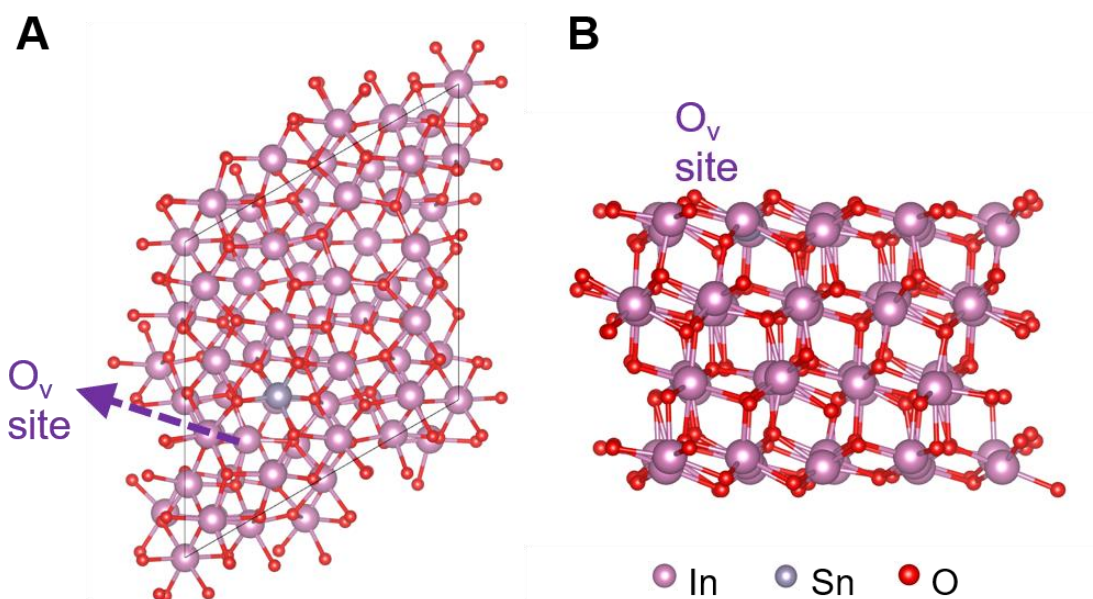


Figure S44. (A) Top view and (B) side view of the optimized ITO (111) surface. The red, pink, and gray balls represent O, In, and Sn atoms, respectively.

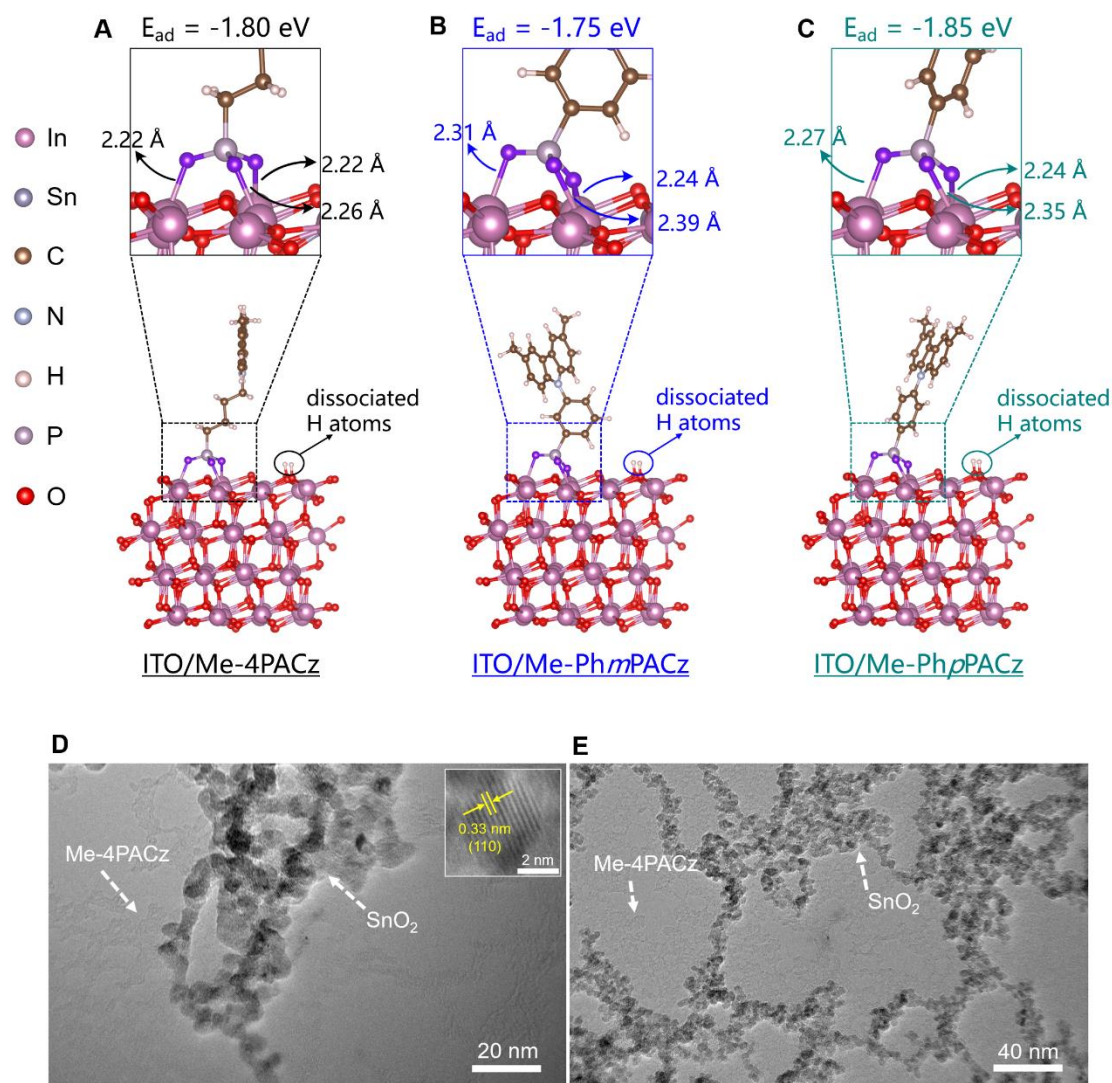


Figure S45. The final state of the dissociative adsorption process of three SAM molecules on the ITO surface: (A) ITO/Me-4PACz, (B) ITO/Me-PhmPACz, and (C) ITO/Me-PhpPACz. TEM images of Me-4PACz with SnO₂ particles at (D) high and (E) low magnifications. The TEM image reveals a lattice spacing of 0.33 nm, consistent with the SnO₂ (110) plane, ruling out the possibility of the light-colored regions being SnO₂.

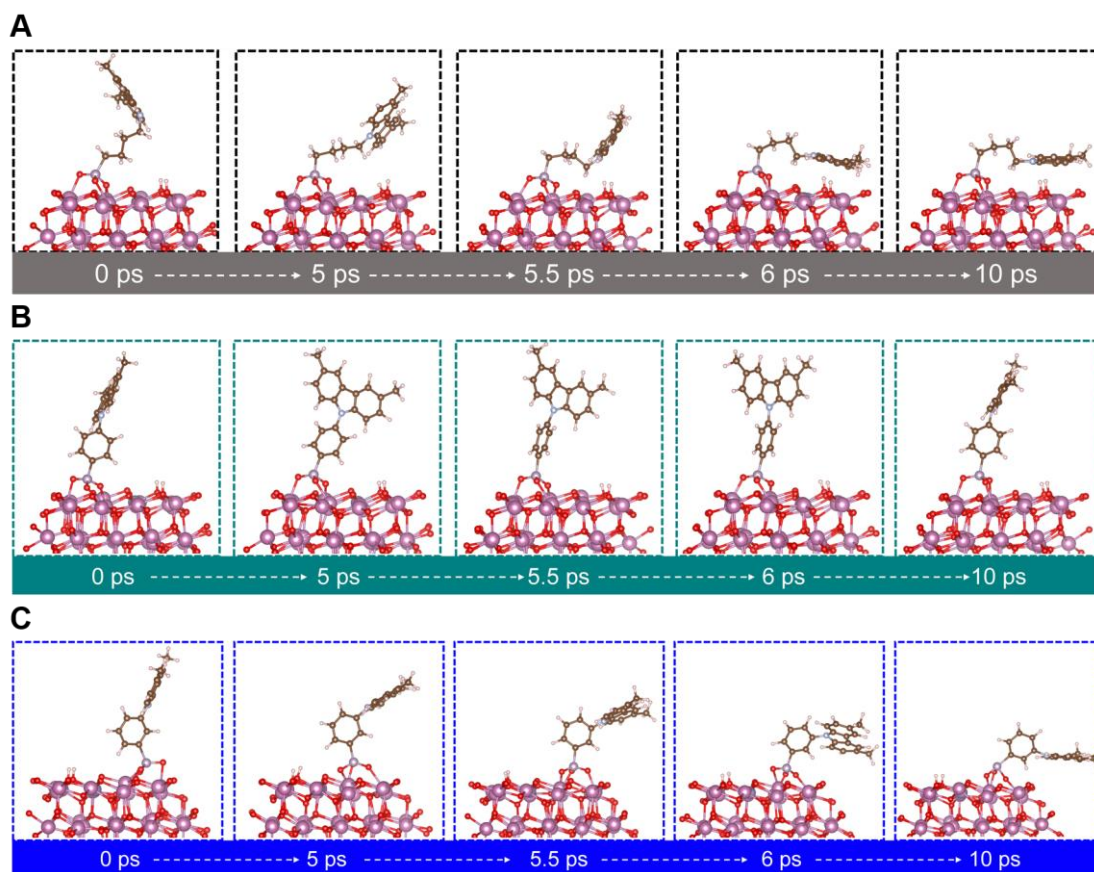


Figure S46. The molecular dynamics simulation trajectory of (A) Me-4PACz, (B) Me-PhpPACz, and (C) Me-PhmPACz molecule on the ITO surface.

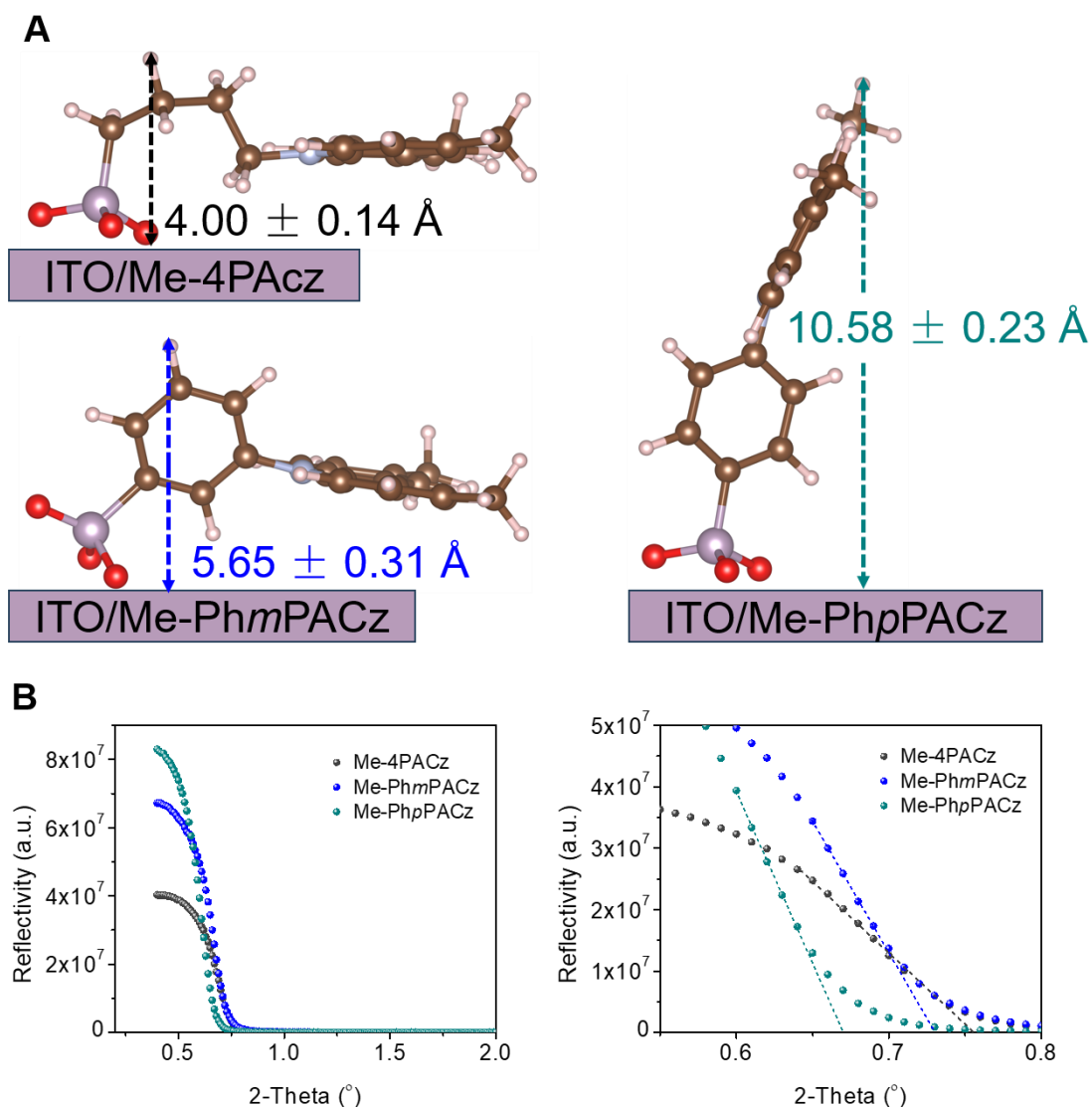


Figure S47. Theoretical calculated and experimental measured molecular height of SAMs

(A) Theoretically calculated single molecular height of SAMs to the ITO surface. The average height was obtained from the final 1 ps AIMD simulation. (B) The X-ray reflectivity (XRR) measurements of self-assembled Me-4PACz, Me-PhmPACz, and Me-PhpPACz on ITO, along with localized enlargements in the 2-theta range of 0.5-0.8°.

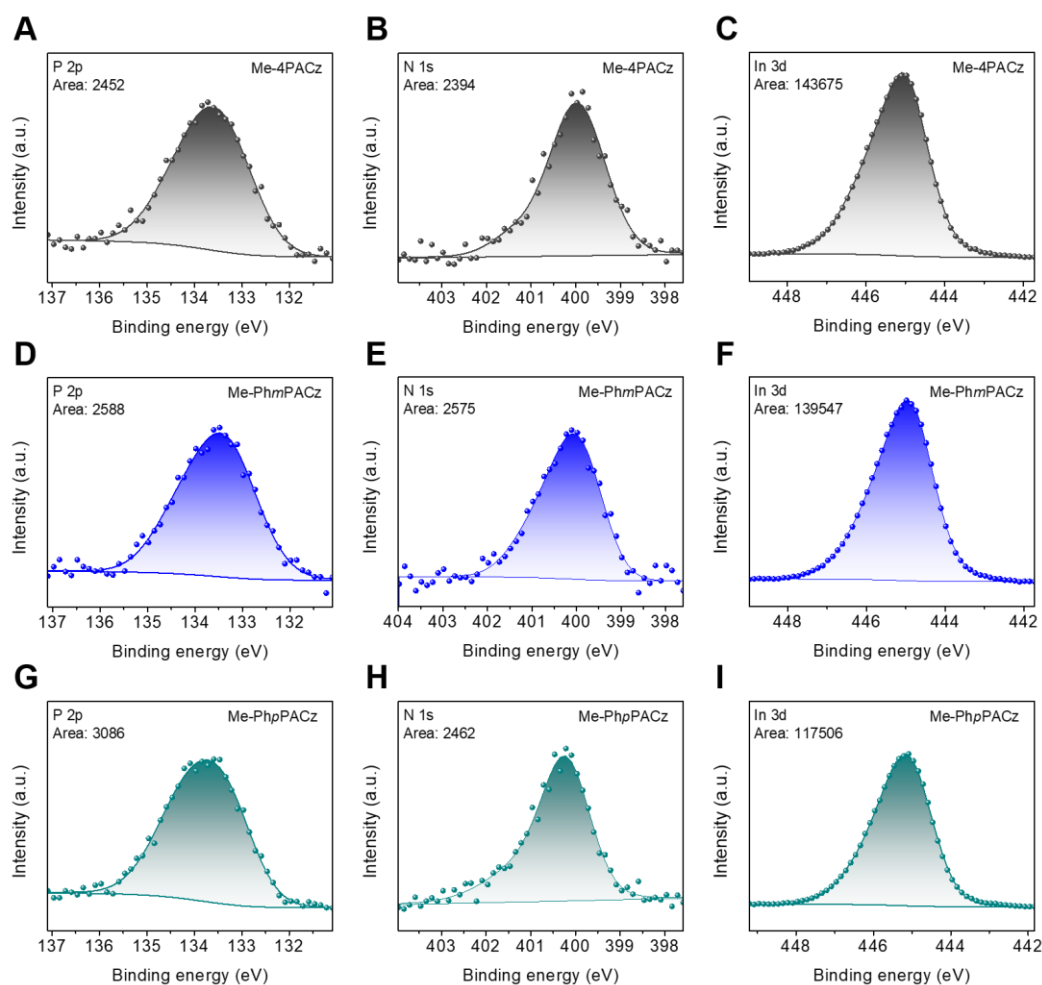


Figure S48. The XPS spectra and corresponding peak areas of the P, N, and In elements for (A-C) Me-4PACz, (D-F) Me-PhmPACz, and (G-I) Me-PhpPACz on ITO.

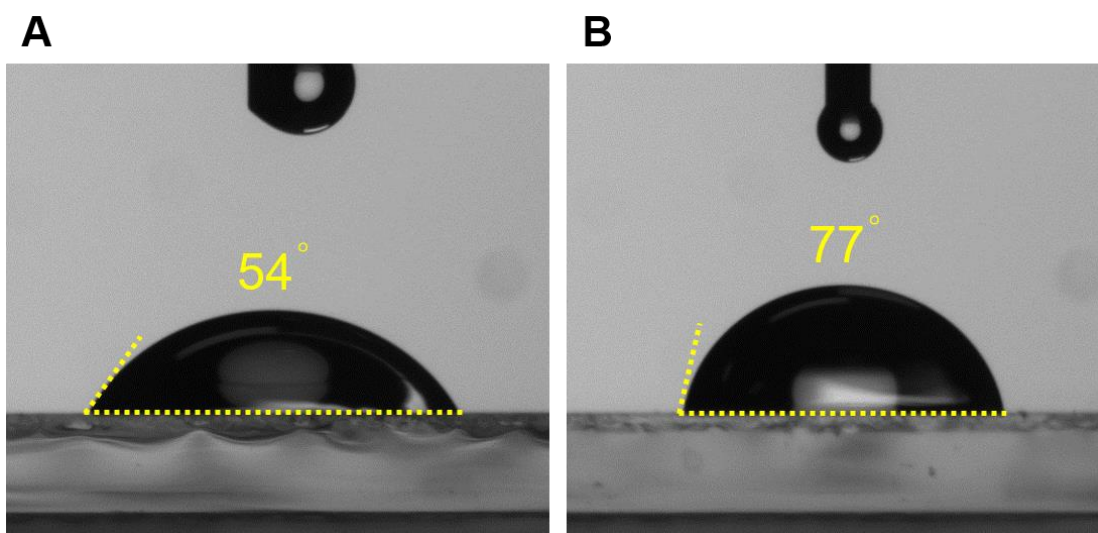


Figure S49. The contact angles of water droplets formed on (A) Me-4PACz and (B) Me-PhpPACz films.

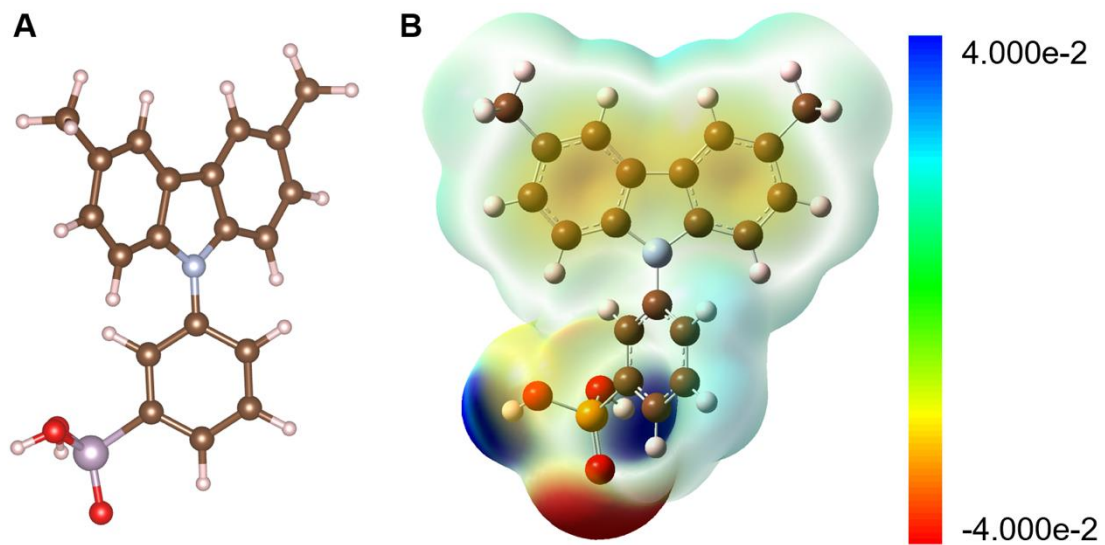


Figure S50. (A) Optimized structure and (B) electrostatic potential surface (EPS) for the Me-PhmPACz molecule.

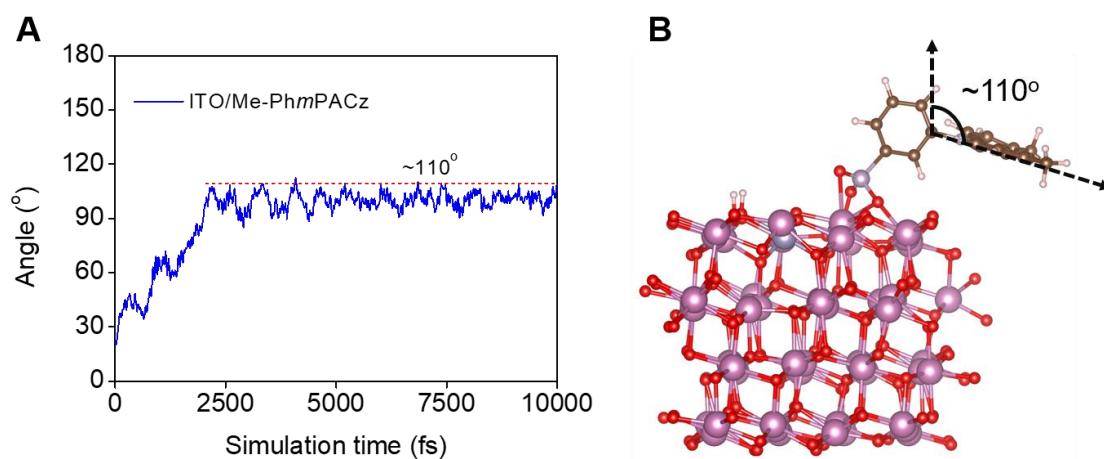


Figure S51. (A) The angular evolution of Me-PhmPACz molecule on the ITO surface during a 10 ps AIMD simulation at a typical temperature of 300 K, along with the final configuration of (B) Me-PhmPACz molecule on the ITO surface after AIMD simulation.

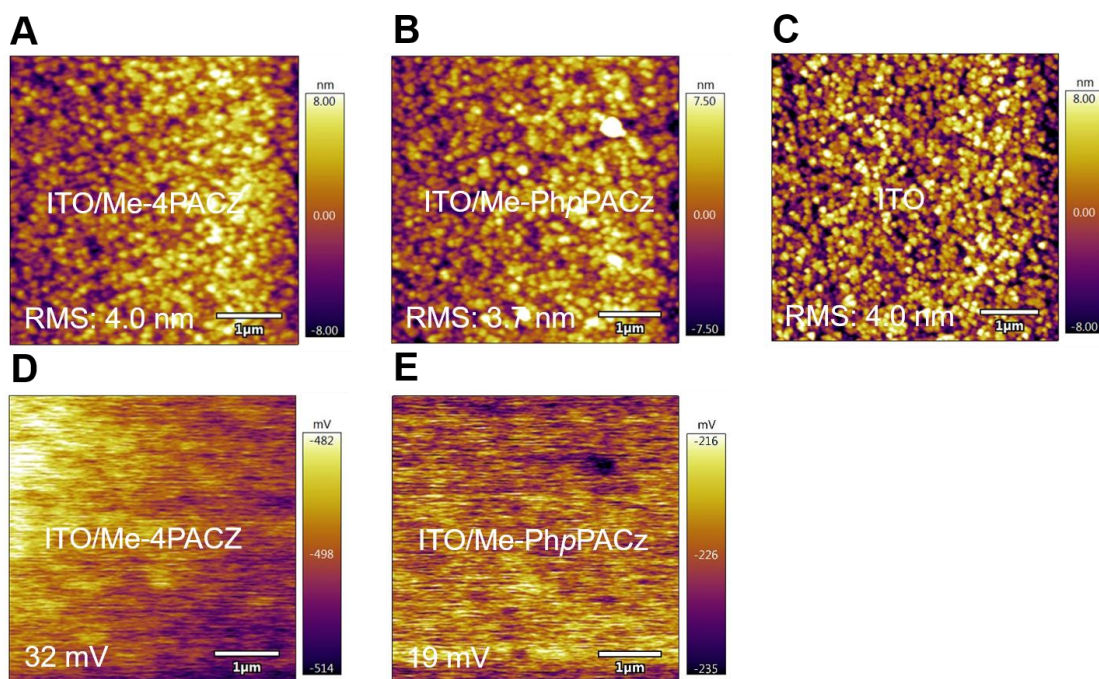


Figure S52. AFM images and root mean square (RMS) values for (A) ITO/Me-4PACz, (B) ITO/Me-PhpPACz and (C) ITO; The KPFM images of (D) Me-4PACz and (E) Me-PhpPACz films on ITO.

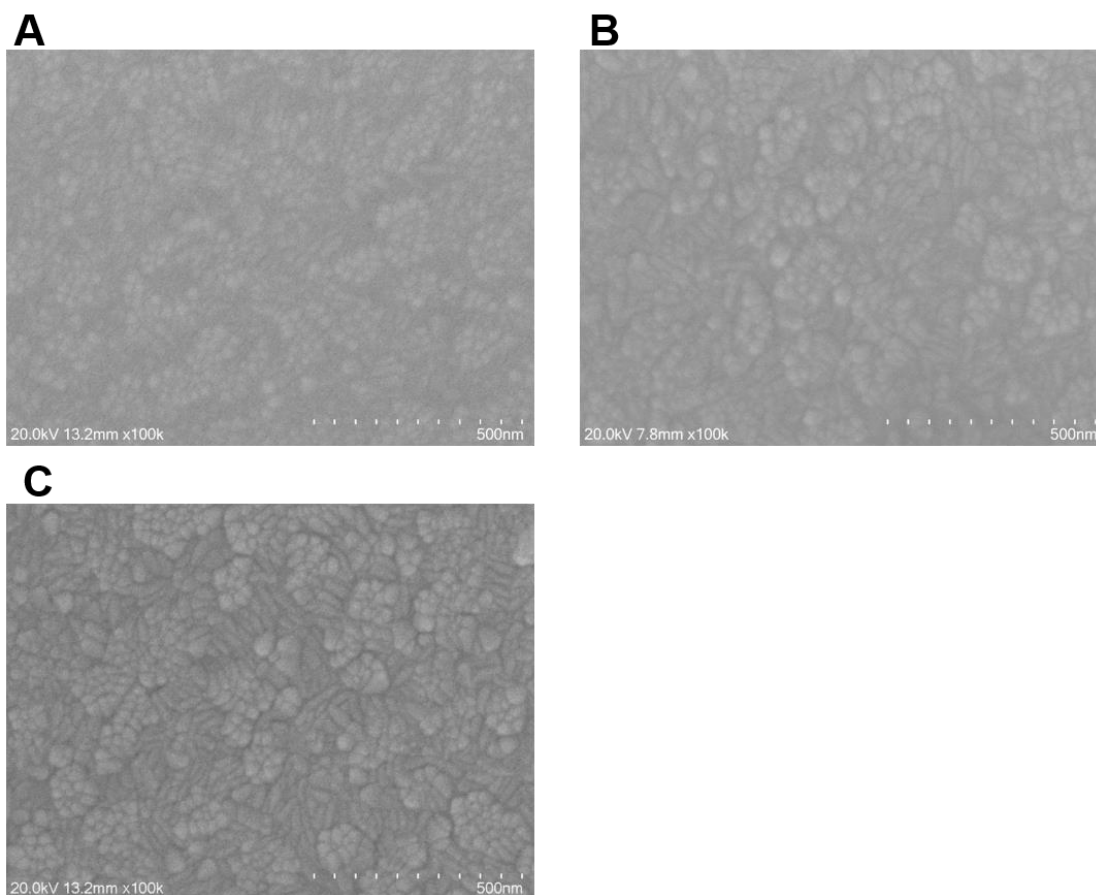
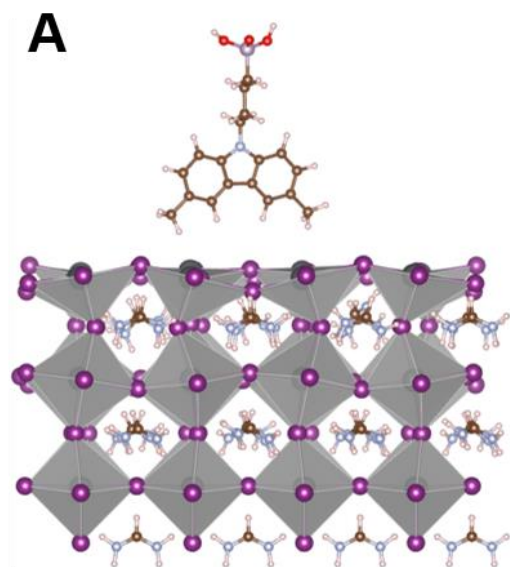
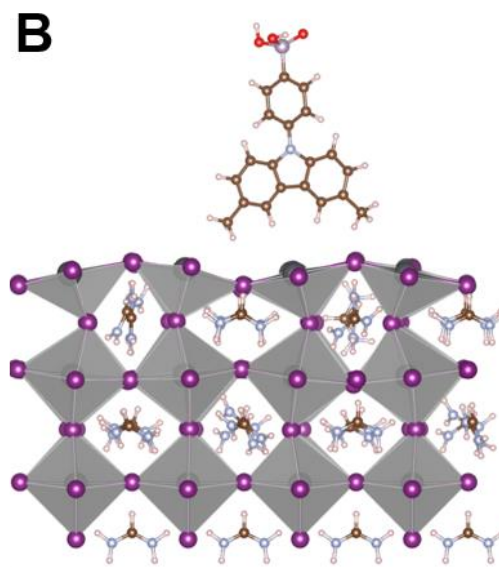


Figure S53. SEM images of (A) bare ITO and (B) Me-4PACz, (C) Me-PhpPACz films deposited on ITO. Obtaining clear SEM images at a 100,000x magnification can be challenging. SEM Images demonstrate that only Me-PhpPACz allows a clear view of the ITO surface, implying an improvement in ITO conductivity.



$$E_{ad} = -0.23 \text{ eV}$$



$$E_{ad} = -1.48 \text{ eV}$$

Figure S54. The configurations of SAM molecules when absorbed on the FAPbI₃ perovskite surface: (A) Me-4PACz and (B) Me-PhpPACz.

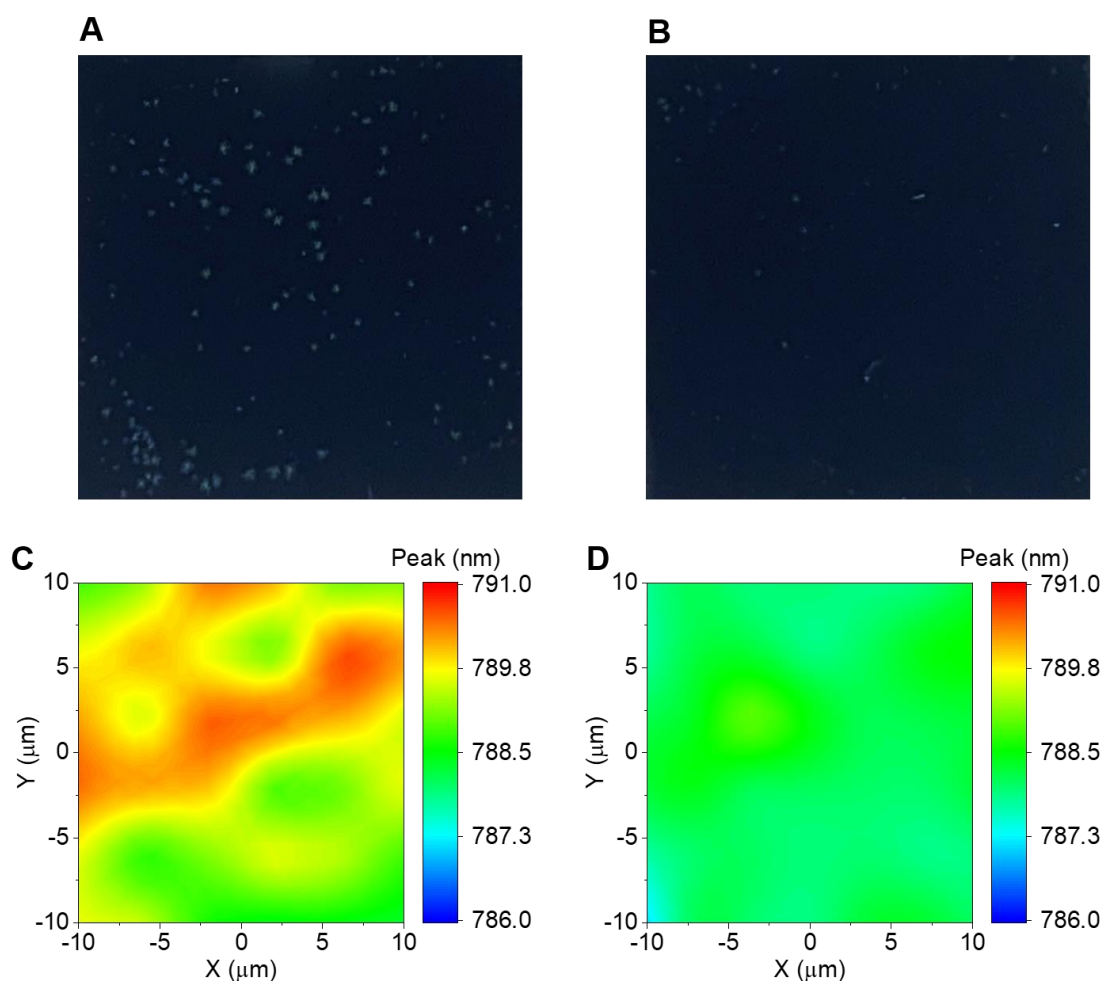


Figure S55. Photographs of the bottom surface (Glass Side) of perovskite prepared on (A) Me-4PACz and (B) Me-PhpPACz. Me-4PACz's perovskite exhibits phase separation (snowflake-like), whereas Me-PhpPACz's perovskite does not. Wavelength-resolved PL mapping images of perovskite deposited on (C) Me-4PACz and (D) Me-PhpPACz.

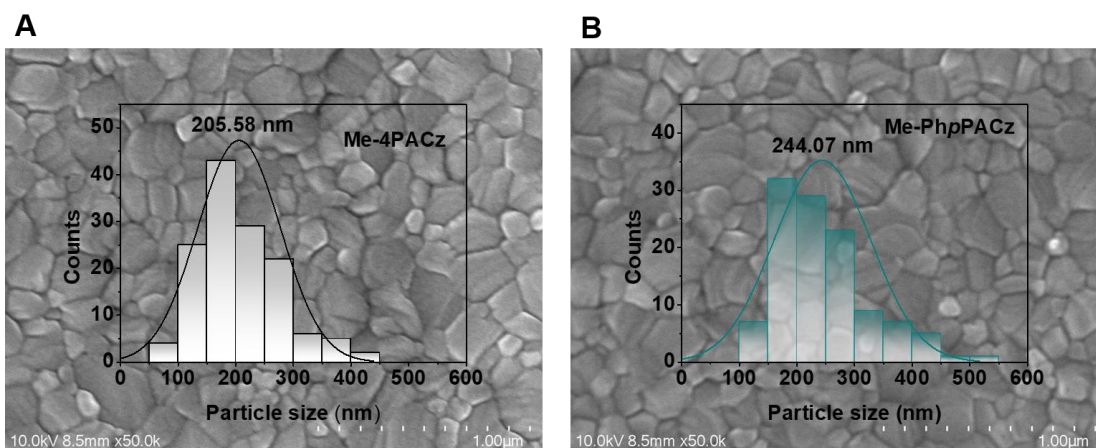


Figure S56. Top-view SEM images and particle-size statistical distributions of perovskite films deposited on (A) Me-4PACz and (B) Me-PhpPACz. SEM images show a slight increase in perovskite grain size for Me-PhpPACz compared to Me-4PACz as the HSL, with sizes of 205.58 nm (Me-4PACz) and 244.07 nm (Me-PhpPACz), respectively.

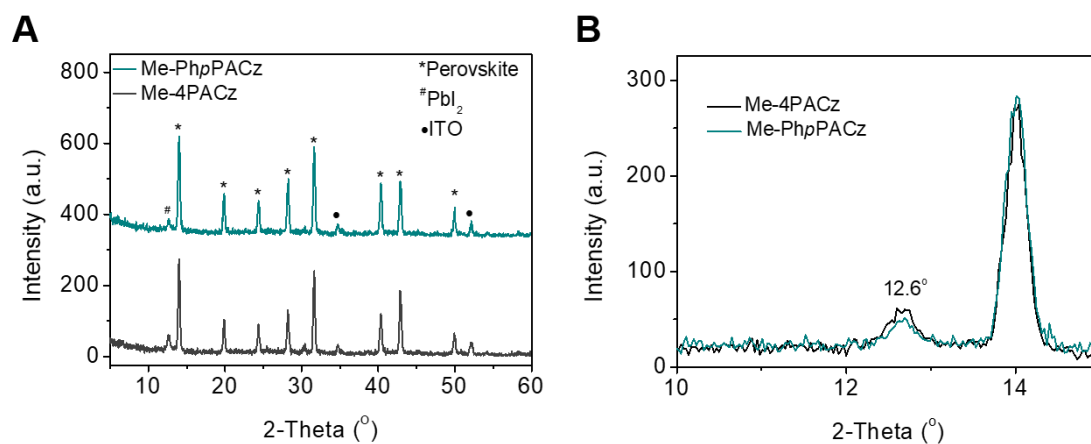


Figure S57. (A) X-ray diffraction (XRD) patterns of perovskite on Me-4PACz and Me-PhpPACz and (B) enlarged pattern with 2-theta value of 10-15°.

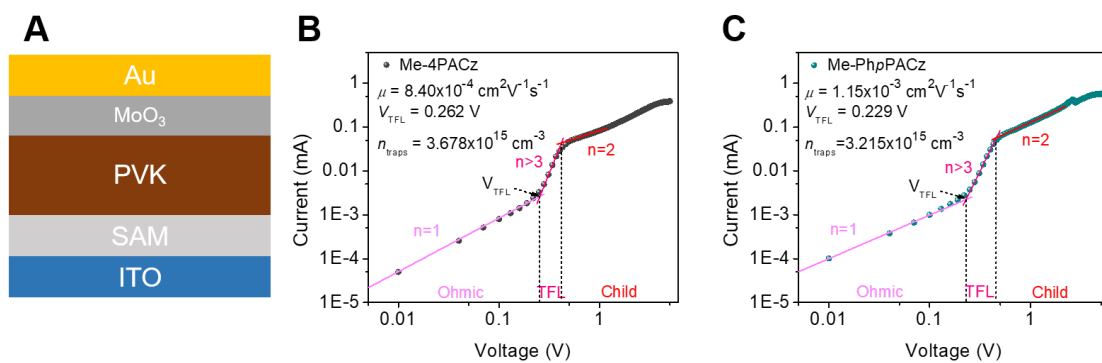


Figure S58. (A) hole-only devices structure and dark J-V curves obtained through SCLC analysis for (B) Me-4PACz and (C) Me-PhpPACz as the HSL.

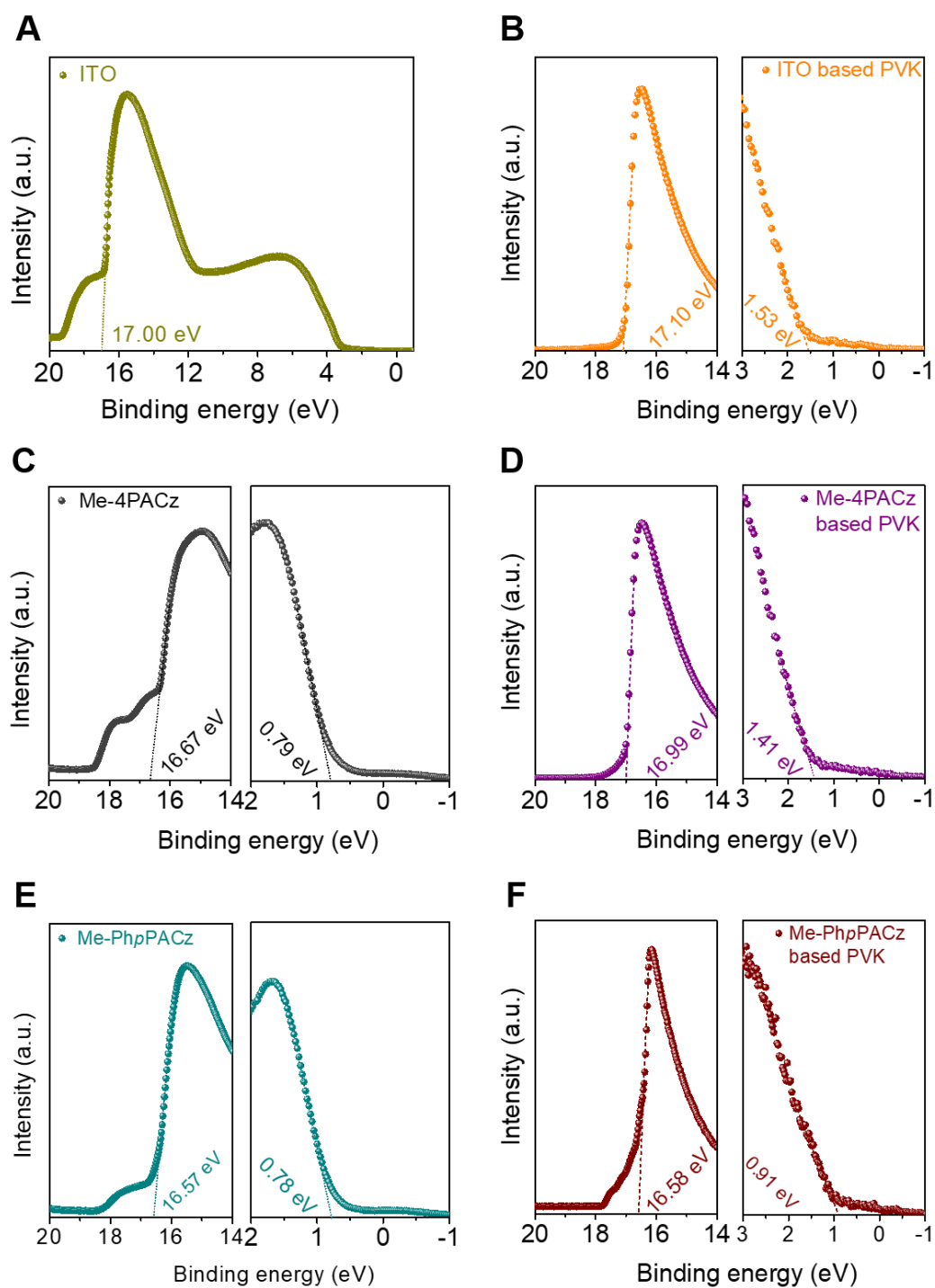


Figure S59. UPS spectra of (A) bare ITO, (B) perovskite deposited on ITO, (C) Me-4PACz modified ITO, (D) perovskite deposited on Me-4PACz, (E) Me-PhpPACz modified ITO, and (F) perovskite deposited on Me-PhpPACz.

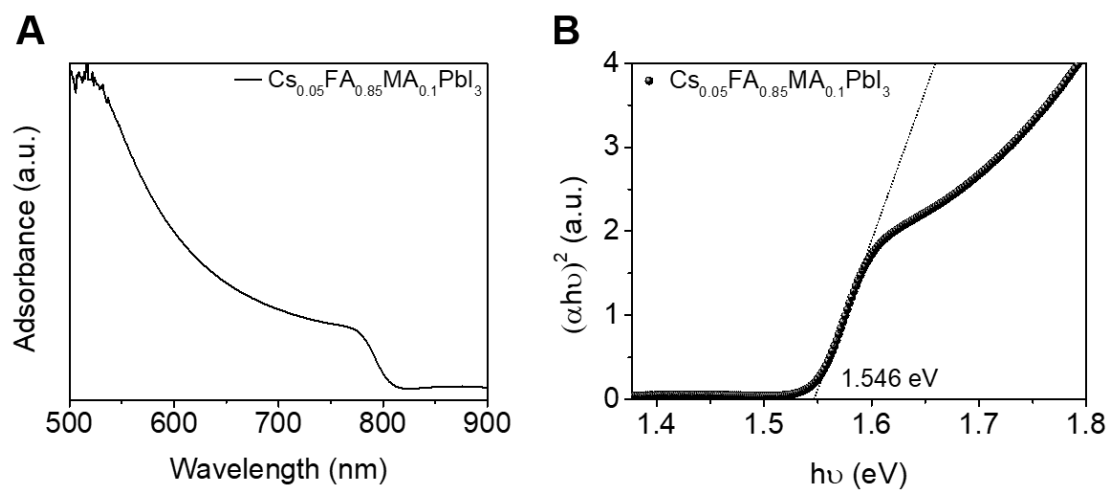


Figure S60. (A) UV-vis absorption spectra and (B) the corresponding Tauc plot of perovskite ($\text{Cs}_{0.05}\text{FA}_{0.85}\text{MA}_{0.1}\text{PbI}_3$).

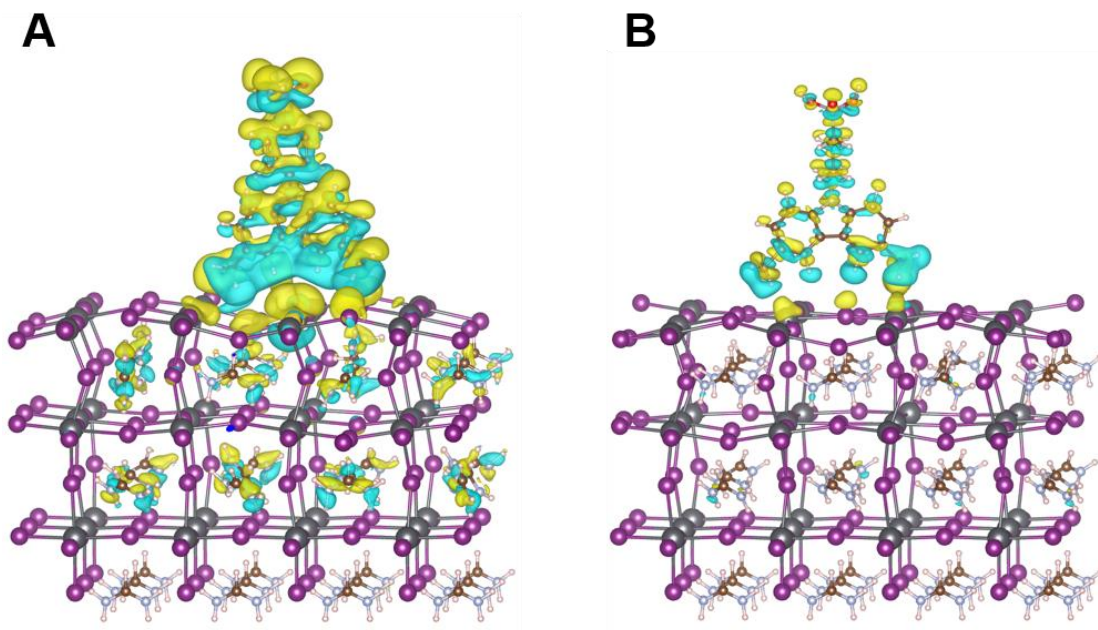


Figure S61. The 3D visual charge density difference between the SAM molecules of (A) Me-PhpPACz and (B) Me-4PACz and perovskite. Electron depletion/accumulation is depicted by the blue/yellow isosurfaces at $\pm 0.001 \text{ |e| \AA}^{-3}$.

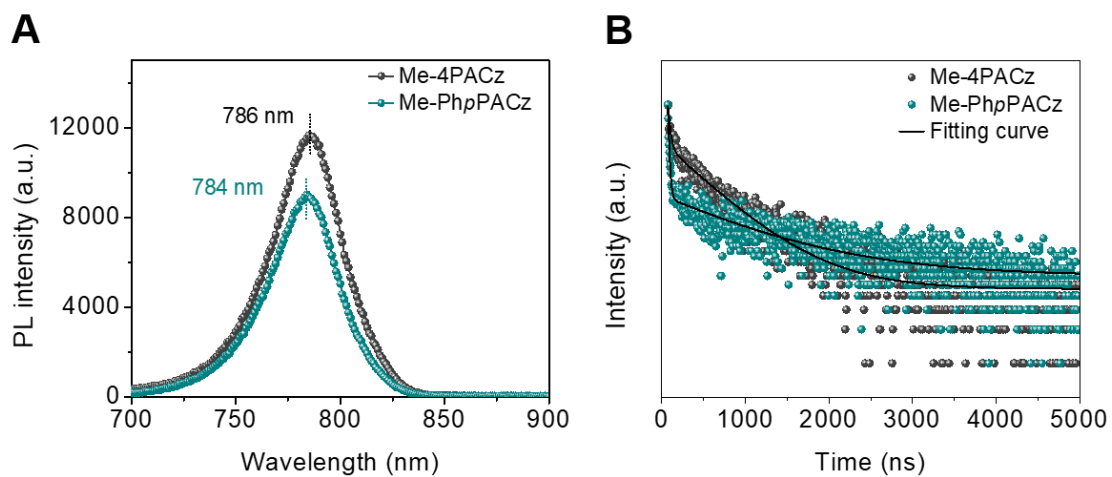
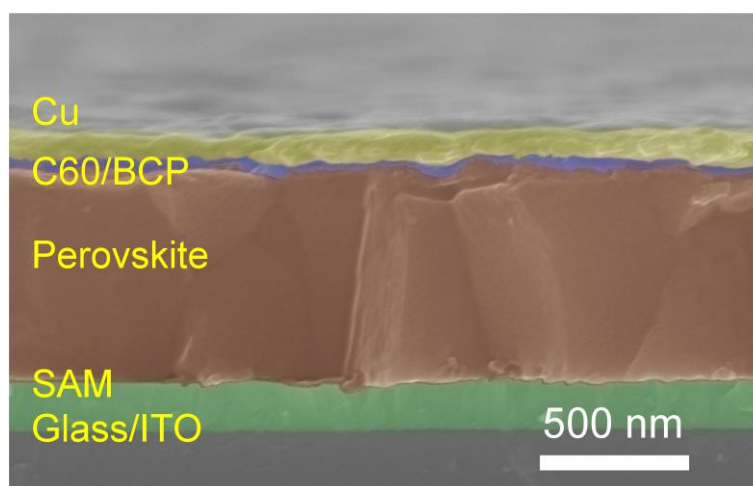


Figure S62. (A) PL spectra and (B) TRPL spectra of ITO/Me-4PACz/PVK and ITO/Me-PhpPACz/PVK.



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Figure S63. Cross-sectional SEM image of PSC based on Me-4PACz.

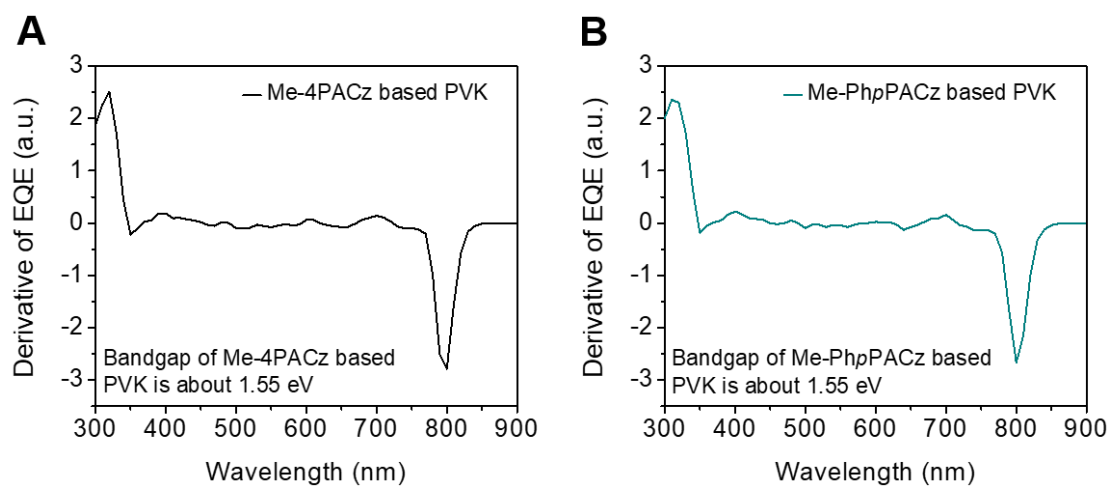
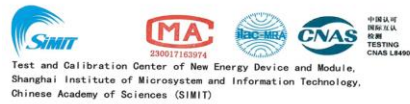


Figure S64. The curve of derivative spectrum of IPCE for (A) Me-4PACz and (B) Me-PhpPACz based PSC devices.

A



Measurement Report

Report No. 24TR012413

Client Name Department of Chemistry, College of Science, Southern University of Science and Technology (SUSTech)

Client Address 1088 Xueyuan Avenue, Nanshan District, Shenzhen, Guangdong, China

Sample Perovskite solar cell

Manufacturer Zong-Xiang Xu's Group, SUSTech

Measurement Date 24th January, 2024

Performed by: Qiang Shi *Qiang Shi* **Date:** 24/01/2024

Reviewed by: Wenjie Zhao *Wenjie Zhao* **Date:** 24/01/2024

Approved by: Yucheng Liu *Yucheng Liu* **Date:** 24/01/2024

Address: No.235 Chengbei Road, Jiading, Shanghai **Post Code:** 201800

E-mail: solarcell@mail.sim.ac.cn **Tel:** +86-021-69976921

The measurement report without signature and seal are not valid.
This report shall not be reproduced, except in full, without the approval of SIMIT.

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Report No. 24TR012413

Measurement Results

	Forward Scan (Isc to Voc)	Reverse Scan (Voc to Isc)
Area	6.72 mm ²	
Isc	1.733 mA	1.737 mA
Voc	1.174 V	1.174 V
Pmax	1.692 mW	1.735 mW
Ipm	1.640 mA	1.642 mA
Vpm	1.031 V	1.044 V
FF	83.16 %	85.08 %
Eff	25.18 %	25.82 %

- Spectral Mismatch Factor SMM=0.9960.
- Designated illumination area defined by a thin metal mask were measured by the measuring microscope.
- Test results listed in this measurement report refer exclusively to the mentioned test sample.
- The results apply only at the time of the test, and do not imply future performance.

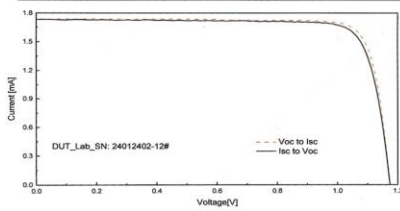


Fig.1 I-V curves of the measured sample

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Report No. 24TR012413

Pmax[mW]	1.692
Ipm[ma]	1.662
Vpm[V]	1.038
Eff[%]	25.18

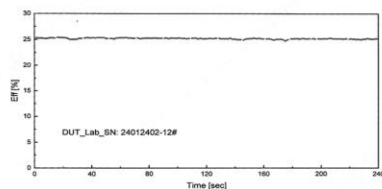


Fig.2 Steady-state maximum power output of the measured sample

—End of Report—

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B

福建省计量科学研究院
FUJIAN METROLOGY INSTITUTE
(国家光伏产业计量测试中心)
National PV Industry Measurement and Testing Center

检测报告
Test Report

报告编号: 2403-00076

客户名称: Xu Zong-Xiang Group, Department of Chemistry, Southern University of Science and Technology

联系信息: Department of Chemistry, College of Science, 1088 Xueyuan Avenue, Nanshan District, Shenzhen, Guangdong, China

物品名称: Perovskite Solar Cell (PV)

型号/规格: 1.5x1.5

物品编号: 20240105 34-2 PSC

制造厂商: Xu Zong-Xiang Group

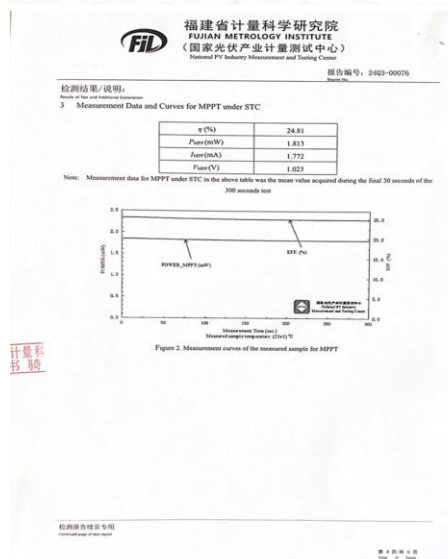
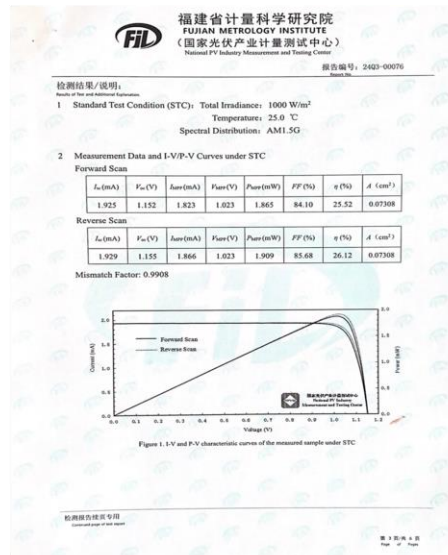
物品验收日期: 2024-01-31

检测日期: 2024-01-31

批准人: 杨建中
核验员: 何翔
检测员: 陈彩云

发布日期: 2024 年 02 月 06 日

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Figure S65. The performance of small area PSCs based on Me-PhpPACz: (A) SIMIT certification report and (B) NPVM certification report.

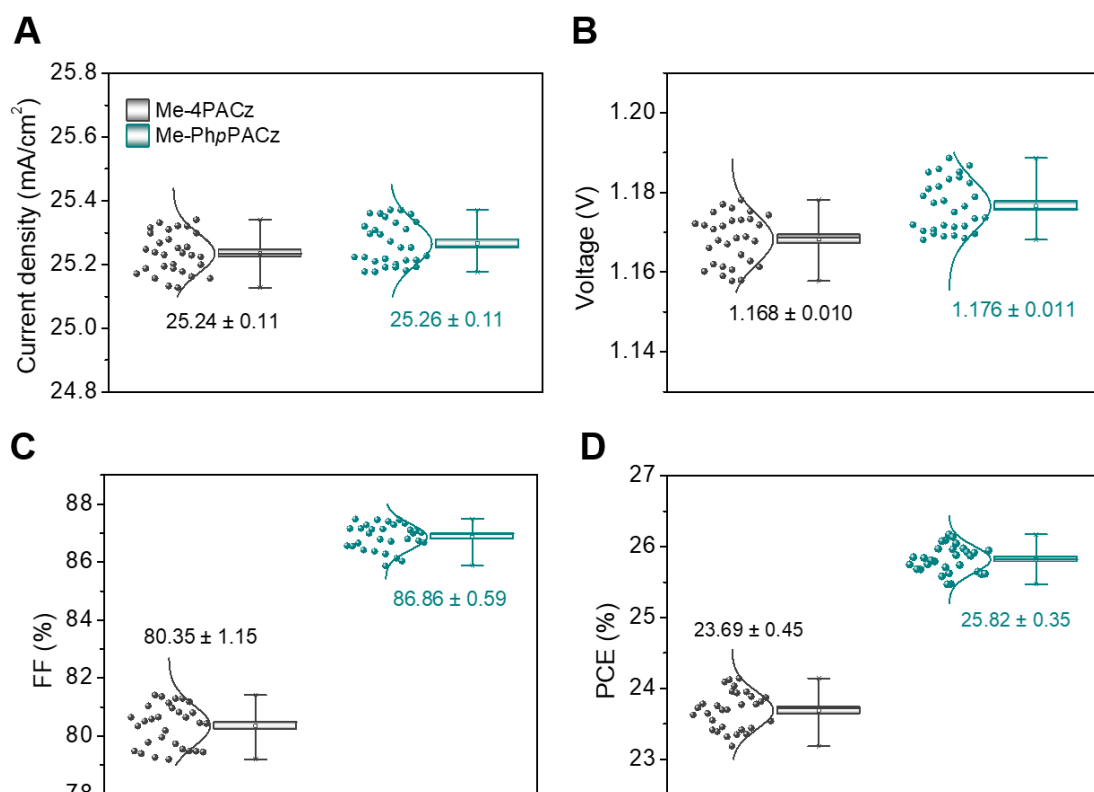


Figure S66. Device statistics of photovoltaic parameters: (A) J_{sc} , (B) V_{oc} , (C) FF, and (D) PCE for Me-4PACz and Me-PhpPACz based PSC devices (30 devices fabricated from the same batch).

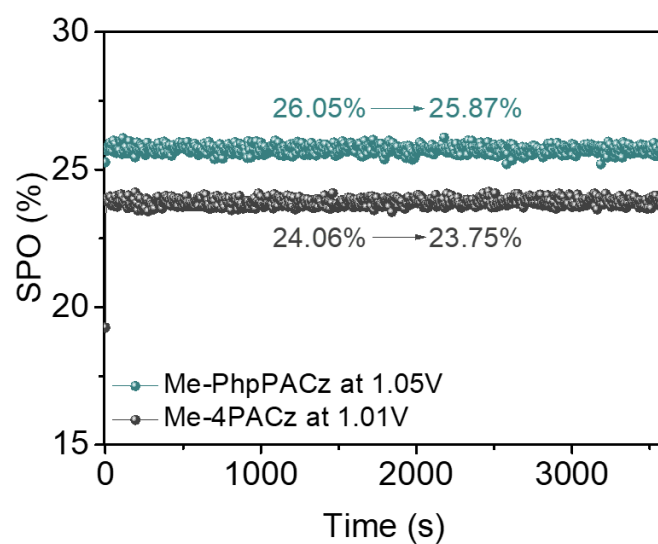


Figure S67. Stabilized power output characteristics of the Me-4PACz and Me-PhpPACz based PSCs.

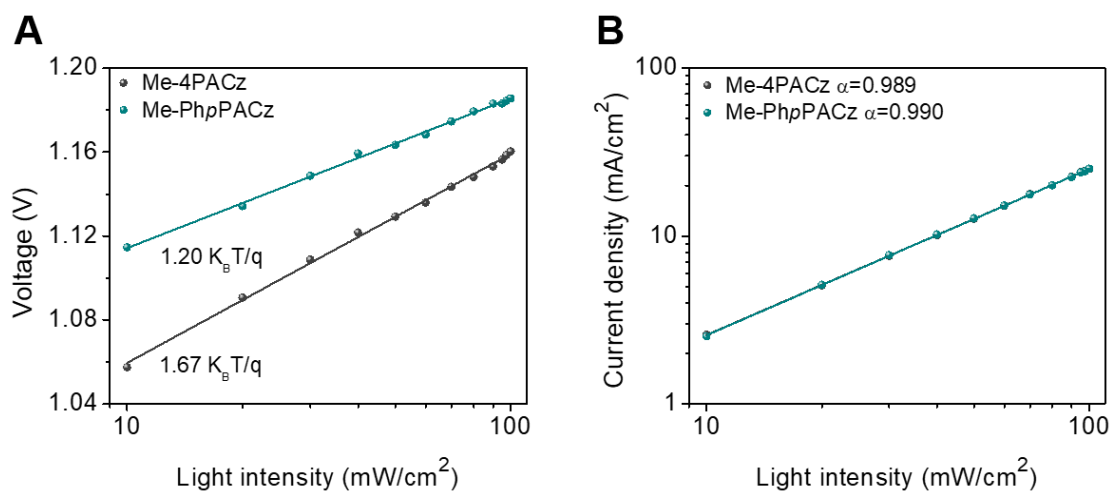


Figure S68. Light intensity-dependent measurements of (A) V_{oc} and (B) J_{sc} of the Me-4PACz and Me-PhpPACz based PSCs.

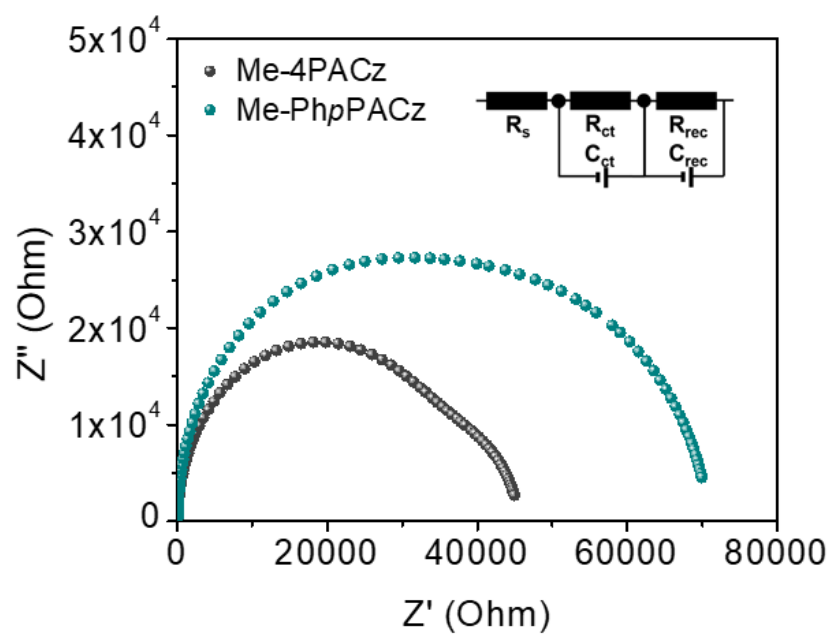
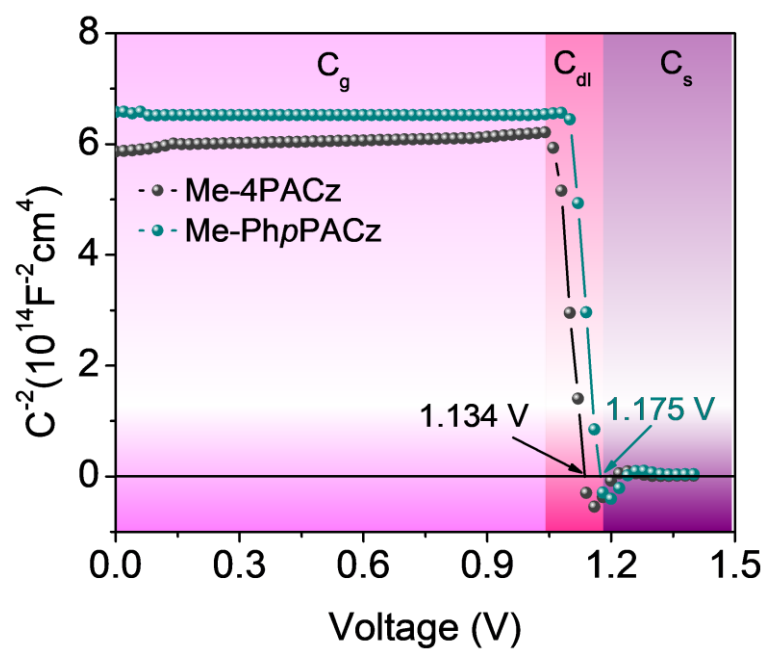


Figure S69. Nyquist plots of the Me-4PACz and Me-PhpPACz based PSCs.



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Figure S70. Mott-Schottky plots of the Me-4PACz and Me-PhpPACz based PSCs at 10 kHz.

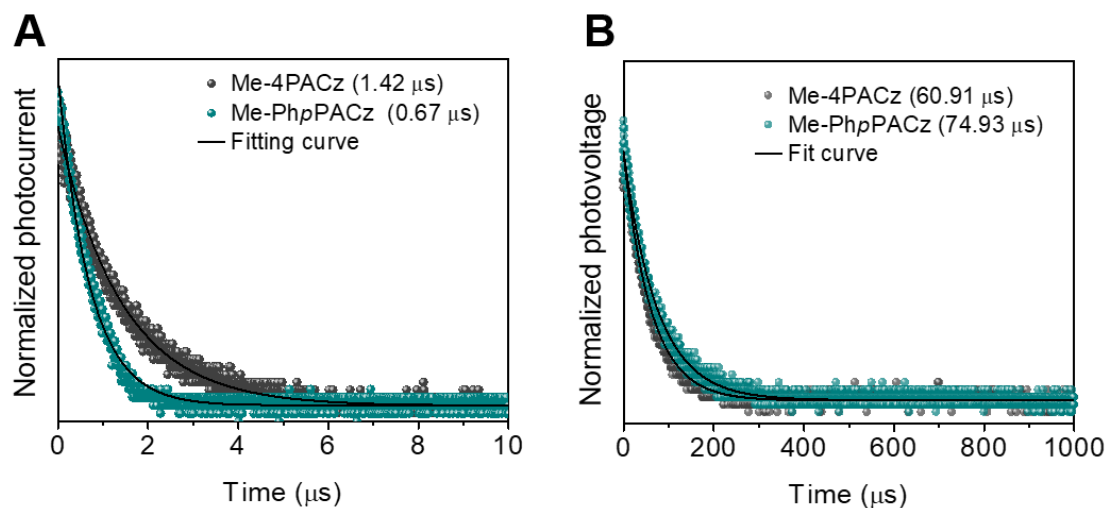


Figure S71. (A) Transient photocurrent and (B) transient photovoltage curves for the Me-4PACz and Me-PhpPACz based PSCs.

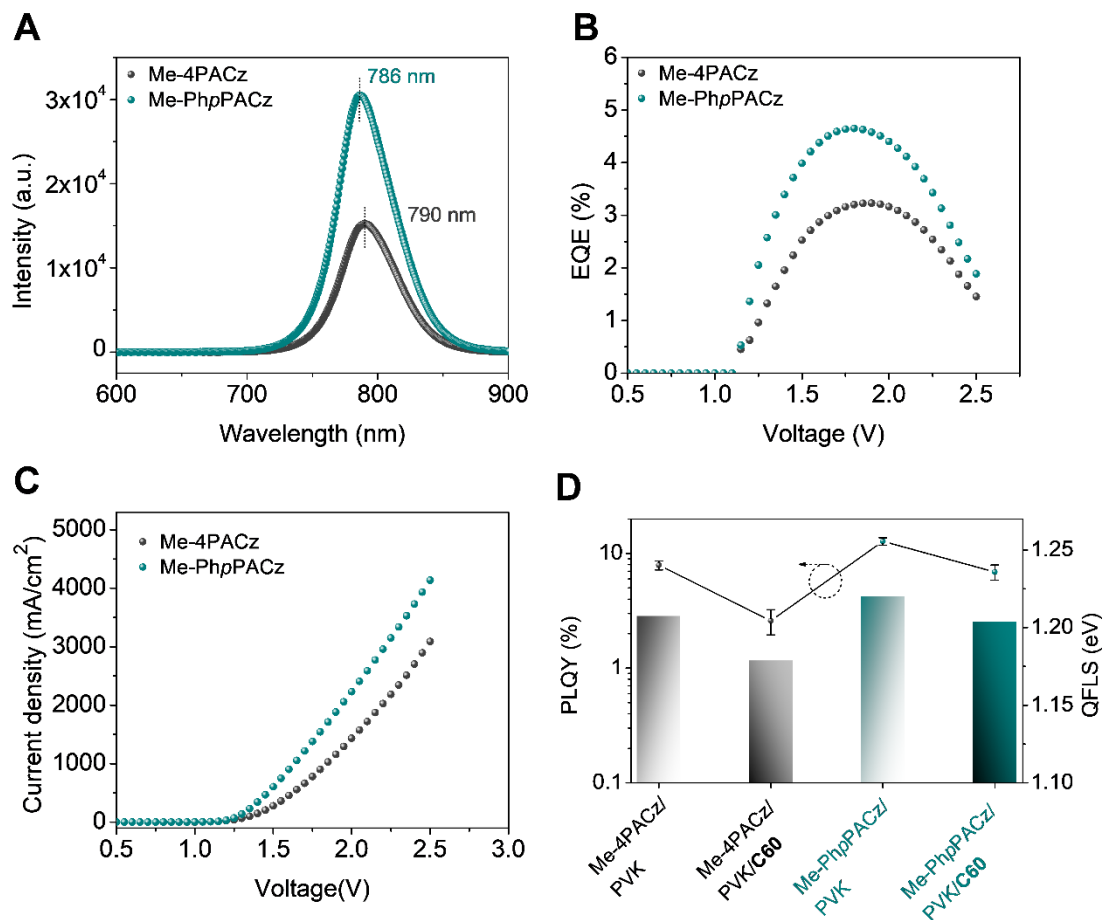


Figure S72. EL and PL measurements of SAM based PSCs

(A) EL emission spectra, (B) EQEEL, (C) current density of PSCs with Me-4PACz and Me-PhpPACz under bias voltages ranging from 0.5 to 2.5 V. (D) PLQY and QFLS of perovskite films on different substrates.

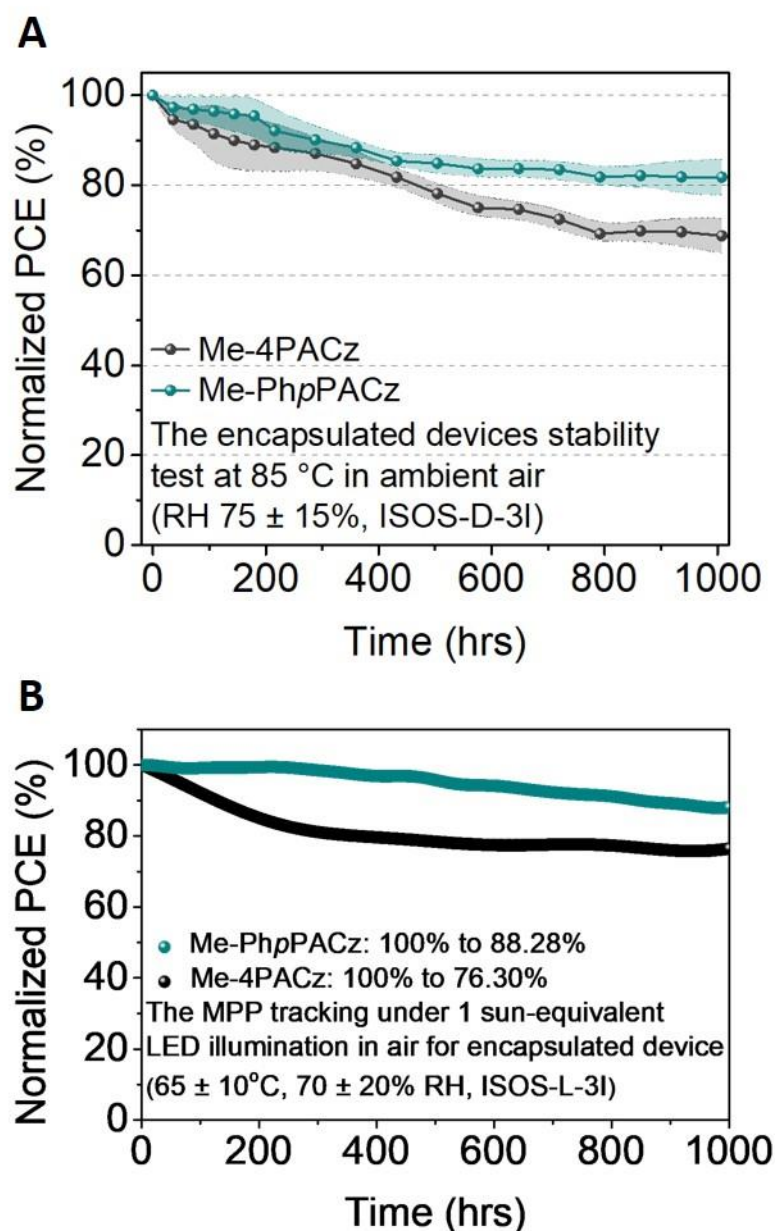


Figure S73. Thermal stability measurements of PSCs

(A) Thermal stability test of the encapsulated PSCs at 85 °C in ambient air (relative humidity 75 ± 15%) and (B) MPP tracking for encapsulated devices in ambient air and under continuous 1 sun-equivalent LED illumination (reaching an operating temperature of 65 ± 10 °C, 70 ± 20% RH).

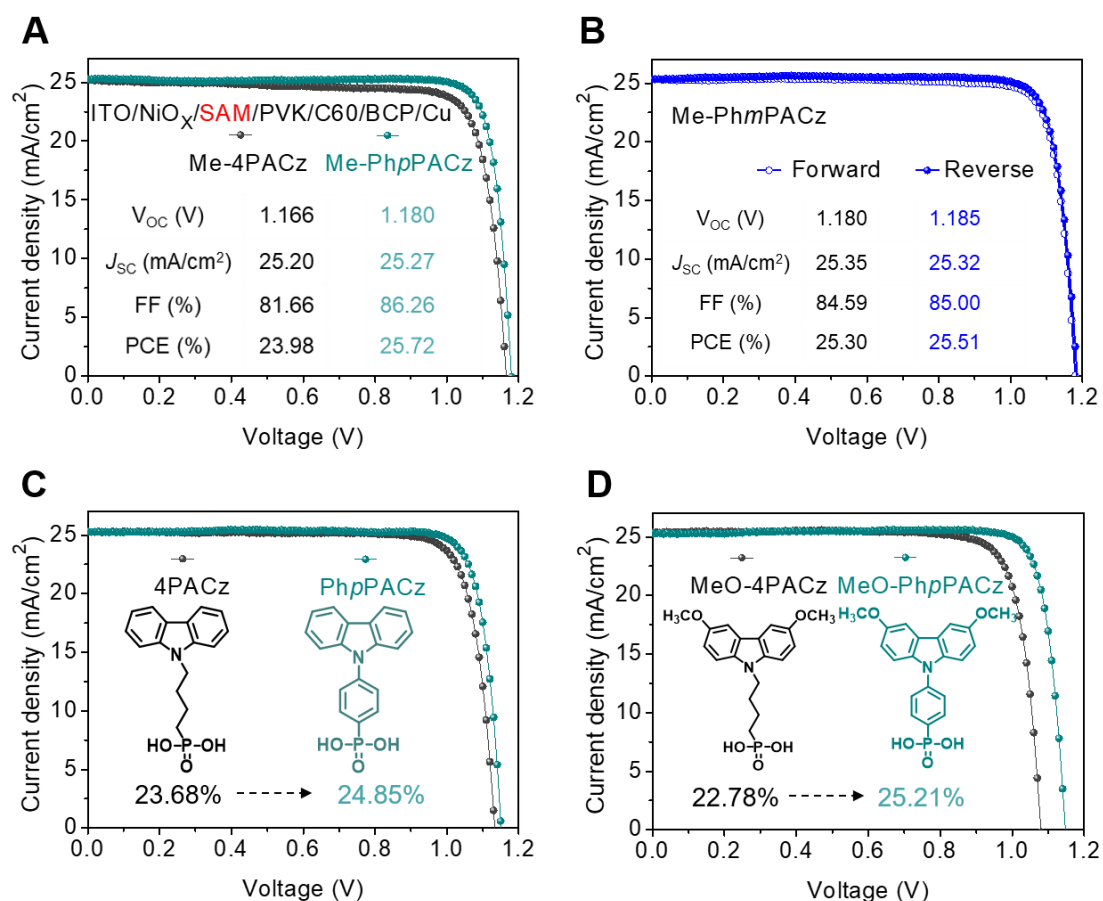


Figure S74. Comparisons of PSCs based on SAMs with different linker

(A) Reverse scan J - V curves of Me-4PACz and Me-PhpPACz based PSCs on NiO_x substrates.
 (B) Forward and reverse scan J - V curves of inverted PSC based on Me-PhmPACz SAM HSL.
 Reverse scan J - V curves of inverted PSC based on SAM HSLs of (C) 4PACz, PhpPAC and (D) MeO-4PACz, MeO-PhpPACz.

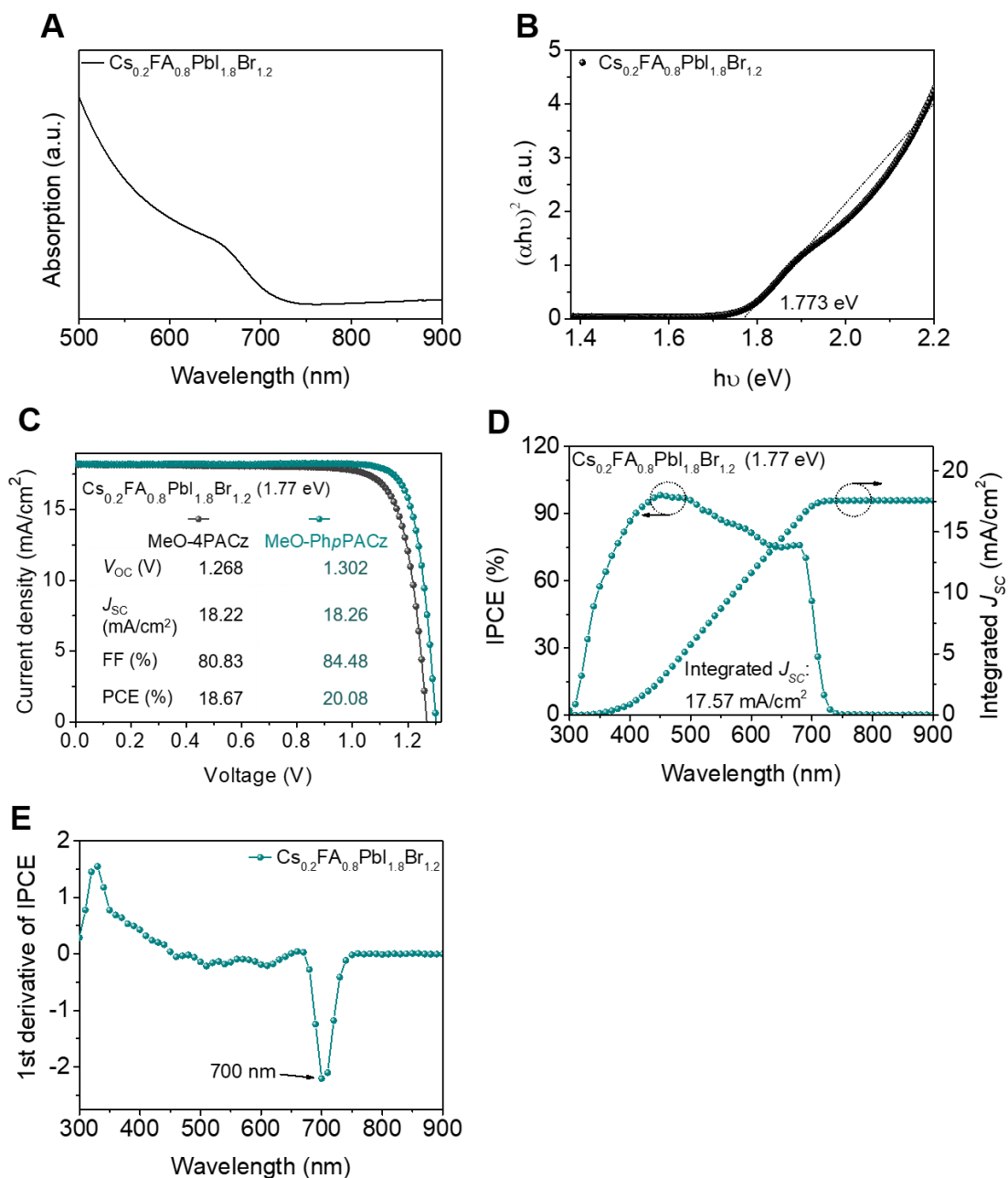


Figure S75. Measurements of PSCs based on SAMs with wide band gap perovskite

(A) UV-vis absorption spectra and (B) the corresponding Tauc plot of wide-bandgap perovskite ($\text{Cs}_{0.2}\text{FA}_{0.8}\text{PbI}_{1.8}\text{Br}_{1.2}$). (C) Reverse scan J - V curves of inverted PSCs based wide band-gap perovskite using MeO-4PACz and MeO-PhpPACz as HSLs. (D) IPCE curve of MeO-PhpPACz based wide-bandgap PSC over the wavelength range from 300 to 900 nm and integrated J_{SC} over the AM1.5G standard spectrum and (E) the corresponding derivative spectrum.

1 **Table S1.** The energy level parameters of SAMs from cyclic voltammograms measurements.

Compound	E_{onset}^{ox} [V]	$Fc_{1/2}$ [V]	HOMO [eV]
Me-4PACz	1.115		-5.393
PhpPACz	1.360		-5.638
Me-PhpPACz	1.238	0.5225	-5.516
Me-PhmPACz	1.220		-5.498
MeO-PhpPACz	1.057		-5.335

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1 **Table S2.** The elemental content and ratio of C, N, P, and In in the XPS full spectra of Me-
2 4PACz, Me-PhmPACz, and Me-PhpPACz films.

Sample	Atomic (%)				ratio	
	C 1s	N 1s	P 2p	In 3d	N/In	P/In
Me-4PACz	47.0	2.2	0.2	19.0	0.116	0.011
Me-PhmPACz	44.7	2.1	0.2	19.1	0.110	0.010
Me-PhpPACz	58.6	2.9	0.4	13.9	0.208	0.029

Table S3. The fitted peak areas of the O 1s XPS spectra for ITO, powders and films of Me-4PACz and Me-PhpPACz, including the following components: H₂O, P-O-H, P-O-In, P=O, V_O, and In/Sn-O.

Sample	H ₂ O	P-O-H	P-O-In	P=O	V _O	In/Sn-O
Me-4PACz powder	/	12819 (80%)	/	3186 (20%)	/	/
Me-4PACz film	2162 (6%)	/	5572 (14%)	/	10360 (27%)	19507+1028 (53%)
Me-PhpPACz powder	/	12117 (73%)	/	4510 (27%)	/	/
Me-PhpPACz film	1220 (3%)	/	5395 (14%)	/	7473 (19%)	23688+1012 (64%)
ITO	/	/	/	/	15863 (33%)	31248+798 (67%)

Table S4. The area and ratio of the fine XPS spectra for the P, N, and In elements on ITO for Me-4PACz, Me-PhmPACz, and Me-PhpPACz.

Sample	Area of N 1s	Area of P 2p	Area of In 3d	Ratio of N/In	Ratio of P/In
Me-4PACz	2394	2452	143675	0.0167	0.0171
Me-PhmPACz	2575	2588	139547	0.0184	0.0185
Me-PhpPACz	3086	2462	117506	0.0263	0.0210

1 **Table S5.** The energy level parameters of ITO, SAMs, and perovskites obtained from UPS.

Sample	$E_{\text{cut-off}}$ (eV)	E_{onset} (eV)	W_F (eV)	HOMO (eV)	gap (eV)	LUMO (eV)
ITO	17.00	/	4.20	/	/	/
ITO/PVK	17.10	1.53	4.12	5.65	1.55	4.10
ITO/Me-4PACz	16.67	0.79	4.55	5.34	3.39	1.95
Me-4PACz/PVK	16.99	1.41	4.23	5.64	1.55	4.09
ITO/Me-PhpPACz	16.57	0.78	4.65	5.43	3.45	1.98
Me-PhpPACz/PVK	16.58	0.91	4.64	5.55	1.55	4.00

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1 **Table S6.** Summary of the parameters used to fit the TRPL measurement data of Me-
 2 4PACz/PVK and Me-PhpPACz/PVK.

Sample	τ_1 (ns)	A ₁	τ_2 (ns)	A ₂	τ_{avg} (ns)
Me-4PACz	27.14	95.36%	609.41	4.64%	331.15
Me-PhpPACz	15.65	99.86%	1139.24	0.14%	82.96

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1 **Table S7.** Photovoltaic parameters for champion devices based on Me-4PACz, Me-PhpPACz,
 2 Me-PhmPACz.

Sample	Scan Direction	V_{oc} (V)	J_{sc} (mA/cm ²)	FF (%)	PCE (%)	HI (%)
Me-4PACz	Forward	1.172	25.25	80.72	23.89	1.04
	Reverse	1.177	25.23	81.30	24.14	
Me-PhpPACz	Forward	1.185	25.39	86.16	25.92	0.96
	Reverse	1.189	25.37	86.79	26.17	
Me-PhmPACz	Forward	1.180	25.35	84.59	25.30	0.82
	Reverse	1.185	25.32	85.00	25.51	

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Supplemental Experimental Procedures

Materials

General

The chemical reagents and solvents used for SAM molecules synthesis are all of reagent grade. 9H-carbazole, 3,6-dimethyl-9H-carbazole, 3,6-dimethoxy-9H-carbazole, 1-bromo-4-iodobenzene, 1-bromo-3-iodobenzene, copper(I) iodide, cyclohexane-1,2-diamine, potassium phosphate, palladium(II) acetate, 1,1-bis(diphenylphosphino)ferrocene (dppf), and potassium acetate were purchased from Bide Pharmatech LTD. Diethyl phosphite, triethylamine, and bromotrimethylsilane (TMSBr) were purchased from Energy Chemical Co. Ltd. Solvents and deuterium reagents were purchased from Adamas-beta Ltd. For the materials used in perovskite devices, unless otherwise specified, all materials are used as received. The materials used in the experiments included: (4-(3,6-Dimethoxy-9H-carbazol-9-yl)butyl)phosphonic acid (MeO-4PACz, 98%, Luminescence Technology), (4-(3,6-Dimethyl-9H-carbazol-9-yl)butyl)phosphonic acid (Me-4PACz, 98%, Luminescence Technology), (4-(9H-carbazol-9-yl)butyl)phosphonic acid (4PACz, 98%, Luminescence Technology), formamidine bromide (FABr; 99.99%, Greatcell Solar Materials Pty. Ltd.), formamidine iodide (FAI; 99.99%, Greatcell Solar Materials Pty. Ltd.), methylammonium iodide (MAI, 99.99%, Greatcell Solar Materials Pty. Ltd.), methylammonium chloride (MACl, 99.99%, Greatcell Solar Materials Pty. Ltd.), lead iodide (PbI_2 ; 99.99%, TCI), lead(II) bromide (PbBr_2 ; 99.999%, Xi'an Polymer Light Technology Corp.), cesium iodide (CsI; 99.999%, Alfa Aesar), C_{60} (99.99%, Xi'an Polymer Light Technology Corp.), bathocuproine (BCP, 99.99%, Xi'an Polymer Light Technology Corp.), piperazinium iodide (PI, 99.99%, Xi'an Polymer Light Technology Corp.), 2-phenylethanamine iodide (PEAI, 99.99%, Xi'an Polymer Light Technology Corp.), nano aluminium oxide (Al_2O_3 , 99.99%, Macklin), molybdenum trioxide (MoO_3 , 99.9995%, Alfa Aesar), water (H_2O ; Alfa Aesar), nickel(II) acetylacetonate ($\text{Ni}(\text{acac})_2$, 95%, Sigma-Aldrich), ammonium hydroxide (NH_4OH , 28.0–30.0% NH_3 basis, Sigma-Aldrich), *N,N*-dimethylformamide (DMF; 99.8%, Sigma-Aldrich), dimethyl sulfoxide (DMSO; 99.9%, Sigma-Aldrich), ethanol (99.5%; Sigma-Aldrich), isopropanol (IPA; 99.5%; Sigma-Aldrich), ethyl acetate (EA, 99.5%; Sigma-Aldrich), chlorobenzene (CB; 99.8%; Sigma-Aldrich), acetone (99.8%; Merck).

SAM molecules synthesis

The detailed synthetic methods of the SAM molecules, except commercialized ones used in this work, are depicted in Figure S1.

General procedure for synthesis of Cz-PhpBr:

To an oven-dried 250 mL two-neck round-bottom flask was added 9H-carbazole derivatives (10 mmol), 1-bromo-4-iodobenzene (5.66 g, 20 mmol, 2 eq.), copper(I) iodide (0.190 g, 1 mmol, 0.1 eq.), 1,2-diaminocyclohexane (DACH, 0.228 g, 2 mmol, 0.2 eq.), and potassium phosphate (6.37 g, 30 mmol, 3 eq.). Upon filling with argon, 50 mL toluene was added to the flask, and the mixture was heated to reflux for 12 h. After cooling, the mixture was extracted with dichloromethane, dried over anhydrous sodium sulfate, and concentrated under reduced pressure. The crude product was then purified by column chromatography on silica gel (eluent: dichloromethane/petroleum ether) and recrystallized from dichloromethane/hexane to give white solids as pure products.

General procedure for synthesis of MeCz-PhmBr:

Following the same procedure for synthesis of Cz-PhpBr using 1-bromo-3-iodobenzene instead

of 1-bromo-4-iodobenzene.

Cz-PhpBr: 2.67 g, 82.9%. ¹H NMR (400 MHz, CDCl₃) δ/ppm 8.16 (d, *J* = 7.7 Hz, 2H), 7.78 – 7.69 (m, 2H), 7.49 – 7.41 (m, 4H), 7.39 (dt, *J* = 8.1, 1.2 Hz, 2H), 7.32 (ddd, *J* = 8.0, 6.8, 1.4 Hz, 2H). ¹³C NMR (101 MHz, CDCl₃) δ/ppm 140.74, 136.94, 133.25, 128.85, 126.22, 123.62, 121.02, 120.53, 120.35, 109.68 (Figure S2 and S3).

MeCz-PhpBr: 3.32 g, 95%. ¹H NMR (400 MHz, CDCl₃) δ/ppm 7.93 – 7.88 (m, 2H), 7.74 – 7.68 (m, 2H), 7.47 – 7.41 (m, 2H), 7.28 (d, *J* = 8.3 Hz, 2H), 7.22 (dd, *J* = 8.5, 1.7 Hz, 2H), 2.55 (s, 6H). ¹³C NMR (101 MHz, CDCl₃) δ/ppm 139.15, 137.37, 133.11, 129.54, 128.51, 127.33, 123.66, 120.45, 120.39, 109.32, 21.52 (Figure S4 and S5).

MeCz-PhmBr: 3.47 g, 99%. ¹H NMR (400 MHz, CDCl₃) δ/ppm 7.88 – 7.85 (m, 2H), 7.70 (t, *J* = 1.9 Hz, 1H), 7.55 – 7.49 (m, 1H), 7.46 (ddd, *J* = 8.0, 2.0, 1.3 Hz, 1H), 7.40 (t, *J* = 7.9 Hz, 1H), 7.28 (d, *J* = 8.3 Hz, 2H), 7.19 (dd, *J* = 8.2, 1.7 Hz, 2H), 2.52 (s, 6H). ¹³C NMR (101 MHz, CDCl₃) δ/ppm 139.66, 139.04, 131.13, 130.11, 129.85, 129.65, 127.35, 125.45, 123.69, 123.23, 120.38, 109.38, 21.52 (Figure S6 and S7).

MeOCz-PhpBr: 3.55 g, 92.9%. ¹H NMR (400 MHz, CDCl₃) δ/ppm 7.75 – 7.65 (m, 2H), 7.55 (d, *J* = 2.5 Hz, 2H), 7.45 – 7.38 (m, 2H), 7.30 (d, *J* = 8.9 Hz, 2H), 7.04 (dd, *J* = 8.9, 2.5 Hz, 2H), 3.95 (s, 6H). ¹³C NMR (101 MHz, CDCl₃) δ/ppm 154.32, 137.39, 136.07, 133.13, 128.32, 123.92, 120.32, 115.40, 110.62, 103.09, 56.23 (Figure S8 and S9).

General procedure for synthesis of Cz-PhPO(OEt)₂:

This synthesis was based on the literature method.^{S1} An oven-dried 100 mL two-neck round-bottom flask was charged with palladium acetate (16.8 mg, 0.075 mmol, 0.025 eq.), 1,1'-bis(diphenylphosphino)ferrocene (83.2 mg, 0.15 mmol, 0.05 eq.), potassium acetate (29.4 mg, 0.3 mmol, 0.1 eq.), and aryl bromides (3 mmol, 1 eq.). Degassed THF (30 mL) and triethylamine (364 mg, 0.5 mL, 3.60 mmol, 1.2 eq.) were injected by a syringe, and the mixture was heated to reflux for 15 min. Diethyl phosphite (414 mg, 0.39 mL, 3 mmol, 1 eq.) was then added and the mixture was allowed to reflux for 16 h. After cooling down to room temperature, the mixture was filtered through a celite-pad and washed with THF (3 × 5 mL) to remove the triethylamine hydrobromide. The filtrate was concentrated under reduced pressure and the crude product was purified by column chromatography on silica gel (eluent: dichloromethane/ethyl acetate). Recrystallization in dichloromethane/hexane gave white solids as products.

Cz-PhpPO(OEt)₂: 0.962 g, 84.5%. ¹H NMR (400 MHz, CDCl₃) δ/ppm 8.15 (dt, *J* = 7.8, 1.0 Hz, 2H), 8.10 – 8.01 (m, 2H), 7.76 – 7.68 (m, 2H), 7.48 (dt, *J* = 8.2, 1.0 Hz, 2H), 7.43 (ddd, *J* = 8.2, 6.9, 1.2 Hz, 2H), 7.32 (ddd, *J* = 7.9, 6.9, 1.2 Hz, 2H), 4.33 – 4.13 (m, 4H), 1.41 (t, *J* = 7.1 Hz, 6H). ¹³C NMR (101 MHz, CDCl₃) δ/ppm 140.34, 133.72, 133.61, 126.77, 126.61, 126.30, 123.89, 120.68, 120.58, 109.83, 62.54, 62.49, 16.61, 16.54. ³¹P NMR (162 MHz, CDCl₃) δ/ppm 17.85 (Figure S10-S12).

MeCz-PhpPO(OEt)₂: 1.15 g, 94%. ¹H NMR (400 MHz, CDCl₃) δ/ppm 8.07 – 7.98 (m, 2H), 7.89 (d, *J* = 1.8 Hz, 2H), 7.74 – 7.65 (m, 2H), 7.37 (d, *J* = 8.4 Hz, 2H), 7.22 (dd, *J* = 8.5, 1.8 Hz, 2H), 4.35 – 4.09 (m, 4H), 2.54 (s, 6H), 1.40 (t, *J* = 7.1 Hz, 6H). ¹³C NMR (101 MHz, CDCl₃) δ/ppm 142.21, 138.73, 133.63, 133.53, 129.94, 127.51, 127.40, 126.33, 126.17, 125.60, 123.98, 120.43, 109.51, 62.49, 62.43, 21.51, 16.59, 16.53. ³¹P NMR (162 MHz, CDCl₃) δ/ppm 18.07 (Figure S13-S15).

MeCz-PhmPO(OEt)₂: 0.88 g, 72%. ¹H NMR (400 MHz, CDCl₃) δ/ppm 8.00 (dt, *J* = 14.1, 1.8 Hz, 1H), 7.93 – 7.84 (m, 3H), 7.75 (ddt, *J* = 8.0, 2.4, 1.3 Hz, 1H), 7.69 (td, *J* = 7.6, 4.7 Hz, 1H),

1 7.29 (d, $J = 8.3$ Hz, 2H), 7.22 (dd, $J = 8.5, 1.7$ Hz, 2H), 4.28 – 4.14 (m, 4H), 2.55 (s, 6H), 1.37
2 (t, $J = 7.0$ Hz, 6H). ^{13}C NMR (101 MHz, CDCl_3) δ /ppm 139.06, 138.70, 138.52, 131.68, 130.67,
3 130.63, 130.33, 130.24, 130.16, 129.76, 129.65, 127.34, 123.71, 120.39, 109.26, 62.56, 62.51,
4 21.50, 16.54, 16.47. ^{31}P NMR (162 MHz, CDCl_3) δ /ppm 17.25 (Figure S16-S18).

5 **MeOCz-PhpPO(OEt)₂**: 1.04 g, 79.1%. ^1H NMR (400 MHz, CDCl_3) δ /ppm 8.02 (dd, $J = 12.9,$
6 8.3 Hz, 2H), 7.68 (dd, $J = 8.3, 3.5$ Hz, 2H), 7.54 (d, $J = 2.4$ Hz, 2H), 7.40 (d, $J = 8.9$ Hz, 2H),
7 7.04 (dd, $J = 8.9, 2.5$ Hz, 2H), 4.34 – 4.05 (m, 4H), 3.95 (s, 6H), 1.39 (t, $J = 7.1$ Hz, 6H). ^{13}C
8 NMR (101 MHz, CDCl_3) δ /ppm 154.56, 142.23, 142.19, 135.61, 133.67, 133.56, 127.37, 126.10,
9 125.95, 125.47, 124.32, 115.41, 110.83, 103.13, 62.49, 62.43, 56.19, 16.58, 16.51. ^{31}P NMR
10 (162 MHz, CDCl_3) δ /ppm 18.03 (Figure S19-S21).

11 **General procedure for synthesis of Cz-PhPA:**

12 In an oven-dried 50 mL two-neck round-bottom flask, Cz-PhPO(OEt)₂ (1 mmol, 1 eq.) was
13 dissolved in 20 mL dichloromethane. Bromotrimethylsilane (TMSBr, 0.612 g, 0.528 mL, 4 mmol,
14 4 eq.) was dropwise added to the well-stirred mixture at room temperature. After 12 h, the
15 mixture was carefully transferred to a 100 mL round-bottom flask and concentrated under
16 reduced pressure. Then a magnetic stir bar and 20 mL methanol were added in sequence. The
17 mixture was allowed to be stirred at room temperature for 4 h. Filtrating the mixture, rinsing with
18 methanol (3 $\times$ 5 mL), and concentrating the combined filtrate under reduced pressure gave
19 solids as crude products. Pure products were obtained from recrystallization in
20 methanol/dichloromethane/diethyl ether.

21 **PhpPACz**: white solid (0.310 g, 95.8%). ^1H NMR (400 MHz, $\text{DMSO}-d_6$) δ /ppm 8.26 (d, $J = 7.7$
22 Hz, 2H), 7.97 (dd, $J = 12.6, 8.3$ Hz, 2H), 7.75 (dd, $J = 8.3, 2.9$ Hz, 2H), 7.48 – 7.41 (m, 4H),
23 7.31 (ddd, $J = 7.9, 5.1, 3.0$ Hz, 2H). ^{13}C NMR (101 MHz, $\text{DMSO}-d_6$) δ /ppm 139.74, 132.57,
24 132.47, 126.43, 126.14, 126.00, 122.98, 120.63, 120.42, 109.72. ^{31}P NMR (162 MHz, $\text{DMSO}-$
25 d_6) δ /ppm 11.79. HRMS-ESI (m/z): $[\text{M}-\text{H}]^-$ calcd. for $\text{C}_{18}\text{H}_{14}\text{NO}_3\text{P}$ 322.0639, found 322.0632
26 (Figure S22-S25).

27 **Me-PhpPACz**: white solid (0.350 g, 99.5%). ^1H NMR (400 MHz, $\text{DMSO}-d_6$) δ /ppm 8.05 – 7.90
28 (m, 4H), 7.70 (dd, $J = 8.4, 2.9$ Hz, 2H), 7.35 (d, $J = 8.4$ Hz, 2H), 7.23 (dd, $J = 8.5, 1.3$ Hz, 2H),
29 2.48 (s, 6H). ^{13}C NMR (101 MHz, $\text{DMSO}-d_6$) δ /ppm 139.57, 139.54, 138.16, 133.53, 132.53,
30 132.42, 131.72, 129.17, 127.49, 125.67, 125.52, 123.08, 120.31, 109.47, 21.03. ^{31}P NMR (162
31 MHz, $\text{DMSO}-d_6$) δ /ppm 11.96. HRMS-ESI (m/z): $[\text{M}-\text{H}]^-$ calcd. for $\text{C}_{20}\text{H}_{18}\text{NO}_3\text{P}$ 350.0951, found
32 350.0945 (Figure S26-S29).

33 **Me-PhmPACz**: white solid (0.308 g, 87.6%). ^1H NMR (400 MHz, $\text{DMSO}-d_6$) δ /ppm 8.04 – 7.96
34 (m, 2H), 7.86 – 7.70 (m, 4H), 7.28 (d, $J = 8.3$ Hz, 2H), 7.23 (dd, $J = 8.4, 1.7$ Hz, 2H), 2.47 (s,
35 6H). ^{13}C NMR (101 MHz, $\text{DMSO}-d_6$) δ /ppm 138.40, 137.32, 137.24, 137.07, 135.54, 130.43,
36 130.28, 129.12, 129.08, 128.70, 127.89, 127.78, 127.50, 123.01, 120.42, 109.23, 21.09. ^{31}P
37 NMR (162 MHz, $\text{DMSO}-d_6$) δ /ppm 11.20. HRMS-ESI (m/z): $[\text{M}-\text{H}]^-$ calcd. for $\text{C}_{20}\text{H}_{18}\text{NO}_3\text{P}$
38 350.0951, found 350.0951 (Figure S30-S33).

39 **MeO-PhpPACz**: white solid (0.310 g, 80.9%). ^1H NMR (400 MHz, $\text{DMSO}-d_6$) δ /ppm 7.93 (dd, J
40 $= 12.5, 8.1$ Hz, 2H), 7.83 (d, $J = 2.1$ Hz, 2H), 7.70 (dd, $J = 8.2, 2.4$ Hz, 2H), 7.38 (d, $J = 8.9$ Hz,
41 2H), 7.04 (dd, $J = 8.9, 2.0$ Hz, 2H), 3.88 (s, 6H). ^{13}C NMR (101 MHz, $\text{DMSO}-d_6$) δ /ppm 154.01,
42 139.75, 139.71, 134.86, 133.23, 132.52, 132.41, 131.41, 125.40, 125.25, 123.70, 115.37,
43 110.65, 103.45, 55.70. ^{31}P NMR (162 MHz, $\text{DMSO}-d_6$) δ /ppm 11.99. HRMS-ESI (m/z): $[\text{M}-\text{H}]^-$
44 calcd. for $\text{C}_{20}\text{H}_{18}\text{NO}_5\text{P}$ 382.0849, found 382.0845 (Figure S34-S37).

Materials and thin films characterization

^1H NMR, ^{13}C NMR, and ^{31}P NMR spectra were recorded on Bruker AVANCE III 400M. HR-ESI: High-resolution electrospray ionization (HR-ESI) mass spectra were obtained from Thermo Scientific Q Exactive mass spectrometer, operated in heated electrospray ionization (HESI) mode, and coupled with Thermo Scientific Ultimate 3000 system. The thermogravimetric measurements were performed on SDT Q600 instrument at a heating rate of 10 K min^{-1} in the range of $25\text{ }^\circ\text{C}$ to $800\text{ }^\circ\text{C}$ under a nitrogen atmosphere. Cyclic voltammetry and differential pulse voltammetry were performed on CH Instruments Electrochemical Analyzer CHI760E using a three-compartment electrochemical cell, with samples in DMF containing $0.1\text{ M }[\text{nBu}_4\text{N}]\text{PF}_6$ as electrolyte and the ferrocene/ferrocenium (Fc/Fc^+) as internal standard under scan rate of 0.1 V/s . Ag/AgCl, glassy carbon, and platinum wire were used as the reference electrode, working electrode, and counter electrode, respectively. Fourier-transform infrared spectra (FTIR) were measured using a Bruker Vertex 70v spectrometer. The UV-vis spectra were recorded using a PerkinElmer Lambda750S spectrophotometer. The morphologies of the prepared samples were analyzed via SEM (SEM; Zeiss Merlin) and high-resolution transmission electron microscopy (HRTEM, JEM-2100Plus, JEOL) imaging. The contact angle measurements were performed using a drop shape analyzer (DSA 25S, Kruss). Atomic force microscopy (AFM), conductive atomic force microscopy (c-AFM), and Kelvin probe force microscopy (KPFM) measurements were conducted using an MFP-3D Stand Alone (Asylum Research) Atomic Force Microscope equipped with Q-control technology. For c-AFM and KPFM measurements, conductive probes coated with Pt/Ir were utilized.

Ultraviolet photo-electron spectroscopy (UPS) and X-ray photoelectron spectroscopy (XPS) measurements were performed using an ultrahigh vacuum surface analysis system equipped with an ULVAC-PHI 5000 VersaProbe III spectrometer. UPS was performed with He I (21.22 eV) acting as the excitation source at an energy resolution of 50 meV . XPS and XPS mapping used monochromatic Al K_α radiation (1486.6 eV). All experiments were calibrated with respect to the Fermi edge of an atomically clean gold surface.

X-ray diffraction (XRD) measurements of the perovskite were performed on a Smartlab 9 kW diffractometer with a Göbel mirror attachment. The irradiation by the parallel $\text{CuK}_{\alpha 1,2}$ X-ray beams was fixed at a grazing angle of incidence (θ) of 0.50° . The detector was moved independently to collect diffraction data within the 2θ range (5° – 60°) with a step size of 0.03° (2θ) at a fixed speed of 0.5 s/step . The X-ray reflectivity (XRR) measurements of the SAMs were conducted using a Smartlab 9 kW diffractometer. X-ray was incident on the SAM's surface at a grazing angle of 0.05° , and data were collected within the 2θ range of 0° to 2° with a step size of 0.01° (2θ).

The PL measurement of the perovskite thin film (ITO/SAMs/perovskite) was conducted on the FLS1000 spectrometer (Edinburgh Instruments) with excitation using a 405 nm xenon lamp. The TRPL and PLQY measurements for the perovskite films (ITO/SAMs/perovskite for TRPL and PLQY, ITO/SAMs/perovskite/ C_{60} for PLQY) were performed on an FLS1000 spectrometer (Edinburgh Instruments) with excitation provided by a 475 nm pulsed laser. The samples were excited from the ITO side under ambient conditions. PL mapping was acquired using a laser scanning confocal Raman spectrometer (LabRAM HR Evolution) with excitation at 532 nm on the perovskite thin film.

The ultrafast spectroscopic experiments involve splitting the fundamental beam (800 nm, pulse duration ~60 fs, ~5 mJ/pulse, 1 kHz repetition rate, Coherent Astrella) into two parts. One beam is sent to an optical parametric amplifier (TOPAS, Light conversion) to generate the pump pulse. The pump is chopped at a frequency of 500 Hz and its intensity is attenuated by two neutral density filter wheels. The other 800 nm beam is focused into a sapphire crystal to generate white light probe (450-810 nm) and detected with a CMOS detector (Timetech, TA-100). The probe pulses are delayed in time with respect to the pump pulses using a motorized translation stage mounted with a retroreflecting mirror. Probe delay can be up to ~5 ns, tuned by a delay line. The pump and probe are focused and overlapped on the surface of the sample spatially. For transient reflection (TR) measurement, the incident angle for pump is around 0° and probe is around 45°.

Device fabrication

Preparation of perovskite precursor solutions

For perovskite solutions with small areas PSCs and NiO_x-based PSCs, the perovskite precursor solution comprised 1.6 mM Cs_{0.05}MA_{0.1}FA_{0.85}PbI₃ (MA: methyl-ammonium; FA: formamidinium), specifically, 20.78 mg CsI, 233.88 mg FAI, 774.50 mg PbI₂ (with 5% mol excess), 25.44 mg MAI, and 16.20 mg MACl in 1 mL of DMF/DMSO (4:1, v/v).

For perovskite solutions with wide-bandgap PSCs, the perovskite precursor solution comprised 1.4 mM Cs_{0.2}FA_{0.8}PbI_{1.8}Br_{1.2}, specifically, 72.75 mg CsI, 96.30 mg FAI, 69.98 mg FABr, 378.25 mg PbI₂, and 205.53 mg PbBr₂ in 1 mL of DMF/DMSO (3:1, v/v).

Small areas PSCs and NiO_x-based PSCs fabrication

Planar p-i-n PSCs were fabricated using indium tin oxide (ITO) or fluorine doped tin oxide (FTO), SAM, perovskite, C₆₀, BCP, and copper (Cu) as the anode, HSL, light absorber layer, electron transport layer (ETL) and cathode, respectively. ITO glasses (7–10 Ω, Yingkou OPV Tech New Energy Co., Ltd, China) were washed with cleaning fluid, deionized water, isopropanol, acetone, ethanol, and sequentially. The ITO was then treated with UV-ozone for 20 min.

For devices using Me-4PACz, 4PACz, PhpPACz, and Me-PhpPACz as the HSLs, a 0.5 mg/mL ethanol solution was spun-coated statically at 5000 rpm for 30 s onto ITO substrates. The substrates were then annealed at 150 °C for 10 min, followed by rinsing off the uncoordinated SAM after annealing using a 0.5 mg/mL solution of Al₂O₃ in ethyl acetate.

For devices employing MeO-4PACz and MeO-PhpPACz as the HSLs, a 0.5 mg/mL ethanol solution was spun-coated statically at 5000 rpm for 30 s onto ITO substrates. The substrates were annealed at 150 °C for 10 min and then rinsed off to remove the uncoordinated SAM using an ethyl acetate solution.

For devices with NiO_x-based SAM as the HSL, NiO_x nanoparticles were synthesized following previous work.^{S2} Subsequently, a 20 mg/mL NiO_x aqueous solution was spun-coated statically at 3000 rpm for 30 s onto ITO substrates and annealed at 140 °C for 30 min. A 0.5 mg/mL ethanol solution of Me-4PACz and Me-PhpPACz was spun-coated statically at 5000 rpm for 30 s onto ITO substrates and annealed at 150 °C for 10 min. The annealed SAM layer was rinsed off to remove uncoordinated SAM using a chlorobenzene solution. The perovskite film was deposited on the HSLs by a solvent engineering method. Spread perovskite solution over HSL and spin-coated in two steps, namely, 1000 rpm for 10 s and 5400 rpm for 30 s. Then, with 8–9 s of spin time remaining, chlorobenzene (200 μL for ITO area of 15 mm × 15 mm) was slowly dispensed onto the middle of the substrate and the films were annealed at 100 °C for 0.5 h. For

the passivation treatment, the IPA solution of PI (0.1 mM) was spin-coated on the perovskite surface at 5,000 rpm for 30 s, followed by annealing at 100 °C for 5 min. The thickness of the as-fabricated perovskite layer was determined to be ~700 nm by cross-sectional SEM images. C₆₀ (40 nm), BCP (8 nm) and a metallic copper (100 nm) electrode were thermally evaporated in a vacuum chamber at a pressure set at $<4 \times 10^{-5}$ Pa through a shadow mask. Finally, the anti-reflection film (Mitsubishi) is attached to the glass surface.

Wide-bandgap PSCs fabrication

The device structure is the same as described above. For the wide-bandgap perovskite devices, a SAM system consisting of MeO-4PACz or MeO-PhPACz was used as the HSLs. A 0.5 mg/mL ethanol SAM mixed solution was spun-coated statically at 5000 rpm for 30 s onto ITO substrates and annealed at 150 °C for 10 min. The annealed SAM layer was rinsed off to remove uncoordinated SAM using the ethyl acetate. The perovskite film was deposited on the HSLs by a solvent engineering method. Spread perovskite solution over SAM and spin-coated in two steps, namely, 1000 rpm for 5 s and 4000 rpm for 40 s. Then, with 8–9 s of spin time remaining, chlorobenzene (180 μ L for ITO area of 15 mm $\times$ 15 mm) was slowly dispensed onto the middle of the substrate and the films were annealed at 100 °C for 0.5 h. For the passivation treatment, the IPA solution of PI and PEAI (0.1 mM) was spin-coated on the perovskite surface at 5,000 rpm for 30 s, followed by annealing at 100 °C for 5 min. C₆₀ (40 nm), BCP (8 nm) and a metallic copper (100 nm) electrode were thermally evaporated in a vacuum chamber at a pressure set at $<4 \times 10^{-5}$ Pa through a shadow mask. Finally, the anti-reflection film (Mitsubishi) is attached to the glass surface.

Device characterization

A Keithley 2400 SourceMeter measurement unit was used along with an Oriel xenon lamp (450 W) with an AM 1.5 filter to characterize the solar cells in air under AM 1.5G illumination at 100 mW/cm² (Newport Oriel 1 kW solar simulator) after calibration with a KG5 filter certified by the National Renewable Energy Laboratory. The *J*–*V* curves for all devices (active area: 0.0397 cm²) were measured at a scanning speed of 0.01 V/step with dwell time of 100 ms under forward and reverse direction. The stabilized power output was measured at the maximum power point (MPP) voltage. The incident-photon-to-current-efficiency (IPCE) spectra were recorded using a constant 100 W xenon lamp source with an automated monochromatic filter and a rectangular spot size of 0.76 mm $\times$ 1.0 mm. The IPCE measurements were conducted within the wavelength range from 300–900 nm using a chopping frequency of 4 Hz (IQE-200B model). The light intensity dependences of the *J*_{sc} and *V*_{oc} characteristics were measured using a 10–97.5% light reduction film to adjust the light intensity of the solar simulator in air under AM 1.5G illumination.

Electrical impedance spectroscopy (EIS) measurements were conducted using an electrochemical workstation (IM6ex, Zahner) with a frequency range from 10 Hz to 4 MHz under an applied voltage of 0.4 V in the dark. For the Mott-Schottky measurements, the capacitance-voltage measurements were conducted at a fixed frequency (10 kHz) using an IM6ex (Zahner) instrument. The Mott-Schottky plot was obtained in the dark, ranging from 0 V to 1.4 V in the positive direction, with an active area of 0.1 cm².

Hole-only devices (ITO/SAMs/perovskite/MoO₃/Au) were fabricated according to the PSC fabrication procedure. MoO₃ (8 nm) and Au (80 nm) were thermally evaporated in a vacuum chamber at a pressure set below 4×10^{-5} Pa. The dark current density-voltage (*J*–*V*)

characteristics of the hole-only devices were measured using a Keithley 2400 SourceMeter. For electroluminescence testing, a Keithley 2400 SourceMeter was employed in conjunction with the XPQY-EQE-200-1100 (Xi Pu Optoelectronics Technology Co., Ltd.) connected to an integrating sphere. TPV and TPC were measured by SouthPort SP-TPVC with a wavelength of 337 nm. Operational stability tests were performed at the MPP for the unencapsulated devices under AM 1.5G illumination (100 mW/cm²) at 50 ± 10 °C in a N₂ atmosphere within a light-emitting diode (LED)-based solar simulator. The bias at the MPP was calculated and then applied automatically. For thermal stability tests, every set of 6 encapsulated devices was exposed to an 85 °C air atmosphere (RH 75 ± 15%) and tested at 72-hour intervals. The light intensity was calibrated versus a standard silicon reference cell supplied by Newport.

Computation

The spin-polarized DFT calculations were performed using the Vienna Ab Initio Simulation Package (VASP) code,^{S3,S4} employing periodic boundary conditions. The projector augmented wave (PAW) pseudopotential was used to describe the ion-core electron interaction,^{S5,S6} while the exchange-correlation energy of valence electrons was described by the Perdew-Burke-Ernzerhof (PBE) functional with a plane-wave cutoff energy of 400 eV.^{S7} Grimme's D3 corrections were also used to describe the dispersion interaction.^{S8} The energy convergence criteria of structure optimization (including lattice parameters and internal atomic positions) were set to 10⁻⁶ eV for each self-consistent field (SCF) iteration step, and the maximum force convergence criterion for geometry optimization was set to 0.02 eV/Å.

The ITO model was constructed using bulk In₂O₃ doped with Sn atom, with the (111) surface selected as the most stable surface for ITO model. The 1 × 1 supercell of cubic In₂O₃ (111) surface is organized with the slab consisting of the fourth O–In–O layers, while the bottom layer is kept frozen. For the ITO surface, our simulation indicates the introduction of one oxygen vacancy (O_v) on the surface following previous literature.^{S9} A vacuum of 20.0 Å was inserted along the [001] direction to avoid unphysical interactions between the slabs. The size of the ITO model is 14.57 × 14.57 × 31.01 Å³. For ITO model, the mean-field DFT+U (with 7 eV for U parameters) method was used to reasonably treat strong electron correlation effect associated with In-4*d* states.^{S10,S11} In this model, the Gamma point k-point sampling was used for the Brillouin-zone integration. The size of the upper three-layer perovskite model is 19.47 × 26.01 × 42.26 Å³, with a vacuum of 20.0 Å inserted along the [001] direction to avoid any unphysical interactions between the slab layers.

The starting point for the ab initio molecular dynamics (AIMD) simulations was the DFT-optimized structure. These AIMD simulations were conducted to investigate the dynamic behavior of SAM molecules on the ITO surface. The simulations were performed in a canonical (NVT) ensemble using Nose-Hoover thermostats with a time step of 1.0 fs at a typical temperature of 300 K. The DFT-optimized structures served as the initial configurations, and the system then underwent approximately 10 ps of equilibration to obtain the most stable configuration of SAM molecules on ITO surface. The angular evolution was then calculated to analyze the inversion of SAM configuration on ITO surface, thus enabling investigation of the dynamic self-assembly behavior of SAM molecules on the ITO and perovskite surface. The charge difference between the SAM molecules and the perovskite is represented as $\Delta\rho = \rho_{\text{SAM/perovskite}} - \rho_{\text{SAM}} - \rho_{\text{Perovskite}}$.

Supplemental Notes

Note S1: FTIR spectroscopy of Me-4PACz and Me-PhpPACz powders and films.

FTIR spectra of Me-4PACz and Me-PhpPACz powders and films on a diamond attenuated total reflection (ATR) are shown in Figure S43. In comparison to the ATR spectra of SAM powder, Me-4PACz film sample exhibits the disappearance of peaks at 1108 and 943 cm^{-1} , while Me-PhpPACz film sample also shows the vanishing of peaks at 1013 and 931 cm^{-1} . These peaks are attributed to the antisymmetric $\nu_a(\text{P-O-H})$ and symmetric $\nu_s(\text{P-O-H})$ stretching vibrations,^{S12} thus explaining the intensity loss due to deprotonation of the phosphonic acid groups. Furthermore, the P=O stretching vibration $\sim 1230 \text{ cm}^{-1}$ also disappear in the SAM films compared to the powder ATR spectra.^{S13} However, the SAM films exhibited smaller stretching vibration peaks at 1013 cm^{-1} and 1014 cm^{-1} , which can be attributed to the P-O bonds formed between phosphonic acid and ITO.^{S14} The emergence of this peak, along with the disappearance of the P-O-H and P=O peaks, provides evidence of tridentate anchoring of phosphonic acid groups on ITO.

Note S2: DFT calculation between SAMs and ITO.

XPS and infrared spectroscopy data have confirmed the binding of SAM molecules to the ITO surface by tridentate coordination interactions. Based on above-mentioned experimental evidence, the chemical adsorption of phosphonic acid on the ITO surface was studied by using DFT calculations. We chose the (111) plane to build the ITO crystal model to investigate the adsorption as it has been proven to be the most stable surface (Figure S44). The DFT static calculations indicate that the binding energies of SAM molecules (Me-PhpPACz, Me-4PACz, and Me-PhmPACz) on the ITO surface are nearly identical, at around -1.80 eV (Figure S45). For the adsorption of SAM molecules on the ITO surface, our simulations indicate that the length of the three O-In bonds on the phosphate group fall within the range of 2.20-2.40 Å, which is comparable to the bond length of the ITO bulk material (calculated as 2.24 Å). These results indicate a strong chemical bonding between phosphonic acid and the ITO surface.

Note S3: AIMD simulations of SAMs on ITO surface: molecular orientation and molecular packing.

Ab initio molecular dynamics (AIMD) simulations were used to further investigate the dynamic behavior of SAM molecules on the ITO surfaces and to obtain stable configurations of SAM molecules on the ITO surface. When examining the SAM molecular structures, we observed Me-4PACz exhibited a more flexible structure and tended to undergo distortion on the ITO surface when compared with Me-PhpPACz. The angular evolution of SAM molecules on the ITO surface during a 10 ps AIMD simulation at a typical temperature of 300 K is depicted in manuscript Figure 2D. We define the angle between the carbazole group and the Z-axis as the orientation of SAM molecules relative to the ITO surface. The results indicate that after 7 ps, SAM molecules reach a stable configuration on the ITO surface. Due to the presence of the phenylene ring, Me-PhpPACz remains rigid-assembled on ITO surface maintaining an angle of approximately 20° relative to the ITO surface. In contrast, Me-4PACz maintains a twisted-assembled on ITO surface, stabilizing at around 90°, while Me-PhmPACz stabilizes at around 110° (Figure S46, S47, S50, and S51). At this time, the dipole moment of the stable SAM molecule on ITO is 3.62 D for Me-PhpPACz and 1.36 D for Me-4PACz. The detailed 3D configuration changes of SAM molecules (Me-PhpPACz, Me-4PACz, and Me-PhmPACz) on

ITO surface during AIMD simulations are shown in Figure S46. By comparing the theoretically calculated molecular lengths (the effective distances between the anchoring group and terminal group of SAM molecules) obtained from AIMD simulations in Figure S47A, which are 10.58 ± 0.23 Å for Me-PhpPACz, 4.00 ± 0.14 Å for Me-4PACz, and 5.65 ± 0.31 Å for Me-PhmPACz, respectively, it can be inferred that Me-PhpPACz can arrange more densely on the ITO surface, thus forming an approximate monolayer film.

To further understand the molecular arrangement and orientation of SAMs on ITO, we conducted theoretical simulations of multiple SAM molecules' dynamic behavior on the ITO surface. By placing multiple SAM molecules (Me-PhpPACz, Me-4PACz) on a larger ITO surface, as shown in manuscript Figure 4A, AIMD simulations revealed that Me-PhpPACz maintains a tighter intermolecular arrangement compared to Me-4PACz. Additionally, Me-PhpPACz exhibited a rigid-assembled, attributed to the presence of the rigid phenylene ring structure.

Note S4: XRR analysis.

To further confirm the thickness of the SAM, X-ray reflectivity (XRR) measurements were conducted to obtain the reflectivity curves for three SAM films:^{S15} Me-4PACz, Me-PhmPACz, and Me-PhpPACz. The Kiessig fringes in XRR arise from interference between reflections at the top and bottom interfaces of the SAM, and the position of the minimum fringe corresponds to the thickness of the SAM.^{S16} For larger layer thicknesses, it shifts to smaller 2θ values. In Figure S47B, a turning point in the attenuated reflectivity curve was observed as the incident angle (2θ) increased. Upon magnification, at the turning point, the height (d) of the SAM layer was calculated using the equation:^{S17} $\theta^2 - \alpha_c^2 = m^2(\lambda/2d)^2$, where α_c (0.325°) is the critical angle for total reflection at the ITO substrate interface, m is the diffraction fringe order ($m = 1$), and λ (0.15406 nm) is the wavelength of the X-rays (Cu K α). As a result, the thickness values for Me-4PACz, Me-PhmPACz, and Me-PhpPACz were calculated to be 3.99 Å, 4.36 Å, and 10.00 Å, respectively, which align closely with the results obtained from AIMD simulations. It can be inferred that Me-4PACz and Me-PhmPACz tend to lie flat on the ITO surface, while Me-PhpPACz exhibits a preference for an upright arrangement on ITO.

Note S5: XPS fine spectra of O, In, N and P elements and XPS mapping.

Manuscript Figure 2A reveals that after the interaction between SAM and ITO, a significant number of oxygen vacancies still exist, indicating that there are ample oxygen vacancies on the ITO surface available for phosphonic acid interactions, and the phosphonic acid on SAM has reacted completely. The oxygen vacancy area ratio (area: oxygen vacancies/InSn-O, Table S2 and S3) for Me-4PACz and Me-PhpPACz films is 51% and 30%, respectively. Excess SAM materials were used in the work, and uncoordinated SAMs were washed away. Results show that the Me-4PACz film exhibits more oxygen vacancies, suggesting that fewer Me-4PACz molecules interact with ITO, and more Me-PhpPACz molecules interact with ITO. This reflects that during the spin-coating process, the molecular orientation of Me-PhpPACz molecules is superior to that of Me-4PACz, leading to a greater number of molecules distributed on the ITO surface, while the arrangement of Me-4PACz molecules exhibits steric hindrance effects affecting the stacking of molecules.

Furthermore, we quantified the content of SAM molecules on ITO through XPS full-spectrum analysis (Figure S41).^{S18} Since each SAM contains one phosphorus atom and one nitrogen atom, the amount of SAM molecules on ITO was quantified based on the content of phosphorus

or nitrogen atoms. According to the XPS full-spectrum analysis results shown in Table S2, the phosphorus content on ITO for Me-4PACz, Me-PhmPACz, and Me-PhpPACz was 0.2%, 0.2%, and 0.4%, respectively, while the nitrogen content was 2.2%, 2.1%, and 2.9%, respectively. The phosphorus-to-indium ratio (P/In) on the surfaces of Me-4PACz, Me-PhmPACz, and Me-PhpPACz films was 0.011, 0.010, and 0.029, respectively, while the nitrogen-to-indium ratio (N/In) was 0.116, 0.110, and 0.208, respectively. The fine spectral areas of nitrogen, phosphorus, and indium were further quantitatively analyzed to determine the content of SAM molecules on ITO, as shown in Figure S48. Table S4 summarizes the area ratios of N and P to In, with the fine spectral area ratios of N/In for Me-4PACz, Me-PhmPACz, and Me-PhpPACz on ITO films being 0.0167, 0.0184, and 0.0263, respectively, and P/In ratios being 0.0171, 0.0185, and 0.0210, respectively. The quantitative analysis of nitrogen and phosphorus indicates that there is a greater amount of Me-PhpPACz molecules anchored to ITO, approximately 2-3 times as that of Me-4PACz and Me-PhmPACz.

Note S6: Contact angle analysis.

The water contact angles of Me-4PACz and Me-PhpPACz are shown in Figure S49, measuring 54° and 77°, respectively. Since the head of the SAM is composed of a carbazole moiety, a larger contact angle indicates a denser film. This denser SAM can effectively prevent direct contact between perovskite (PVK) and the ITO substrate, while substantially improving the surface properties of ITO.

Note S7: AFM and KPFM analysis.

The surface morphology of the SAM on ITO was characterized by atomic force microscopy (AFM). As shown in Figure S52A-C, all SAMs exhibit a morphology consistent with ITO, indicating that the molecules on ITO form only a very thin layer. Figure S52D, E display the results of Kelvin probe force microscopy (KPFM) for Me-4PACz and Me-PhpPACz on ITO. We observe that Me-PhpPACz has a higher potential compared to Me-4PACz. The potential peak for Me-4PACz is -514 mV to -482 mV, with an average surface potential of approximately -498 mV. In contrast, the potential peak for Me-PhpPACz was determined be -235 mV to -216 mV, with an average surface potential of about -226 mV. The overall surface potential of Me-PhpPACz is significantly higher, indicating a deeper work function and more p-type semiconductor-like compared to Me-4PACz.^{S19} The lower potential difference for Me-PhpPACz (19 mV) compared to Me-4PACz (32 mV) is related to the packing density, and it also reflects the superior uniformity of Me-PhpPACz.^{S20}

Note S8: DFT calculation on perovskites: adsorption and charge density differences.

We successfully obtained the initial configurations of SAM molecules on the ITO surface by performing geometric optimization via DFT calculations. After getting the configurations, the binding energies of SAM molecules were calculated using the formula $E_{ad} = E_{SAM/ITO} - E_{SAM} - E_{ITO}$, where $E_{SAM/ITO}$ is the energy of adsorbed configuration, while the E_{SAM} and E_{ITO} are the energies of the SAM molecules and the ITO substrate, respectively. This binding energy is employed for comparing the bindings of various SAM molecules on the ITO surface. The static DFT calculations show that the adsorption energy of SAM molecules (Me-PhpPACz and Me-4PACz) on the surface of perovskite (FAPbI₃) is -1.48 eV and -0.23 eV, respectively (Figure S54). These results indicate that Me-PhpPACz molecules exhibit a higher affinity for adsorption onto the perovskite surface. The charge density difference provides a clear explanation for the stronger

interaction observed between the Me-PhpPACz molecule and the perovskite surface (Figure S61). The charge density transfer between Me-PhpPACz and the perovskite is distributed throughout the entire interface; in contrast, the electronic interactions between Me-4PACz and the perovskite show obvious localized features.

Note S9: PL mapping analysis.

Wavelength-resolved PL mapping shows the PL emission peak positions within a 20 μm x 20 μm scanning area (Figure S55C, D). Both perovskite samples on Me-4PACz and Me-PhpPACz substrates produce PL emissions at 788.5 nm. However, the perovskite on the Me-4PACz substrate exhibits emissions at 791 nm in some regions, indicating non-uniform perovskite growth. The perovskite on the Me-4PACz substrate shows a partial redshift compared to Me-PhpPACz, suggesting an increase in perovskite defects.^{S21,S22} The redshift of the perovskite is attributed to emissions occurring from energy states closer to the average energy due to the occupancy of capture states in highly trapped films. However, if fewer defects are present, these trap states cannot be filled, and emissions occur between states with a greater average energy separation, resulting in a blue shift in the PL spectrum.^{S21,S22}

Note S10: XRD analysis.

The influence of Me-4PACz and Me-PhpPACz as substrates on perovskite thin film growth was studied using X-ray diffraction (XRD). Except for the ITO substrate, all perovskite samples exhibited identical diffraction peaks without any second phase, indicating that the crystal structure of the perovskite remained unchanged (Figure S57). The diffraction peak at $2\theta = 12.6^\circ$ corresponds to the PbI_2 (001) crystal lattice plane, which is formed due to excess lead iodide. The perovskite peak intensity of (100) crystal plane for Me-PhpPACz was slightly higher than that for Me-4PACz, suggesting that the high-density ordered SAM substrate is favorable for perovskite crystal growth. As previously reported,^{S23-S25} the enhancement of the (100) plane and orientation improvement are widely associated with reduced defect density and improved carrier lifetime in bulk perovskite, indicating that a better SAM HSL can lead to higher-quality perovskite formation.

Note S11: Spatial charge-limited current (SCLC) analysis.

Considering the distinct crystallization and growth of perovskite on different SAM HSL, we conducted SCLC measurements using a unipolar diode structure (hole-only transport, Figure S58A). This special structure consists of ITO/SAM/PVK/ MoO_3 /Au and is employed to determine charge carrier mobility and trap state density (n_{traps}). As shown in Figure S58B and S58C, the trap-filled limit voltage (V_{TFL}) for the Me-4PACz and Me-PhpPACz samples is 0.262 V and 0.229 V, respectively. Consequently, we can calculate the corresponding trap state densities (n_{traps}) as $3.678 \times 10^{15} \text{ cm}^{-3}$ and $3.215 \times 10^{15} \text{ cm}^{-3}$, respectively. It is well-known that defects are a major cause of non-radiative recombination within the absorbing layer, leading to reduced efficiency in solution-processed thin-film solar cells. High-quality bottom interfaces can effectively reduce the density of defects generated during these processes. Furthermore, the charge carrier mobilities for Me-4PACz and Me-PhpPACz samples are $8.40 \times 10^{-4} \text{ cm}^2\text{V}^{-1}\text{s}^{-1}$ and $1.15 \times 10^{-3} \text{ cm}^2\text{V}^{-1}\text{s}^{-1}$, respectively, which align with the results from conductive AFM. These findings confirm that using Me-PhpPACz as the perovskite substrate can yield high-quality perovskites, with devices exhibiting remarkably high charge carrier mobility characteristics and an extremely low defect density. These features are crucial for achieving high-performance

PSCs.

The trap-filling limiting voltage (V_{TFL}) was defined as the voltage applied at the knee point and was determined based on the trap state density (n_{traps}), which was calculated according to the following formula:^{S26}

$$n_{traps} = \frac{2\varepsilon_0\varepsilon V_{TFL}}{eL^2}$$

where ε represents the elementary charge, L denotes the absorption layer thickness (~700 nm by SEM), e is the relative dielectric constant of perovskite (62.23),^{S27} and ε_0 is the vacuum dielectric constant (8.8542×10^{-14} F/cm). When the applied voltage is increased further, the current follows a linear relationship with the voltage squared ($n=2$), and application of the Mott-Gurney law allows us to obtain the carrier mobility (μ) using the following formula:^{S28}

$$\mu = \frac{8J_D L^3}{9\varepsilon\varepsilon_0 V^2}$$

where J_D represents the current density and V represents the applied voltage, and the relative dielectric constants, vacuum dielectric constants, and thicknesses of the various perovskite films are represented by ε , ε_0 , and L , respectively.

Note S12: UV-vis and UPS analysis.

The energy levels at the interface are also one of the main mechanisms influencing charge carrier transport. Ultraviolet-visible (UV-vis) spectroscopy indicates that the bandgaps of Me-4PACz and Me-PhpPACz molecules are 3.39 eV and 3.45 eV, respectively, while the perovskite exhibits a bandgap of ~1.55 eV (Figure S59 and S60). Ultraviolet photoelectron spectroscopy (UPS) measurements show that the highest occupied molecular orbitals (HOMO) of the bottom perovskite grown on Me-4PACz and Me-PhpPACz surfaces are at 5.64 eV and 5.55 eV, respectively, with Fermi levels at 4.23 and 4.64 eV (Table S5). This indicates that perovskite growth on different substrates affects its Fermi level. We also measured the HOMO and Fermi levels of the perovskite grown on ITO, which were found to be 5.65 eV and 4.12 eV, respectively. It's evident that the Fermi level of the bottom interface perovskite on both ITO and Me-4PACz is close to the conduction band, whereas on Me-PhpPACz, it is farther from the conduction band, which enhances hole transport.^{S29,S30} The UPS results show that the Fermi energy levels of Me-4PACz and Me-PhpPACz are 4.55 eV and 4.65 eV, respectively, indicating that the Fermi energy level between the HSL modified ITO and the perovskite is better matched in the case of Me-PhpPACz.

Note S13: Ultrafast spectroscopic analysis.

The measured surface carrier kinetics can be modelled with diffusion and surface recombination/transfer model. The carriers are generated with profile

$$N(x, 0) = N_0 e^{-\alpha x}$$

where N_0 is the initial surface carrier density. If the traces are normalized, then N_0 is equal to 1. The values for the absorption coefficient at different pump-photon energies for samples are known. The carrier movement obeys the diffusion and surface recombination/transfer equation:

$$\frac{\partial N(x, t)}{\partial t} = D \frac{\partial^2 N(x, t)}{\partial x^2} - \frac{N(x, t)}{\tau_B}$$
$$\frac{\partial N(x, t)}{\partial t} \Big|_{x=0} = \frac{S_F}{D} N(0, t)$$

$$\frac{\partial N(x, t)}{\partial t} \Big|_{x=L} = -\frac{S_L}{D} N(L, t)$$

where $N(x, t)$ is the carrier density as a function of depth (x) and time (t), D is the ambipolar diffusion coefficient and τ_B is the bulk carrier lifetime. Generally, in perovskite material, τ_B is on the order of hundreds of ns, in the 5000 ps probe window, it can be neglected. The boundary condition can be described by the followings, for the front surface, surface recombination/transfer will dominate (S_F). For the back surface, surface recombination/transfer happens with S_L . For pure perovskites, the back surface and front surface can be assumed to have the same surface recombination velocity. For samples with HSL, the front surface transfer velocity is different from the back where the back (S_L) takes the value of pure perovskites. L is the thickness of the film.

Note S14: PL and TRPL analysis.

The charge dynamics during hole extraction and transport processes in PSCs were investigated through measurements of steady-state photoluminescence (PL, Figure S62A) and time-resolved photoluminescence (TRPL, Figure S62B). Perovskite samples were prepared on hole selected layer of Me-4PACz and Me-PhpPACz for PL measurements, with emission peaks at 786 nm and 784 nm, respectively. The blue shift in the emission peak indicates a reduction in defects and the suppression of non-radiative charge recombination.^{S31, S32} In comparison to perovskite on the Me-4PACz substrate, the PL intensity of perovskite on the Me-PhpPACz substrate is significantly quenched, with a 11.23% reduction in PL quenching, suggesting that Me-PhpPACz exhibits enhanced charge extraction capabilities compared to Me-4PACz.^{S31, S32} Further insights into charge carrier extraction and transport characteristics were obtained through TRPL measurements (Figure S62B). The TRPL data were fitted using a biexponential function (Table S6).^{S33, S34} The average lifetime τ_{avg} was calculated using the following formula:

$$\tau_{avg} = \frac{\sum A_i \tau_i^2}{\sum A_i \tau_i}$$

The τ_{avg} for perovskite on the Me-4PACz substrate is 331.15 ns ($\tau_1 = 27.14$ ns, $\tau_2 = 609.41$ ns), whereas for perovskite on the Me-PhpPACz substrate, τ_{avg} is 82.96 ns ($\tau_1 = 15.65$ ns, $\tau_2 = 1139.24$ ns). The fast initial decay represents a combination of charge extraction (quenching) and non-radiative recombination processes (τ_1). While the slower decay lifetime (τ_2) is predominantly linked to bulk and surface trap-assisted recombination, this is widely recognized as a reliable hallmark of high-quality perovskite (Table S6).^{S24, S31, S32, S35} According to this theory, we attribute the similar initial fast decay observed in TRPL for Me-PhpPACz and Me-4PACz to retained charge extraction, and the significantly increased slow decay lifetime is attributed to reduced non-radiative recombination. The PL and TRPL results indicate that Me-PhpPACz can enhance the charge extraction capabilities of the devices and improve the quality of the perovskite film.

Note S15: J_{sc} and V_{oc} under different light intensity, and S-Q limit analysis.

To further investigate the charge carrier recombination mechanisms, we also evaluated the performance of J_{sc} and V_{oc} under different light intensities. Linear dependencies of J_{sc} and V_{oc} on light intensity were obtained based on the original J - V curves (Figure S68). The slope of V_{oc} relative to light intensity is represented as the ideality factor (n_{ID}), slope = $n_{ID} k_B T / q$, where q , k_B , and T respectively denote the elementary charge, Boltzmann's constant, and Kelvin

temperature.^{S36} It is noteworthy that, compared to devices with Me-4PACz as the SAM, the n_{ID} value for devices with Me-PhpPACz decreased from 1.67 kT/q to 1.20 kT/q , indicating that charge carrier recombination in the PSC is suppressed (Figure S68A).^{S37} Trap-assisted non-radiative losses and charge transport losses are the two main reasons for the observed FF being lower than the S-Q limit deficiency, The maximum FF (FF_{max}) can be empirically calculated using the following formula:^{S38,S39}

$$FF_{max} = \frac{v_{oc} - \ln(v_{oc} + 0.72)}{v_{oc} + 1}$$

where

$$v_{oc} = \frac{V_{oc}}{nk_B T/q}$$

where n and V_{oc} represent the ideality factor (n_{ID}) and the open-circuit voltage of the device, respectively. As shown in manuscript Figure 4E, the calculated FF_{max} values for Me-4PACz and Me-PhpPACz are 84.66% and 88.15%, respectively. Trap-assisted non-radiative losses and charge transport losses in Me-PhpPACz devices are significantly suppressed, confirming that the use of Me-PhpPACz as the HSL at the bottom interface can suppress trap-assisted non-radiative recombination and improve charge transport in the device.^{S25,S37} Additionally, we also analyzed the slope of α , which is an index factor related to current intensity and current density. When the α value is close to 1, it indicates the absence of space-charge effects in the device. The α value for devices with Me-4PACz as HSL is 0.989, and for devices with Me-PhpPACz as HSL, the α value is 0.990 (Figure S68B). Both α values are close to 1, suggesting that double-molecule recombination occurs less frequently in both Me-4PACz and Me-PhpPACz-based PSCs.^{S31}

Note S16: EIS analysis.

To investigate the specific charge transfer processes in perovskite devices, we conducted electrochemical impedance spectroscopy (EIS) measurements at an applied voltage of 0.4 V using a simulated circuit diagram (Figure S69). This allowed us to obtain Nyquist plots and results that fit the series resistance (R_s), charge transfer resistance (R_{CT}), and recombination resistance (R_{Rec}). Our research results indicate that compared to the PSC with Me-4PACz as the HSL, the R_s of the PSC with Me-PhpPACz significantly decreased from 30.52 Ω to 23.49 Ω . This suggests that the SAM with conjugated linker structure enhances the migration of charge carriers at the perovskite layer interface, thereby reducing recombination of carriers within the bulk. The R_{CT} decreased from 34,102 Ω to 28,934 Ω , indicating that the self-assembled Me-PhpPACz layer improved the contact at the perovskite/HSL interface. Additionally, we observed that R_{Rec} increased from 11,389 Ω for Me-4PACz to 41,978 Ω for Me-PhpPACz, primarily due to the dense SAM layer forming a high-quality perovskite, thereby suppressing current leakage caused by defects within the perovskite layer.^{S34}

Note S17: Mott-Schottky analysis.

Mott-Schottky plot analysis reveals three regions (Figure S70), C_g , C_{dl} , and C_s , representing geometric capacitance, depletion layer capacitance, and electrode polarization capacitance, respectively.^{S40} The Mott-Schottky plot shows a very narrow C_{dl} region, indicating that the studied devices contain layers with fewer defects, good film quality, and a low intrinsic defect density.^{S41,S42} The built-in potential (V_{bi}) of the PSC is extracted from the C_{dl} region, with Me-

PhpPACz having a V_{bi} of 1.175 V and Me-4PACz devices having a V_{bi} of 1.134 V. The improved V_{bi} will contribute to enhancing the V_{OC} of the device.

Note S18: TPC and TPV analysis.

We also utilized transient photocurrent (TPC) and transient photovoltage (TPV) to study the charge extraction and recombination behavior (Figure S71). TPC determines the overall efficiency of a solar cell by measuring the rate at which carriers generated by light are collected by the electrodes. As shown in Figure S71A, the TPC decay curve for the PSC based on Me-PhpPACz (0.67 μ s) exhibits a faster photocurrent decay response compared to the PSC based on Me-4PACz (1.42 μ s). Slower carrier collection can lead to significant losses due to recombination and diffusion.^{S43} TPV, on the other hand, assesses the efficiency of a solar cell by testing the separation of carriers generated by light. The TPV decay curve lifetime for Me-PhpPACz (74.93 μ s) is longer than that of Me-4PACz-based PSCs (60.91 μ s) (Figure S71B), and a lower TPV can indicate poor absorption layer quality or a mismatch between the bandgap of perovskite.^{S43} These results further confirm that the conjugated linker of SAM contributes to accelerating charge extraction and reducing defects.

Note S19: EQEEL, PLQY, and QFLS analysis.

Manuscript Figure 4F and Supplemental Figure S72 depict the variation curves of external quantum efficiency for electroluminescence (EQEEL) and current density with voltage bias for Me-4PACz and Me-PhpPACz-based PSCs when functioning as LEDs. The electroluminescence (EL) emission spectra at different voltages are presented in Figure S72A, with Me-4PACz and Me-PhpPACz exhibiting EL emissions at 790 nm and 786 nm, respectively. A blue-shift in the EL emission peak is observed for Me-PhpPACz, consistent with the PL results. Additionally, Me-PhpPACz-based PSC devices show higher EL emission intensity, indicating a higher level of carrier injection.^{S44} Furthermore, the turn-on voltage for Me-PhpPACz-based PSC devices is 1.28 V, which is lower than that for Me-4PACz-based PSC devices (1.43 V, Figure S72B). This suggests that Me-PhpPACz-based PSC devices have more balanced carrier injection, reduced leakage current, lower energy losses, and decreased non-radiative recombination.^{S44,S45} Figure S72C illustrates the current density as a function of EQEEL for Me-4PACz and Me-PhpPACz-based PSCs when operating as LEDs. When solar cell devices function as LEDs, their V_{OC} values are positively correlated with EQEEL values^[S24] The voltage variation (ΔV_{OC}) can be determined using the formula $\Delta V_{OC} = (k_B T/q) \ln(EQE_{Me-PhpPACz}/EQE_{Me-4PACz})$.^{S46} The EQEEL values for Me-4PACz and Me-PhpPACz-based PSCs are 3.23% and 4.65%, respectively. According to the formula, ΔV_{OC} is calculated to be 9.41 mV, which is nearly consistent with the J-V results.

To better quantify interface non-radiative recombination losses, we measured the photoluminescence quantum yield (PLQY) for partially stacked and fully stacked perovskite films on different hole transport layers to estimate the quasi-Fermi level splitting (QFLS, Figure S72D).^{S47} We calculated the internal electron-hole QFLS from PLQY results by using the equation:^{S47}

$$QFLS = k_B \times \ln \left(PLQY \times \frac{J_G}{J_{0,rad}} \right)$$

Where J_G represents the generated current density under 1-sun illumination, approximated by the short-circuit current density of the entire solar cell. $J_{0,rad}$ signifies the radiative recombination

current in dark conditions, determined through the integration of the overlap between the external quantum efficiency of the full device (EQE) and the black body radiative spectrum at 300 K across the energy spectrum.

$$J_{0,rad} = \int EQE \phi_{BB} d\epsilon$$

$$\phi_{BB} = \frac{1}{4^2 h^3 c^2} \frac{E^2}{\exp\left(\frac{E}{k_B T}\right) - 1}$$

Where E represents photon energy, h is the Planck constant, and c is the speed of light in a vacuum. The calculated value of $J_{0,rad}$ is $9.72 \times 10^{-20} \text{ A m}^{-2}$, based on the EQE spectrum of the full device and the black body radiative spectrum. The perovskite film on Me-4PACz exhibited a low PLQY (7.91%), which we attributed to severe non-radiative recombination losses. In contrast, the perovskite film on Me-PhpPACz achieved a high PLQY of up to 12.75%. Compared to the QFLS value of 1.207 eV for Me-4PACz-based perovskite, the QFLS value for Me-PhpPACz samples increased slightly to 1.220 eV. When compared to PLQY of partially stacked, the fully stacked structure devices exhibited a significant decrease in PLQY due to non-radiative recombination at the perovskite/C₆₀ interface. The PLQY for Me-PhpPACz-based samples remained around 6.89%, while that of 2PACz decreased to 2.59%. Additionally, the fully stacked QFLS values for Me-4PACz and Me-PhpPACz decreased to 1.179 eV and 1.204 eV, respectively. This indicates that improved energy level alignment and optimized bottom structures contribute to reduced perovskite defects and minimize interface non-radiative recombination losses.^{S48,S49}

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